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Learning Transferable Visual Models From Natural Language Supervision

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Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at `https://github.com/OpenAI/CLIP`.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019). Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of "text-to-text" as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which preformed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

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从自然语言监督中学习可迁移的视觉模型

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摘要

最先进的计算机视觉系统被训练用于预测一组固定的预先定义的对象类别。这种受限的监督形式限制了它们的通用性和可用性，因为若要指定任何其他视觉概念，都需要额外的标注数据。直接从关于图像的原始文本中学习是一种有前景的替代方案，它利用了范围更广的监督来源。我们证明，在从互联网收集的 4 亿个（图像，文本）对数据集上，从零开始学习 SOTA 图像表示时，预测哪条标题与哪张图像匹配这一简单的预训练任务是一种高效且可扩展的方法。预训练之后，使用自然语言来引用已学习的视觉概念（或描述新的概念），从而使模型能够零样本迁移到下游任务。我们通过在 30 多个不同的现有计算机视觉数据集上进行基准测试来研究这种方法的性能，这些数据集涵盖了 OCR、视频中的动作识别、地理定位以及多种细粒度对象分类等任务。该模型在大多数任务上都能实现非平凡的迁移，并且通常无需任何特定数据集的训练就能与完全监督的基线相媲美。例如，在 ImageNet 上进行零样本评估时，我们达到了原始 ResNet-50 的准确率，而无需使用其训练所依赖的 128 万个训练样本中的任何一个。我们在 https://github.com/OpenAI/CLIP 发布了代码和预训练模型权重。

1. 引言与研究动机

直接从原始文本中学习的预训练方法在过去几年彻底改变了自然语言处理领域（Dai & Le, 2015；Peters et al., 2018；Howard & Ruder, 2018；Radford et al., 2018；Devlin et al., 2018；Raffel et al., 2019）。自回归和掩码语言建模等与任务无关的目标，已在计算量、模型容量和数据规模上跨越多个数量级扩展，并持续提升能力。将“文本到文本”发展为标准化的输入-输出接口（McCann 等，2018；Radford 等，2019；Raffel 等，2019），使得与任务无关的架构能够以零样本方式迁移到下游数据集，从而无需专门的输出头或针对特定数据集的定制。像 GPT-3（Brown 等，2020）这样的旗舰系统，如今在许多任务上已能与定制模型相竞争，同时几乎不需要或完全不需要特定数据集的训练数据。

这些结果表明，现代预训练方法能够从网络规模文本集合中获得的聚合监督信号，超过了高质量众包标注 NLP 数据集所提供的监督信号。然而，在计算机视觉等其他领域，仍然普遍采用在 ImageNet（Deng 等，2009）等众包标注数据集上预训练模型的做法。可扩展的预训练方法若能直接从网络文本中学习，是否也能在计算机视觉领域带来类似的突破？先前的工作令人鼓舞。

20多年前，Mori等人（1999）探索了通过训练模型来预测与图像配对的文本文档中的名词和形容词，从而改进基于内容的图像检索。Quattoni等人（2007）证明，可以通过在为预测图像相关说明文字中的词语而训练的分类器的权重空间中进行流形学习，以更高的数据效率学习图像表示。Srivastava和Salakhutdinov（2012）探索了深度表示学习，他们在低层次图像和文本标签特征之上训练多模态深度玻尔兹曼机。Joulin等人（2016）使这一研究方向现代化，并证明了训练用于预测图像说明文字中词语的CNN能够学习到有用的图像表示。他们将YFCC100M数据集（Thomee等人，2016）中图像的标题、描述和话题标签元数据转换为词袋式多标签分类任务，并表明，预训练AlexNet（Krizhevsky等人，2012）来预测这些标签所学习到的表示，在迁移任务上的表现与基于ImageNet的预训练相近。随后，Li等人（2017）将这一方法扩展到除单个词之外还预测短语n-gram，并展示了其系统零样本迁移到其他图像的能力。

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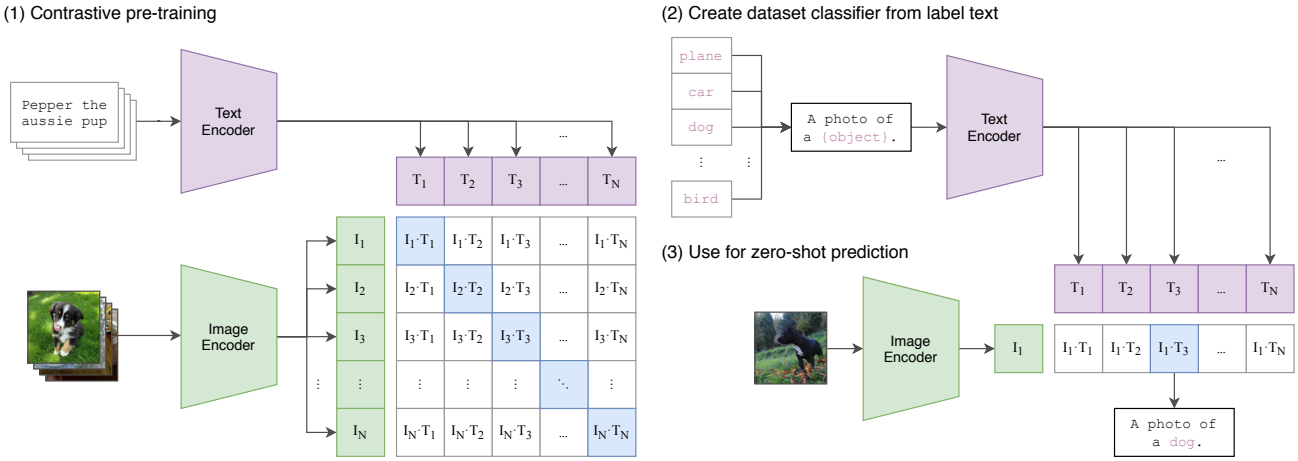


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

classification datasets by scoring target classes based on their dictionary of learned visual n-grams and predicting the one with the highest score. Adopting more recent architectures and pre-training approaches, VirTex (Desai & Johnson, 2020), ICMLM (Bulent Sariyildiz et al., 2020), and ConVIRT (Zhang et al., 2020) have recently demonstrated the potential of transformer-based language modeling, masked language modeling, and contrastive objectives to learn image representations from text.

While exciting as proofs of concept, using natural language supervision for image representation learning is still rare. This is likely because demonstrated performance on common benchmarks is much lower than alternative approaches. For example, Li et al. (2017) reach only 11.5% accuracy on ImageNet in a zero-shot setting. This is well below the 88.4% accuracy of the current state of the art (Xie et al., 2020). It is even below the 50% accuracy of classic computer vision approaches (Deng et al., 2012). Instead, more narrowly scoped but well-targeted uses of weak supervision have improved performance. Mahajan et al. (2018) showed that predicting ImageNet-related hashtags on Instagram images is an effective pre-training task. When fine-tuned to ImageNet these pre-trained models increased accuracy by over 5% and improved the overall state of the art at the time. Kolesnikov et al. (2019) and Dosovitskiy et al. (2020) have also demonstrated large gains on a broader set of transfer benchmarks by pre-training models to predict the classes of the noisily labeled JFT-300M dataset.

This line of work represents the current pragmatic middle ground between learning from a limited amount of supervised "gold-labels" and learning from practically unlimited amounts of raw text. However, it is not without compromises. Both works carefully design, and in the process limit, their supervision to 1000 and 18291 classes respectively. Natural language is able to express, and therefore supervise, a much wider set of visual concepts through its generality. Both approaches also use static softmax classifiers to perform prediction and lack a mechanism for dynamic outputs. This severely curtails their flexibility and limits their "zero-shot" capabilities.

A crucial difference between these weakly supervised models and recent explorations of learning image representations directly from natural language is scale. While Mahajan et al. (2018) and Kolesnikov et al. (2019) trained their models for accelerator years on millions to billions of images, VirTex, ICMLM, and ConVIRT trained for accelerator days on one to two hundred thousand images. In this work, we close this gap and study the behaviors of image classifiers trained with natural language supervision at large scale. Enabled by the large amounts of publicly available data of this form on the internet, we create a new dataset of 400 million (image, text) pairs and demonstrate that a simplified version of ConVIRT trained from scratch, which we call CLIP, for Contrastive Language-Image Pre-training, is an efficient method of learning from natural language supervision. We study the scalability of CLIP by training a series of eight models spanning almost 2 orders of magnitude of compute and observe that transfer performance is a smoothly predictable function of compute (Hestness et al., 2017; Kaplan et al., 2020). We find that CLIP, similar to the GPT family, learns to perform a wide set of tasks during pre-training including OCR, geo-localization, action recognition, and many others. We measure this by benchmarking the zero-shot transfer performance of CLIP on over 30 existing datasets and find

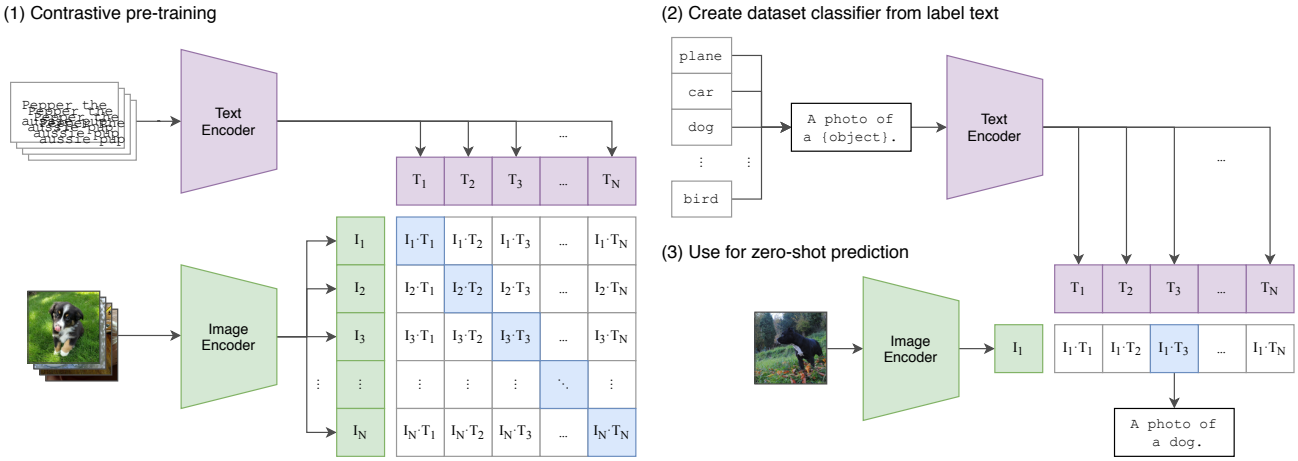


图 1. 我们方法的总结。标准图像模型联合训练图像特征提取器和线性分类器来预测某个标签，而 CLIP 联合训练图像编码器和文本编码器，以预测一批（图像、文本）训练样本中的正确配对。在测试时，学习到的文本编码器通过嵌入目标数据集类别的名称或描述来合成一个零样本线性分类器。

通过基于目标类别的已学习视觉 n-gram 词典为其打分，并预测得分最高的类别，用于分类数据集。采用更新的架构和预训练方法，VirTex（Desai & Johnson，2020）、ICMLM（Bulent Sariyildiz 等，2020）以及 ConVIRT（Zhang 等，2020）近期展示了基于 Transformer 的语言建模、掩码语言建模和对比目标在从文本中学习图像表示方面的潜力。

尽管作为概念验证令人振奋，但使用自然语言监督来进行图像表示学习仍然很少见。这很可能是因为其在常见基准上展示出的性能远低于其他方法。例如，Li 等人（2017）在零样本设置下在 ImageNet 上仅达到 11.5% 的准确率。这远低于当前最先进水平的 88.4% 准确率（Xie 等人，2020）。甚至低于经典计算机视觉方法 50% 的准确率（Deng 等人，2012）。相比之下，范围更窄但目标明确的弱监督用法提升了性能。Mahajan 等人（2018）表明，预测 Instagram 图像上与 ImageNet 相关的标签是一项有效的预训练任务。当在 ImageNet 上进行微调时，这些预训练模型将准确率提高了超过 5%，并提升了当时的整体最先进水平。Kolesnikov 等人（2019）和 Dosovitskiy 等人（2020）也通过预训练模型来预测带有噪声标签的 JFT-300M 数据集的类别，在更广泛的一组迁移基准上展示了显著提升。

这一研究方向代表了当前在从有限数量的有监督“金标准标签”中学习与从几乎无限量的原始文本中学习之间的务实折中。然而，它并非没有妥协之处。

mises。这两项工作都精心设计了它们的监督方式，并在此过程中将其监督范围分别限制在 1000 个和 18291 个类别。自然语言能够表达，并因此监督，范围广泛得多的视觉概念集合，这得益于其通用性。这两种方法还都使用静态 softmax 分类器进行预测，并且缺乏动态输出机制。这严重削弱了它们的灵活性，并限制了它们的“零样本”能力。

这些弱监督模型与近期关于直接从自然语言中学习图像表示的探索之间，一个关键差异在于规模。Mahajan 等人（2018）和 Kolesnikov 等人（2019）在数百万到数十亿张图像上训练其模型，耗费了以加速器年计的计算资源；而 VirTex、ICMLM 和 ConVIRT 则在十万到二十万张图像上训练，耗费的是以加速器天计的计算资源。在这项工作中，我们弥合了这一差距，并研究了在大规模自然语言监督下训练的图像分类器的行为。借助互联网上大量公开可用的此类数据，我们创建了一个包含 4 亿个（图像，文本）对的新数据集，并证明了一种从头训练的简化版 ConVIRT——我们称之为 CLIP，即对比语言-图像预训练（Contrastive Language-Image Pre-training）——是一种高效的从自然语言监督中学习的方法。我们通过训练一系列八个模型来研究 CLIP 的可扩展性，这些模型覆盖了近两个数量级的计算量，并观察到迁移性能是计算量的一个平滑且可预测的函数（Hestness 等，2017；Kaplan 等，2020）。我们发现，类似于 GPT 系列，CLIP 在预训练过程中学会执行广泛的任务，包括 OCR、地理定位、动作识别以及许多其他任务。我们通过在 30 多个现有数据集上评测 CLIP 的零样本迁移性能来衡量这一点，并发现

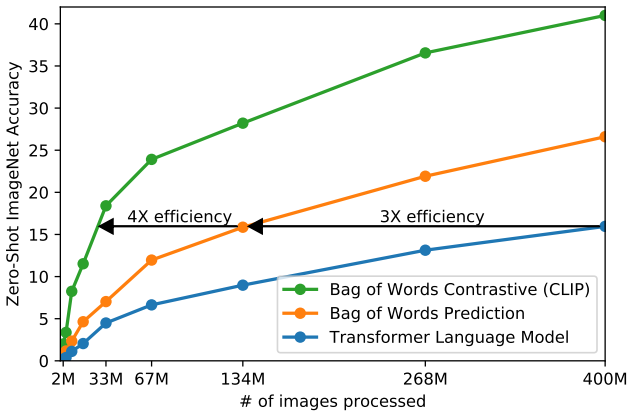


Figure 2. **CLIP is much more efficient at zero-shot transfer than our image caption baseline.** Although highly expressive, we found that transformer-based language models are relatively weak at zero-shot ImageNet classification. Here, we see that it learns 3x slower than a baseline which predicts a bag-of-words (BoW) encoding of the text (Joulin et al., 2016). Swapping the prediction objective for the contrastive objective of CLIP further improves efficiency another 4x.

it can be competitive with prior task-specific supervised models. We also confirm these findings with linear-probe representation learning analysis and show that CLIP outperforms the best publicly available ImageNet model while also being more computationally efficient. We additionally find that zero-shot CLIP models are much more robust than equivalent accuracy supervised ImageNet models which suggests that zero-shot evaluation of task-agnostic models is much more representative of a model's capability. These results have significant policy and ethical implications, which we consider in Section 7.

2. Approach

2.1. Natural Language Supervision

At the core of our approach is the idea of learning perception from supervision contained in natural language. As discussed in the introduction, this is not at all a new idea, however terminology used to describe work in this space is varied, even seemingly contradictory, and stated motivations are diverse. Zhang et al. (2020), Gomez et al. (2017), Joulin et al. (2016), and Desai & Johnson (2020) all introduce methods which learn visual representations from text paired with images but describe their approaches as unsupervised, self-supervised, weakly supervised, and supervised respectively.

We emphasize that what is common across this line of work is not any of the details of the particular methods used but the appreciation of natural language as a training signal. All these approaches are learning from *natural language supervision*. Although early work wrestled with the complexity of natural language when using topic model and n-gram representations, improvements in deep contextual representation learning suggest we now have the tools to effectively leverage this abundant source of supervision (McCann et al., 2017).

Learning from natural language has several potential strengths over other training methods. It's much easier to scale natural language supervision compared to standard crowd-sourced labeling for image classification since it does not require annotations to be in a classic "machine learning compatible format" such as the canonical 1-of-N majority vote "gold label". Instead, methods which work on natural language can learn passively from the supervision contained in the vast amount of text on the internet. Learning from natural language also has an important advantage over most unsupervised or self-supervised learning approaches in that it doesn't "just" learn a representation but also connects that representation to language which enables flexible zero-shot transfer. In the following subsections, we detail the specific approach we settled on.

2.2. Creating a Sufficiently Large Dataset

Existing work has mainly used three datasets, MS-COCO (Lin et al., 2014), Visual Genome (Krishna et al., 2017), and YFCC100M (Thomee et al., 2016). While MS-COCO and Visual Genome are high quality crowd-labeled datasets, they are small by modern standards with approximately 100,000 training photos each. By comparison, other computer vision systems are trained on up to 3.5 *billion* Instagram photos (Mahajan et al., 2018). YFCC100M, at 100 million photos, is a possible alternative, but the metadata for each image is sparse and of varying quality. Many images use automatically generated filenames like `20160716_113957.JPG` as "titles" or contain "descriptions" of camera exposure settings. After filtering to keep only images with natural language titles and/or descriptions in English, the dataset shrunk by a factor of 6 to only 15 million photos. This is approximately the same size as ImageNet.

A major motivation for natural language supervision is the large quantities of data of this form available publicly on the internet. Since existing datasets do not adequately reflect this possibility, considering results only on them would underestimate the potential of this line of research. To address this, we constructed a new dataset of 400 million (image, text) pairs collected form a variety of publicly available sources on the Internet. To attempt to cover as broad a set of visual concepts as possible, we search for (image, text) pairs as part of the construction process whose text includes one of a set of 500,000 queries.[1] We approximately class

[1]The base query list is all words occurring at least 100 times in the English version of Wikipedia. This is augmented with bi-grams

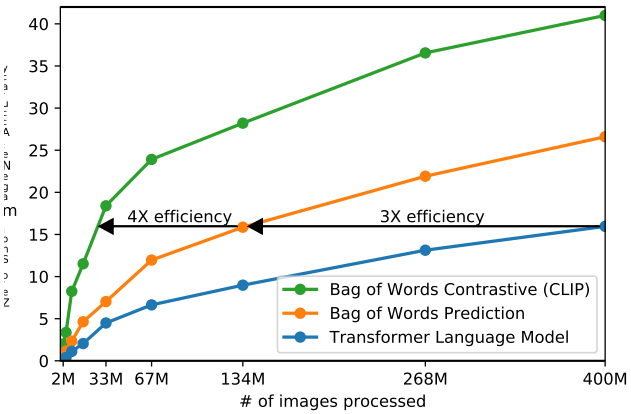


图 2. CLIP 在零样本迁移方面比我们的图像描述基线高效得多。尽管基于 Transformer 的语言模型具有很强的表达能力，但我们发现它们在零样本 ImageNet 分类上相对较弱。在这里，我们看到它的学习速度比一个预测文本词袋（BoW）编码的基线慢 3 倍（Joulin 等，2016）。将预测目标替换为 CLIP 的对比目标后，效率又进一步提高了 4 倍。

它可以与以往针对特定任务的监督模型相竞争。我们还通过线性探针表征学习分析证实了这些发现，并表明 CLIP 优于目前公开可用的最佳 ImageNet 模型，同时计算效率也更高。此外，我们发现零样本 CLIP 模型比具有同等准确率的监督式 ImageNet 模型鲁棒性强得多，这表明对任务无关模型进行零样本评估更能代表模型的能力。这些结果具有重要的政策和伦理影响，我们将在第 7 节中讨论。

2. 方法

2.1. 自然语言监督

我们方法的核心思想是从自然语言中包含的监督信号来学习感知。正如引言中所讨论的，这并不是一个新想法；然而，用于描述这一领域工作的术语各不相同，甚至看似相互矛盾，并且其所陈述的动机也多种多样。Zhang 等（2020）、Gomez 等（2017）、Joulin 等（2016）以及 Desai 和 Johnson（2020）都提出了从与图像配对的文本中学习视觉表征的方法，但他们分别将自己的方法描述为无监督、自监督、弱监督和监督式方法。

我们强调，这一系列工作中的共同点并不是所采用的具体方法的任何细节，而是对自然语言作为训练信号的重视。所有这些方法都在从 *natural language supervision* 中学习。尽管早期工作在使用主题模型和 n-gram 表示时曾努力应对自然语言的复杂性，但深度上下文表示学习的进展表明，我们现在已有工具能够有效利用这一丰富的监督来源（McCann 等，2017）。

从自然语言中学习相较于其他训练方法具有若干潜在优势。与图像分类中标准的众包标注相比，自然语言监督更容易扩展，因为它不要求标注采用经典的“兼容机器学习的格式”，例如规范的 N 选 1 多数投票“黄金标签”。相反，处理自然语言的方法可以从互联网上海量文本中所包含的监督信号中被动学习。从自然语言中学习相较于大多数无监督或自监督学习方法还有一个重要优势：它并不“仅仅”学习一种表示，还将该表示与语言连接起来，从而实现灵活的零样本迁移。在以下小节中，我们将详细介绍最终采用的具体方法。

2.2. 创建足够大的数据集

现有工作主要使用了三个数据集：MS-COCO（Lin 等，2014）、Visual Genome（Krishna 等，2017）和 YFCC100M（Thomee 等，2016）。虽然 MS-COCO 和 Visual Genome 是高质量的众包标注数据集，但按照现代标准来看，它们规模较小，每个数据集约有 100,000 张训练照片。相比之下，其他计算机视觉系统使用多达 3.5 *billion* 张 Instagram 照片进行训练（Mahajan 等，2018）。YFCC100M 拥有 1 亿张照片，是一种可能的替代选择，但每张图像的元数据都很稀疏且质量参差不齐。许多图像使用自动生成的文件名（如 20160716 113957.JPG）作为“标题”，或包含相机曝光设置的“描述”。在筛选后仅保留具有英文自然语言标题和/或描述的图像后，该数据集规模缩小了 6 倍，仅剩 1500 万张照片。这与 ImageNet 的规模大致相同。

自然语言监督的一个主要动机是，互联网上公开可获得大量这种形式的数据。由于现有数据集无法充分反映这种可能性，如果只考虑在这些数据集上的结果，就会低估这一研究方向的潜力。为了解决这一问题，我们构建了一个新的数据集，包含从互联网上各种公开可用来源收集的 4 亿个（图像，文本）对。为了尽可能覆盖广泛的视觉概念，我们在构建过程中搜索其文本包含 500,000 个查询之一的（图像，文本）对。[1] 我们大致分类

[1]The base query list is all words occurring at least 100 times in the English version of Wikipedia. This is augmented with bi-grams

balance the results by including up to 20,000 (image, text) pairs per query. The resulting dataset has a similar total word count as the WebText dataset used to train GPT-2. We refer to this dataset as WIT for WebImageText.

2.3. Selecting an Efficient Pre-Training Method

State-of-the-art computer vision systems use very large amounts of compute. Mahajan et al. (2018) required 19 GPU years to train their ResNeXt101-32x48d and Xie et al. (2020) required 33 TPUv3 core-years to train their Noisy Student EfficientNet-L2. When considering that both these systems were trained to predict only 1000 ImageNet classes, the task of learning an open set of visual concepts from natural language seems daunting. In the course of our efforts, we found training efficiency was key to successfully scaling natural language supervision and we selected our final pre-training method based on this metric.

Our initial approach, similar to VirTex, jointly trained an image CNN and text transformer from scratch to predict the caption of an image. However, we encountered difficulties efficiently scaling this method. In Figure 2 we show that a 63 million parameter transformer language model, which already uses twice the compute of its ResNet-50 image encoder, learns to recognize ImageNet classes three times slower than a much simpler baseline that predicts a bag-of-words encoding of the same text.

Both these approaches share a key similarity. They try to predict the *exact* words of the text accompanying each image. This is a difficult task due to the wide variety of descriptions, comments, and related text that co-occur with images. Recent work in contrastive representation learning for images has found that contrastive objectives can learn better representations than their equivalent predictive objective (Tian et al., 2019). Other work has found that although generative models of images can learn high quality image representations, they require over an order of magnitude more compute than contrastive models with the same performance (Chen et al., 2020a). Noting these findings, we explored training a system to solve the potentially easier proxy task of predicting only which text *as a whole* is paired with which image and not the exact words of that text. Starting with the same bag-of-words encoding baseline, we swapped the predictive objective for a contrastive objective in Figure 2 and observed a further 4x efficiency improvement in the rate of zero-shot transfer to ImageNet.

Given a batch of N (image, text) pairs, CLIP is trained to predict which of the $N \times N$ possible (image, text) pairings across a batch actually occurred. To do this, CLIP learns a multi-modal embedding space by jointly training an image encoder and text encoder to maximize the cosine similarity of the image and text embeddings of the N real pairs in the batch while minimizing the cosine similarity of the embeddings of the $N^2 - N$ incorrect pairings. We optimize a symmetric cross entropy loss over these similarity scores. In Figure 3 we include pseudocode of the core of an implementation of CLIP. To our knowledge this batch construction technique and objective was first introduced in the area of deep metric learning as the *multi-class N-pair loss* Sohn (2016), was popularized for contrastive representation learning by Oord et al. (2018) as the InfoNCE loss, and was recently adapted for contrastive (text, image) representation learning in the domain of medical imaging by Zhang et al. (2020).

Due to the large size of our pre-training dataset, over-fitting is not a major concern and the details of training CLIP are simplified compared to the implementation of Zhang et al. (2020). We train CLIP from scratch without initializing the image encoder with ImageNet weights or the text encoder with pre-trained weights. We do not use the non-linear projection between the representation and the contrastive embedding space, a change which was introduced by Bachman et al. (2019) and popularized by Chen et al. (2020b). We instead use only a linear projection to map from each encoder's representation to the multi-modal embedding space. We did not notice a difference in training efficiency between the two versions and speculate that non-linear projections may be co-adapted with details of current image only in self-supervised representation learning methods. We also remove the text transformation function t_u from Zhang et al. (2020) which samples a single sentence at uniform from the text since many of the (image, text) pairs in CLIP's pre-training dataset are only a single sentence. We also simplify the image transformation function t_v. A random square crop from resized images is the only data augmentation used during training. Finally, the temperature parameter which controls the range of the logits in the softmax, τ, is directly optimized during training as a log-parameterized multiplicative scalar to avoid turning as a hyper-parameter.

2.4. Choosing and Scaling a Model

We consider two different architectures for the image encoder. For the first, we use ResNet-50 (He et al., 2016a) as the base architecture for the image encoder due to its widespread adoption and proven performance. We make several modifications to the original version using the ResNet-D improvements from He et al. (2019) and the antialiased rect-2 blur pooling from Zhang (2019). We also replace the global average pooling layer with an attention pooling mechanism. The attention pooling is implemented as a single layer of "transformer-style" multi-head QKV attention where the query is conditioned on the global average-pooled

with high pointwise mutual information as well as the names of all Wikipedia articles above a certain search volume. Finally all WordNet synsets not already in the query list are added.

通过每个查询最多包含 20,000 个（图像，文本）对来平衡结果。所得数据集的总词数与用于训练 GPT-2 的 WebText 数据集相近。我们将该数据集称为 WIT，即 WebImageText。

2.3. 选择高效的预训练方法

最先进的计算机视觉系统使用非常大量的计算资源。Mahajan 等人（2018）需要 19 个 GPU 年来训练他们的 ResNeXt101-32x48d，而 Xie 等人（2020）需要 33 个 TPUv3 核心年来训练他们的 Noisy Student EfficientNet-L2。考虑到这两个系统都只是被训练用于预测 1000 个 ImageNet 类别，从自然语言中学习开放集合的视觉概念这一任务似乎令人望而生畏。在我们的工作过程中，我们发现训练效率是成功扩展自然语言监督的关键，并且我们基于这一指标选择了最终的预训练方法。

我们最初的方法与 VirTex 类似，从头开始联合训练一个图像 CNN 和一个文本 Transformer，以预测图像的说明文字。然而，我们在高效扩展这种方法时遇到了困难。在图 2 中，我们展示了一个拥有 6300 万参数的 Transformer 语言模型，尽管其计算量已经是其 ResNet-50 图像编码器的两倍，但它学习识别 ImageNet 类别的速度却比一个简单得多的基线模型慢三倍；该基线模型预测的是同一文本的词袋编码。

这两种方法有一个关键的相似之处。它们都试图预测每张图像所配文本中的 *exact* 个词。由于与图像共同出现的描述、评论和相关文本种类繁多，这是一项困难的任务。近期关于图像对比表示学习的工作发现，对比目标可以比其等价的预测目标学习到更好的表示（Tian et al., 2019）。其他工作发现，尽管图像生成模型可以学习到高质量的图像表示，但要达到相同性能，它们所需的计算量比对比模型高出一个数量级以上（Chen et al., 2020a）。注意到这些发现后，我们探索了训练一个系统来解决一个可能更容易的代理任务：只预测哪段文本 *as a whole* 与哪张图像配对，而不是预测该文本的确切词语。我们从相同的词袋编码基线开始，将图 2 中的预测目标替换为对比目标，并观察到向 ImageNet 进行零样本迁移的效率进一步提高了 4 倍。

给定一批 N (图像、文本) 对，CLIP 被训练来预测在一个批次中 $N \times N$ 个可能的（图像、文本）配对中哪些是真实出现的。为此，CLIP 学习一个

with high pointwise mutual information as well as the names of all Wikipedia articles above a certain search volume. Finally all WordNet synsets not already in the query list are added.

通过联合训练图像编码器和文本编码器来学习多模态嵌入空间，使批次中 N 个真实配对的图像与文本嵌入之间的余弦相似度最大化，同时使 $N^2 - N$ 个错误配对的嵌入之间的余弦相似度最小化。我们在这些相似度分数上优化对称交叉熵损失。在图 3 中，我们给出了 CLIP 实现核心部分的伪代码。据我们所知，这种批次构建技术和目标最早是在深度度量学习领域以 *multi-class N-pair loss* Sohn（2016）的形式提出的，随后由 Oord 等人（2018）以 InfoNCE 损失的形式在对比表示学习中推广，并且最近由 Zhang 等人（2020）适用于医学影像领域中的对比（文本、图像）表示学习。

由于我们的预训练数据集规模很大，过拟合并不是主要问题，因此与 Zhang 等人（2020）的实现相比，CLIP 的训练细节得到了简化。我们从头开始训练 CLIP，不使用 ImageNet 权重初始化图像编码器，也不使用预训练权重初始化文本编码器。我们不使用表示与对比嵌入空间之间的非线性投影；这一改动由 Bachman 等人（2019）引入，并由 Chen 等人（2020b）推广。相反，我们只使用线性投影，将每个编码器的表示映射到多模态嵌入空间。我们没有注意到这两个版本在训练效率上存在差异，并推测在自监督表示学习方法中，非线性投影可能会与当前仅图像方法的细节共同适应。我们还移除了 Zhang 等人（2020）中的文本转换函数 t_u，该函数会从文本中均匀采样一个句子，因为 CLIP 预训练数据集中的许多（图像，文本）对只有一个句子。我们还简化了图像转换函数 t_v。训练期间使用的唯一数据增强是从调整大小后的图像中随机裁剪正方形区域。最后，控制 softmax 中 logits 范围的温度参数 τ 在训练过程中作为对数参数化的乘性标量被直接优化，以避免将其作为超参数进行调优。

2.4. 选择和扩展模型

我们考虑了两种不同的图像编码器架构。第一种使用 ResNet-50（He 等，2016a）作为图像编码器的基础架构，因为它被广泛采用且性能已得到验证。我们基于 He 等（2019）的 ResNet-D 改进以及 Zhang（2019）的抗混叠 rect-2 模糊池化，对原始版本进行了若干修改。我们还用注意力池化机制替换了全局平均池化层。注意力池化被实现为单层“transformer 风格”的多头 QKV 注意力，其中查询以全局平均池化结果为条件。

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

representation of the image. For the second architecture, we experiment with the recently introduced Vision Transformer (ViT) (Dosovitskiy et al., 2020). We closely follow their implementation with only the minor modification of adding an additional layer normalization to the combined patch and position embeddings before the transformer and use a slightly different initialization scheme.

The text encoder is a Transformer (Vaswani et al., 2017) with the architecture modifications described in Radford et al. (2019). As a base size we use a 63M-parameter 12-layer 512-wide model with 8 attention heads. The transformer operates on a lower-cased byte pair encoding (BPE) representation of the text with a 49,152 vocab size (Sennrich et al., 2015). For computational efficiency, the max sequence length was capped at 76. The text sequence is bracketed with `[SOS]` and `[EOS]` tokens and the activations of the highest layer of the transformer at the `[EOS]` token are treated as the feature representation of the text which is layer normalized and then linearly projected into the multi-modal embedding space. Masked self-attention was used in the text encoder to preserve the ability to initialize with a pre-trained language model or add language modeling as an auxiliary objective, though exploration of this is left as future work.

While previous computer vision research has often scaled models by increasing the width (Mahajan et al., 2018) or depth (He et al., 2016a) in isolation, for the ResNet image encoders we adapt the approach of Tan & Le (2019) which found that allocating additional compute across all of width, depth, and resolution outperforms only allocating it to only one dimension of the model. While Tan & Le (2019) tune the ratio of compute allocated to each dimension for their EfficientNet architecture, we use a simple baseline of allocating additional compute equally to increasing the width, depth, and resolution of the model. For the text encoder, we only scale the width of the model to be proportional to the calculated increase in width of the ResNet and do not scale the depth at all, as we found CLIP's performance to be less sensitive to the capacity of the text encoder.

2.5. Training

We train a series of 5 ResNets and 3 Vision Transformers. For the ResNets we train a ResNet-50, a ResNet-101, and then 3 more which follow EfficientNet-style model scaling and use approximately 4x, 16x, and 64x the compute of a ResNet-50. They are denoted as RN50x4, RN50x16, and RN50x64 respectively. For the Vision Transformers we train a ViT-B/32, a ViT-B/16, and a ViT-L/14. We train all models for 32 epochs. We use the Adam optimizer (Kingma & Ba, 2014) with decoupled weight decay regularization (Loshchilov & Hutter, 2017) applied to all weights that are not gains or biases, and decay the learning rate using a cosine schedule (Loshchilov & Hutter, 2016). Initial hyper-parameters were set using a combination of grid searches, random search, and manual tuning on the baseline ResNet-50 model when trained for 1 epoch. Hyper-parameters were then adapted heuristically for larger models due to computational constraints. The learnable temperature parameter τ was initialized to the equivalent of 0.07 from (Wu et al., 2018) and clipped to prevent scaling the logits by more than 100 which we found necessary to prevent training instability. We use a very large minibatch size of 32,768. Mixed-precision (Micikevicius et al., 2017) was used to accelerate training and save memory. To save additional memory, gradient checkpointing (Griewank & Walther, 2000; Chen et al., 2016), half-precision Adam statistics (Dhariwal et al., 2020), and half-precision stochastically rounded text encoder weights were used. The calculation of embedding similarities was also sharded with individual GPUs computing only the subset of the pairwise similarities necessary for their local batch of embeddings. The largest ResNet model, RN50x64, took 18 days to train on 592 V100 GPUs while the largest Vision Transformer took 12 days on 256 V100 GPUs. For the ViT-L/14 we also pre-train at a higher 336 pixel resolution for one additional epoch to boost performance similar to FixRes (Touvron et al., 2019). We denote this model as ViT-L/14@336px. Unless otherwise specified, all results reported in this paper as "CLIP" use this model which we found to perform best.

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

图 3. CLIP 实现核心的类 Numpy 伪代码。

图像的表示。对于第二种架构，我们尝试了近期提出的视觉 Transformer（ViT）（Dosovitskiy 等，2020）。我们基本遵循其实现，仅作少量修改：在输入 Transformer 之前，对组合后的图像块嵌入和位置嵌入额外添加一层层归一化，并采用略有不同的初始化方案。

文本编码器是一个 Transformer（Vaswani 等，2017），并采用了 Radford 等（2019）中描述的架构修改。作为基础规模，我们使用了一个拥有 6300 万参数、12 层、宽度为 512 的模型，并配备 8 个注意力头。该 Transformer 对文本的小写字节对编码（BPE）表示进行操作，词表大小为 49,152（Sennrich 等，2015）。为提高计算效率，最大序列长度被限制为 76。文本序列由 [SOS] 和 [EOS] 标记括起来，并将 Transformer 最高层在 [EOS] 标记处的激活视为文本的特征表示；该表示经过层归一化后，再线性投影到多模态嵌入空间中。文本编码器中使用了带掩码的自注意力，以保留使用预训练语言模型进行初始化或将语言建模作为辅助目标的能力，不过对此的探索留待未来工作。

尽管以往的计算机视觉研究通常通过单独增加宽度（Mahajan et al., 2018）或深度（He et al., 2016a）来扩展模型，但对于 ResNet 图像编码器，我们采用了 Tan & Le（2019）的方法，该研究发现，将额外计算量分配到宽度、深度和分辨率这三者上，比仅将其分配给某一项更优。

模型的一个维度。Tan 和 Le（2019）为其 EfficientNet 架构调整分配给各个维度的计算量比例，而我们采用一个简单的基线方法：将额外的计算量平均分配给模型宽度、深度和分辨率的增加。对于文本编码器，我们只按 ResNet 宽度计算出的增幅成比例地扩展模型宽度，而完全不扩展深度，因为我们发现 CLIP 的性能对文本编码器容量的敏感性较低。

2.5. 训练

我们训练了一系列模型，包括 5 个 ResNet 和 3 个 Vision Transformer。对于 ResNet，我们训练了一个 ResNet-50、一个 ResNet-101，随后又训练了 3 个遵循 EfficientNet 风格模型缩放的模型，其计算量约为 ResNet-50 的 4 倍、16 倍和 64 倍。它们分别记为 RN50x4、RN50x16 和 RN50x64。对于 Vision Transformer，我们训练了 ViT-B/32、ViT-B/16 和 ViT-L/14。我们对所有模型训练 32 个 epoch。我们使用 Adam 优化器（Kingma & Ba, 2014），并对所有非增益或偏置的权重应用解耦权重衰减正则化（Loshchilov & Hutter, 2017），同时使用余弦调度来衰减学习率（Loshchilov & Hutter, 2016）。初始超参数是在基线 ResNet-50 模型训练 1 个 epoch 时，通过网格搜索、随机搜索和手动调优相结合的方式设定的。由于计算限制，随后通过启发式方法为更大的模型调整超参数。可学习的温度参数 τ 被初始化为（Wu et al., 2018）中 0.07 的等价值，并进行裁剪，以防止将 logits 缩放超过 100；我们发现这对于防止训练不稳定是必要的。我们使用非常大的小批量大小 32,768。使用混合精度（Micikevicius et al., 2017）来加速训练并节省内存。为了进一步节省内存，我们使用了梯度检查点（Griewank & Walther, 2000; Chen et al., 2016）、半精度 Adam 统计量（Dhariwal et al., 2020）以及采用半精度随机舍入的文本编码器权重。嵌入相似度的计算也被分片处理，各个 GPU 仅计算其本地嵌入批次所需的成对相似度子集。最大的 ResNet 模型 RN50x64 在 592 块 V100 GPU 上训练耗时 18 天，而最大的 Vision Transformer 在 256 块 V100 GPU 上训练耗时 12 天。对于 ViT-L/14，我们还以更高的 336 像素分辨率额外预训练 1 个 epoch，以类似 FixRes（Touvron et al., 2019）的方式提升性能。我们将该模型记为 ViT-L/14@336px。除非另有说明，本文中报告为“CLIP”的所有结果均使用该模型，我们发现它表现最佳。

3. Experiments

3.1. Zero-Shot Transfer

3.1.1. Motivation

In computer vision, zero-shot learning usually refers to the study of generalizing to unseen object categories in image classification (Lampert et al., 2009). We instead use the term in a broader sense and study generalization to unseen datasets. We motivate this as a proxy for performing unseen tasks, as aspired to in the zero-data learning paper of Larochelle et al. (2008). While much research in the field of unsupervised learning focuses on the *representation learning* capabilities of machine learning systems, we motivate studying zero-shot transfer as a way of measuring the *task-learning* capabilities of machine learning systems. In this view, a dataset evaluates performance on a task on a specific distribution. However, many popular computer vision datasets were created by the research community primarily as benchmarks to guide the development of generic image classification methods rather than measuring performance on a specific task. While it is reasonable to say that the SVHN dataset measures the task of street number transcription on the distribution of Google Street View photos, it is unclear what "real" task the CIFAR-10 dataset measures. It is clear, however, what distribution CIFAR-10 is drawn from - TinyImages (Torralba et al., 2008). On these kinds of datasets, zero-shot transfer is more an evaluation of CLIP's robustness to distribution shift and domain generalization rather than task generalization. Please see Section 3.3 for analysis focused on this.

To our knowledge, Visual N-Grams (Li et al., 2017) first studied zero-shot transfer to existing image classification datasets in the manner described above. It is also the only other work we are aware of that has studied zero-shot transfer to standard image classification datasets using a generically pre-trained model and serves as the best reference point for contextualizing CLIP. Their approach learns the parameters of a dictionary of 142,806 visual n-grams (spanning 1- to 5- grams) and optimizes these n-grams using a differential version of Jelinek-Mercer smoothing to maximize the probability of all text n-grams for a given image. In order to perform zero-shot transfer, they first convert the text of each of the dataset's class names into its n-gram representation and then compute its probability according to their model, predicting the one with the highest score.

Our focus on studying zero-shot transfer as an evaluation of task learning is inspired by work demonstrating task learning in the field of NLP. To our knowledge Liu et al. (2018) first identified task learning as an "unexpected side-effect" when a language model trained to generate Wikipedia articles learned to reliably transliterate names between languages. While GPT-1 (Radford et al., 2018) focused on pre-training as a transfer learning method to improve supervised fine-tuning, it also included an ablation study demonstrating that the performance of four heuristic zero-shot transfer methods improved steadily over the course of pre-training, without any supervised adaption. This analysis served as the basis for GPT-2 (Radford et al., 2019) which focused exclusively on studying the task-learning capabilities of language models via zero-shot transfer.

3.1.2. Using CLIP for Zero-Shot Transfer

CLIP is pre-trained to predict if an image and a text snippet are paired together in its dataset. To perform zero-shot classification, we reuse this capability. For each dataset, we use the names of all the classes in the dataset as the set of potential text pairings and predict the most probable (image, text) pair according to CLIP. In a bit more detail, we first compute the feature embedding of the image and the feature embedding of the set of possible texts by their respective encoders. The cosine similarity of these embeddings is then calculated, scaled by a temperature parameter τ, and normalized into a probability distribution via a softmax. Note that this prediction layer is a multinomial logistic regression classifier with L2-normalized inputs, L2-normalized weights, no bias, and temperature scaling. When interpreted this way, the image encoder is the computer vision backbone which computes a feature representation for the image and the text encoder is a hypernetwork (Ha et al., 2016) which generates the weights of a linear classifier based on the text specifying the visual concepts that the classes represent. Lei Ba et al. (2015) first introduced a zero-shot image classifier of this form while the idea of generating a classifier from natural language dates back to at least Elhoseiny et al. (2013). Continuing with this interpretation, every step of CLIP pre-training can be viewed as optimizing the performance of a randomly created proxy to a computer vision dataset which contains 1 example per class and has 32,768 total classes defined via natural language descriptions. For zero-shot evaluation, we cache the zero-shot classifier once it has been computed by the text encoder and reuse it for all subsequent predictions. This allows the cost of generating it to be amortized across all the predictions in a dataset.

3.1.3. Initial Comparison to Visual N-Grams

In Table 1 we compare Visual N-Grams to CLIP. The best CLIP model improves accuracy on ImageNet from a proof of concept 11.5% to 76.2% and matches the performance of the original ResNet-50 despite using none of the 1.28 million crowd-labeled training examples available for this dataset. Additionally, the top-5 accuracy of CLIP models are noticeably higher than their top-1, and this model has a 95% top-5 accuracy, matching Inception-V4 (Szegedy et al., 2016). The ability to match the performance of a strong, fully supervised baselines in a zero-shot setting suggests

3. 实验

3.1. 零样本迁移

3.1.1. 动机

在计算机视觉中，零样本学习通常指研究在图像分类中泛化到未见过的对象类别（Lampert 等，2009）。我们则以更广义的方式使用这一术语，并研究对未见过数据集的泛化。我们将其作为执行未见任务的代理来加以动机化，这也是 Larochelle 等人（2008）的零数据学习论文所追求的目标。尽管无监督学习领域的许多研究关注机器学习系统的 *representation learn- ing* 能力，我们则将研究零样本迁移作为衡量机器学习系统 *task- learning* 能力的一种方式。在这种观点下，一个数据集评估的是某个任务在特定分布上的性能。然而，许多流行的计算机视觉数据集主要是由研究社区作为基准创建的，用于指导通用图像分类方法的发展，而不是衡量在某个特定任务上的性能。虽然可以合理地说，SVHN 数据集衡量的是在 Google 街景照片分布上的街道门牌号转录任务，但并不清楚 CIFAR-10 数据集衡量的是什么“真实”任务。不过，很清楚的是 CIFAR-10 来源于什么分布——TinyImages（Torralba 等，2008）。在这类数据集上，零样本迁移更多是在评估 CLIP 对分布偏移和领域泛化的鲁棒性，而不是任务泛化。关于这一点的分析请参见第 3.3 节。

据我们所知，Visual N-Grams（Li 等，2017）首次以上述方式研究了向现有图像分类数据集的零样本迁移。据我们所知，它也是唯一另一项使用通用预训练模型研究向标准图像分类数据集进行零样本迁移的工作，并且是用于理解 CLIP 背景的最佳参照点。他们的方法学习一个包含 142,806 个视觉 n-gram（涵盖 1-gram 到 5-gram）的词典参数，并使用 Jelinek-Mercer 平滑的微分版本来优化这些 n-gram，以最大化给定图像的所有文本 n-gram 的概率。为了执行零样本迁移，他们首先将数据集中每个类别名称的文本转换为其 n-gram 表示，然后根据他们的模型计算其概率，并预测得分最高的类别。

我们专注于研究零样本迁移，将其作为对任务学习的评估，这一思路受到自然语言处理领域中展示任务学习的工作的启发。据我们所知，Liu 等人（2018）最早将任务学习识别为一种“意外的副作用”：一个经过训练用于生成维基百科文章的语言模型，学会了在不同语言之间可靠地音译姓名。虽然 GPT-1（Radford 等人，2018）专注于预-训练作为一种迁移学习方法，用于改进有监督微调；其中还包括一项消融研究，表明在没有任何有监督适配的情况下，四种启发式零样本迁移方法的性能在预训练过程中稳步提升。该分析成为 GPT-2（Radford 等，2019）的基础，后者专注于通过零样本迁移研究语言模型的任务学习能力。

3.1.2. 使用 CLIP 进行零样本迁移

CLIP 经过预训练，用于预测一张图像和一段文本片段是否在其数据集中配对在一起。为了进行零样本分类，我们复用了这一能力。对于每个数据集，我们将数据集中所有类别的名称作为潜在文本配对集合，并根据 CLIP 预测最可能的（图像，文本）配对。更具体地说，我们首先分别通过各自的编码器计算图像的特征嵌入，以及可能文本集合的特征嵌入。然后计算这些嵌入的余弦相似度，用温度参数 τ 进行缩放，并通过 softmax 归一化为概率分布。请注意，这一预测层是一个多项逻辑回归分类器，其输入经过 L2 归一化，权重经过 L2 归一化，无偏置，并使用温度缩放。从这种角度解释时，图像编码器是计算机视觉骨干网络，用于为图像计算特征表示；而文本编码器则是一个超网络（Ha 等，2016），它基于指定类别所代表的视觉概念的文本，生成线性分类器的权重。Lei Ba 等（2015）首次提出了这种形式的零样本图像分类器，而从自然语言生成分类器的思想至少可以追溯到 Elhoseiny 等（2013）。继续沿用这一解释，CLIP 预训练的每一步都可以被视为在优化一个随机创建的、针对计算机视觉数据集的代理任务的性能；该数据集每个类别包含 1 个样本，总共有 32,768 个通过自然语言描述定义的类别。对于零样本评估，我们在文本编码器计算出零样本分类器后将其缓存，并将其复用于所有后续预测。这使得生成该分类器的成本能够在一个数据集中的所有预测之间摊销。

3.1.3. 与视觉 N-gram 的初步比较

在表 1 中，我们将 Visual N-Grams 与 CLIP 进行了比较。最佳的 CLIP 模型将 ImageNet 上的准确率从概念验证阶段的 11.5% 提升到 76.2%，并且在没有使用该数据集可用的 128 万个众包标注训练样本的情况下，达到了原始 ResNet-50 的性能。此外，CLIP 模型的 top-5 准确率明显高于其 top-1 准确率，并且该模型的 top-5 准确率为 95%，与 Inception-V4（Szegedy 等，2016）相当。在零样本设置下能够达到强大的全监督基线的性能，这表明

	aYahoo	ImageNet	SUN
Visual N-Grams	72.4	11.5	23.0
CLIP	**98.4**	**76.2**	**58.5**

Table 1. Comparing CLIP to prior zero-shot transfer image classification results. CLIP improves performance on all three datasets by a large amount. This improvement reflects many differences in the 4 years since the development of Visual N-Grams (Li et al., 2017).

CLIP is a significant step towards flexible and practical zero-shot computer vision classifiers. As mentioned above, the comparison to Visual N-Grams is meant for contextualizing the performance of CLIP and should not be interpreted as a direct methods comparison between CLIP and Visual N-Grams as many performance relevant differences between the two systems were not controlled for. For instance, we train on a dataset that is 10x larger, use a vision model that requires nearly 100x more compute per prediction, likely used over 1000x their training compute, and use a transformer-based model which did not exist when Visual N-Grams was published. As a closer comparison, we trained a CLIP ResNet-50 on the same YFCC100M dataset that Visual N-Grams was trained on and found it matched their reported ImageNet performance within a V100 GPU day. This baseline was also trained from scratch instead of being initialized from pre-trained ImageNet weights as in Visual N-Grams.

CLIP also outperforms Visual N-Grams on the other 2 reported datasets. On aYahoo, CLIP achieves a 95% reduction in the number of errors, and on SUN, CLIP more than doubles the accuracy of Visual N-Grams. To conduct a more comprehensive analysis and stress test, we implement a much larger evaluation suite detailed in Appendix A. In total we expand from the 3 datasets reported in Visual N-Grams to include over 30 datasets and compare to over 50 existing computer vision systems to contextualize results.

3.1.4. Prompt Engineering and Ensembling

Most standard image classification datasets treat the information naming or describing classes which enables natural language based zero-shot transfer as an afterthought. The vast majority of datasets annotate images with just a numeric id of the label and contain a file mapping these ids back to their names in English. Some datasets, such as Flowers102 and GTSRB, don't appear to include this mapping at all in their released versions preventing zero-shot transfer entirely.[2] For many datasets, we observed these labels may be

[2]Alec learned much more about flower species and German traffic signs over the course of this project than he originally anticipated.

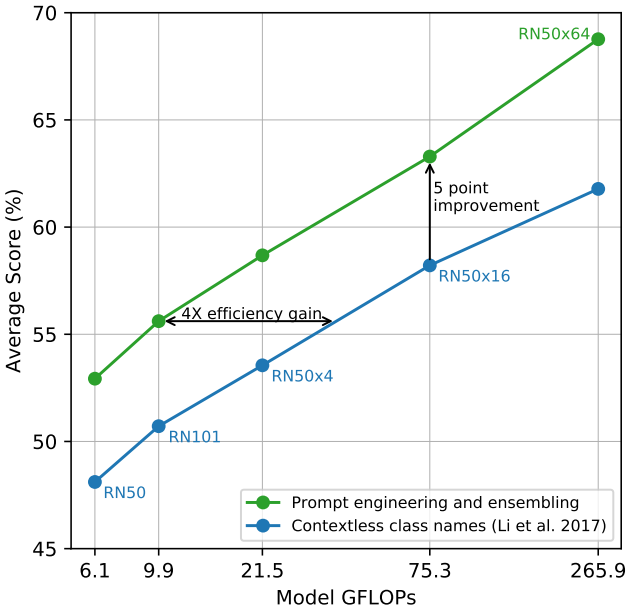


Figure 4. **Prompt engineering and ensembling improve zero-shot performance.** Compared to the baseline of using contextless class names, prompt engineering and ensembling boost zero-shot classification performance by almost 5 points on average across 36 datasets. This improvement is similar to the gain from using 4 times more compute with the baseline zero-shot method but is "free" when amortized over many predictions.

chosen somewhat haphazardly and do not anticipate issues related to zero-shot transfer which relies on task description in order to transfer successfully.

A common issue is polysemy. When the name of a class is the only information provided to CLIP's text encoder it is unable to differentiate which word sense is meant due to the lack of context. In some cases multiple meanings of the same word might be included as different classes in the same dataset! This happens in ImageNet which contains both construction cranes and cranes that fly. Another example is found in classes of the Oxford-IIIT Pet dataset where the word boxer is, from context, clearly referring to a breed of dog, but to a text encoder lacking context could just as likely refer to a type of athlete.

Another issue we encountered is that it's relatively rare in our pre-training dataset for the text paired with the image to be just a single word. Usually the text is a full sentence describing the image in some way. To help bridge this distribution gap, we found that using the prompt template "`A photo of a {label}.`" to be a good default that helps specify the text is about the content of the image. This often improves performance over the baseline of using only the label text. For instance, just using this prompt improves accuracy on ImageNet by 1.3%.

	aYahoo	ImageNet	SUN
Visual N-Grams	72.4	11.5	23.0
CLIP	**98.4**	**76.2**	**58.5**

表1。将 CLIP 与先前的零样本迁移图像分类结果进行比较。CLIP 在所有三个数据集上的性能都有大幅提升。这一提升反映了自 Visual N-Grams（Li 等，2017）开发以来 4 年间的诸多差异。

CLIP 是朝着灵活且实用的零样本计算机视觉分类器迈出的重要一步。如上所述，与 Visual N-Grams 的比较旨在为 CLIP 的性能提供背景，不应被解读为 CLIP 与 Visual N-Grams 之间的直接方法比较，因为两个系统之间许多与性能相关的差异并未得到控制。例如，我们在一个大 10 倍的数据集上进行训练，使用的视觉模型在每次预测时所需的计算量接近 100 倍，训练计算量可能超过其 1000 倍，并且使用了基于 Transformer 的模型，而在 Visual N-Grams 发表时这种模型尚不存在。作为更接近的比较，我们在 Visual N-Grams 训练所用的同一个 YFCC100M 数据集上训练了一个 CLIP ResNet-50，并发现它在一个 V100 GPU 日内达到了他们报告的 ImageNet 性能。这个基线模型也是从头开始训练的，而不是像 Visual N-Grams 那样从预训练的 ImageNet 权重初始化。

在另外两个报告的数据集上，CLIP 也优于 Visual N-Grams。在 aYahoo 上，CLIP 将错误数量减少了 95%；在 SUN 上，CLIP 的准确率是 Visual N-Grams 的两倍多。为了进行更全面的分析和压力测试，我们实现了一个规模大得多的评估套件，详见附录 A。总体而言，我们从 Visual N-Grams 报告的 3 个数据集扩展到包含 30 多个数据集，并与 50 多个现有计算机视觉系统进行比较，以便对结果进行背景化分析。

3.1.4. 提示工程与集成

大多数标准图像分类数据集都将用于命名或描述类别的信息视为事后补充，而这些信息本可支持基于自然语言的零样本迁移。绝大多数数据集只是用标签的数字 ID 来标注图像，并包含一个文件将这些 ID 映射回其英文名称。一些数据集，如 Flowers102 和 GTSRB，在其发布版本中似乎根本不包含这种映射，从而完全阻碍了零样本迁移。[2] 对于许多数据集，我们观察到这些标签可能会

[2]Alec learned much more about flower species and German traffic signs over the course of this project than he originally anticipated.

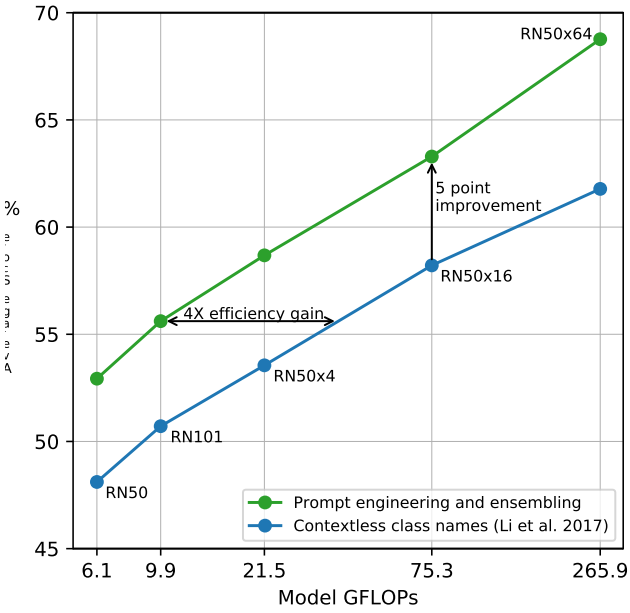


图 4. 提示工程和集成提升了零样本性能。与使用无上下文类别名称的基线相比，提示工程和集成使零样本分类性能在 36 个数据集上平均提高了近 5 个百分点。这一提升与在基线零样本方法中使用 4 倍计算量所带来的增益相当，但在分摊到大量预测上时几乎是“免费”的。

选择得有些随意，并未预见到与零样本迁移相关的问题，而零样本迁移需要依赖任务描述才能成功迁移。

一个常见问题是多义性。当提供给 CLIP 文本编码器的唯一信息是类别名称时，由于缺乏上下文，它无法区分所指的是哪个词义。在某些情况下，同一个词的多个含义甚至可能作为不同类别出现在同一个数据集中！这种情况出现在 ImageNet 中，其中既包含建筑起重机，也包含会飞的鹤。另一个例子来自 Oxford-IIIT Pet 数据集的类别，其中 boxer 这个词从上下文来看显然指的是一种犬种，但对于缺乏上下文的文本编码器来说，它同样很可能指的是一种运动员。

我们遇到的另一个问题是，在我们的预训练数据集中，与图像配对的文本仅为单个词的情况相对少见。通常，文本是以某种方式描述图像的完整句子。为了帮助弥合这种分布差距，我们发现使用提示模板“A photo of a {label}.”是一个不错的默认选择，有助于明确文本是关于图像内容的。与仅使用标签文本的基线相比，这通常能提升性能。例如，仅使用这个提示就能将 ImageNet 上的准确率提高 1.3%。

Similar to the "prompt engineering" discussion around GPT-3 (Brown et al., 2020; Gao et al., 2020), we have also observed that zero-shot performance can be significantly improved by customizing the prompt text to each task. A few, non exhaustive, examples follow. We found on several fine-grained image classification datasets that it helped to specify the category. For example on Oxford-IIIT Pets, using "`A photo of a {label}, a type of pet.`" to help provide context worked well. Likewise, on Food101 specifying *a type of food* and on FGVC Aircraft *a type of aircraft* helped too. For OCR datasets, we found that putting quotes around the text or number to be recognized improved performance. Finally, we found that on satellite image classification datasets it helped to specify that the images were of this form and we use variants of "`a satellite photo of a {label}.`".

We also experimented with ensembling over multiple zero-shot classifiers as another way of improving performance. These classifiers are computed by using different context prompts such as '`A photo of a big {label}`" and "`A photo of a small {label}`". We construct the ensemble over the embedding space instead of probability space. This allows us to cache a single set of averaged text embeddings so that the compute cost of the ensemble is the same as using a single classifier when amortized over many predictions. We've observed ensembling across many generated zero-shot classifiers to reliably improve performance and use it for the majority of datasets. On ImageNet, we ensemble 80 different context prompts and this improves performance by an additional 3.5% over the single default prompt discussed above. When considered together, prompt engineering and ensembling improve ImageNet accuracy by almost 5%. In Figure 4 we visualize how prompt engineering and ensembling change the performance of a set of CLIP models compared to the contextless baseline approach of directly embedding the class name as done in Li et al. (2017).

3.1.5. Analysis of Zero-Shot CLIP Performance

Since task-agnostic zero-shot classifiers for computer vision have been understudied, CLIP provides a promising opportunity to gain a better understanding of this type of model. In this section, we conduct a study of various properties of CLIP's zero-shot classifiers. As a first question, we look simply at how well zero-shot classifiers perform. To contextualize this, we compare to the performance of a simple off-the-shelf baseline: fitting a fully supervised, regularized, logistic regression classifier on the features of the canonical ResNet-50. In Figure 5 we show this comparison across 27 datasets. Please see Appendix A for details of datasets and setup.

Zero-shot CLIP outperforms this baseline slightly more often than not and wins on 16 of the 27 datasets. Looking at individual datasets reveals some interesting behavior. On fine-grained classification tasks, we observe a wide spread in performance. On two of these datasets, Stanford Cars and Food101, zero-shot CLIP outperforms logistic regression on ResNet-50 features by over 20% while on two others, Flowers102 and FGVCAircraft, zero-shot CLIP underperforms by over 10%. On OxfordPets and Birdsnap, performance is much closer. We suspect these difference are primarily due to varying amounts of per-task supervision between WIT and ImageNet. On "general" object classification datasets such as ImageNet, CIFAR10/100, STL10, and PascalVOC2007 performance is relatively similar with a slight advantage for zero-shot CLIP in all cases. On STL10, CLIP achieves 99.3% overall which appears to be a new state of the art despite not using any training examples. Zero-shot CLIP significantly outperforms a ResNet-50 on two datasets measuring action recognition in videos. On Kinetics700, CLIP outperforms a ResNet-50 by 14.5%. Zero-shot CLIP also outperforms a ResNet-50's features by 7.7% on UCF101. We speculate this is due to natural language providing wider supervision for visual concepts involving verbs, compared to the noun-centric object supervision in ImageNet.

Looking at where zero-shot CLIP notably underperforms,

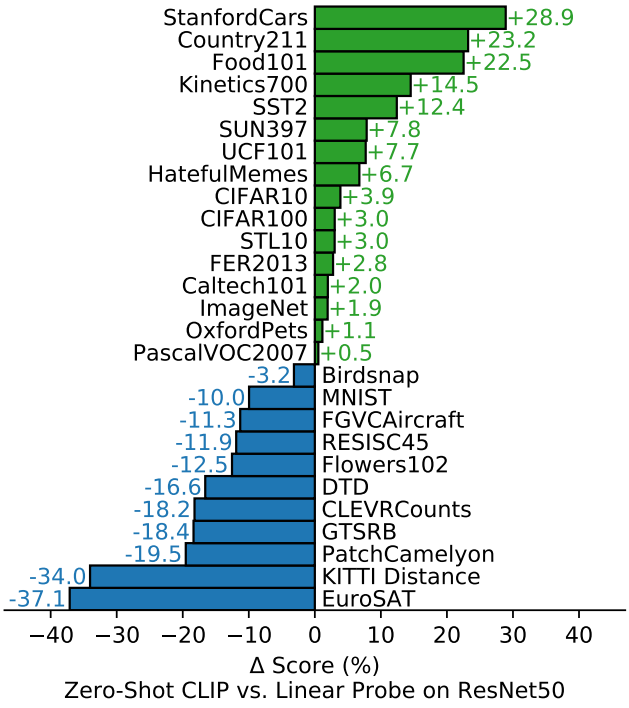


Figure 5. **Zero-shot CLIP is competitive with a fully supervised baseline.** Across a 27 dataset eval suite, a zero-shot CLIP classifier outperforms a fully supervised linear classifier fitted on ResNet-50 features on 16 datasets, including ImageNet.

类似于围绕 GPT-3 的“提示工程”讨论（Brown et al., 2020；Gao et al., 2020），我们也观察到，通过针对每项任务定制提示文本，可以显著提升零样本性能。以下是一些非穷尽的例子。我们在多个细粒度图像分类数据集上发现，明确指定类别会有所帮助。例如，在 Oxford-IIIT Pets 上，使用“一张{标签}的照片，一种宠物。”来帮助提供上下文，效果很好。同样，在 Food101 上指定 *a type of food*，以及在 FGVC Aircraft 上指定 *a type of aircraft* 也有帮助。对于 OCR 数据集，我们发现给要识别的文本或数字加上引号可以提升性能。最后，我们发现，在卫星图像分类数据集上，明确说明图像属于这种形式会有所帮助，因此我们使用了“一张{标签}的卫星照片。”的变体。

我们还尝试了对多个零样本分类器进行集成，作为提升性能的另一种方式。这些分类器是通过使用不同的上下文提示来计算的，例如“A photo of a big {label}”和“A photo of a small {label}”。我们在嵌入空间而不是概率空间中构建集成。这样我们就可以缓存一组平均后的文本嵌入，因此当在大量预测上摊销时，集成的计算成本与使用单个分类器相同。我们观察到，对许多生成的零样本分类器进行集成能够可靠地提升性能，并将其用于大多数数据集。在 ImageNet 上，我们集成了 80 个不同的上下文提示，相比上述单一默认提示，性能额外提升了 3.5%。综合来看，提示工程和集成使 ImageNet 准确率提高了近 5%。在图 4 中，我们可视化了提示工程和集成如何改变一组 CLIP 模型的性能，并将其与无上下文的基线方法进行比较；该基线方法直接将类别名称嵌入，正如 Li 等人（2017）所做的那样。

3.1.5. 零样本 CLIP 性能分析

由于用于计算机视觉的任务无关零样本分类器研究不足，CLIP 提供了一个很有前景的机会，使我们能够更好地理解这类模型。在本节中，我们研究 CLIP 零样本分类器的各种属性。作为第一个问题，我们首先简单考察零样本分类器的表现如何。为了提供背景参考，我们将其与一个简单的现成基线进行比较：在标准 ResNet-50 的特征上拟合一个全监督、正则化的逻辑回归分类器。图 5 展示了在 27 个数据集上的这一比较。有关数据集和设置的详细信息，请参见附录 A。

零样本 CLIP 的表现略微优于这一基线，幅度更大一些-

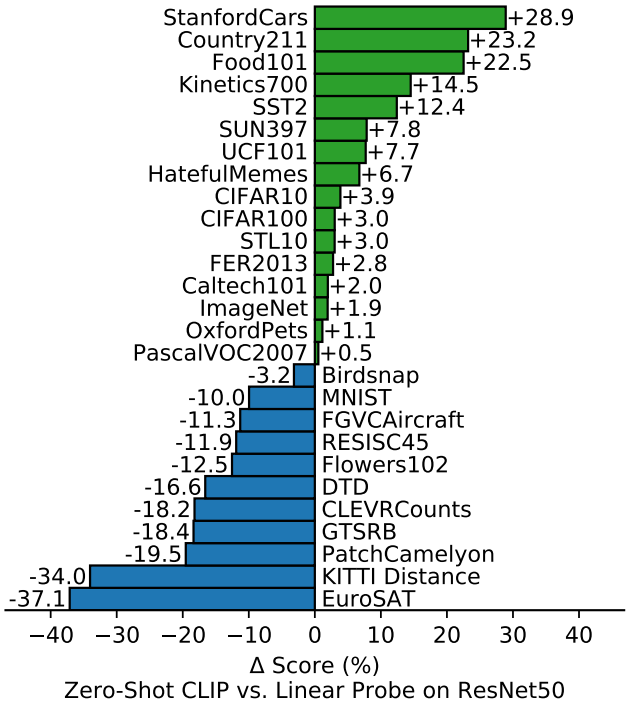


图 5. 零样本 CLIP 可与全监督基线相媲美。在包含 27 个数据集的评估套件中，零样本 CLIP 分类器在 16 个数据集上优于基于 ResNet-50 特征拟合的全监督线性分类器，其中包括 ImageNet。

超过不及，并在 27 个数据集中的 16 个上获胜。观察单个数据集会发现一些有趣的现象。在细粒度分类任务上，我们观察到性能差异很大。在其中两个数据集 Stanford Cars 和 Food101 上，零样本 CLIP 比基于 ResNet-50 特征的逻辑回归高出 20% 以上；而在另外两个数据集 Flowers102 和 FGVCAircraft 上，零样本 CLIP 则低于其 10% 以上。在 OxfordPets 和 Birdsnap 上，性能则接近得多。我们怀疑这些差异主要源于 WIT 和 ImageNet 之间每项任务监督量的不同。在 ImageNet、CIFAR10/100、STL10 和 PascalVOC2007 等“通用”目标分类数据集上，性能相对相似，且在所有情况下零样本 CLIP 都略占优势。在 STL10 上，CLIP 的总体准确率达到 99.3%，尽管没有使用任何训练样本，但这似乎是新的最先进水平。零样本 CLIP 在两个用于衡量视频动作识别的数据集上显著优于 ResNet-50。在 Kinetics700 上，CLIP 比 ResNet-50 高出 14.5%。在 UCF101 上，零样本 CLIP 也比 ResNet-50 的特征高出 7.7%。我们推测，这是因为与 ImageNet 中以名词为中心的对象监督相比，自然语言为涉及动词的视觉概念提供了更广泛的监督。

观察零样本 CLIP 明显表现不佳的情况，

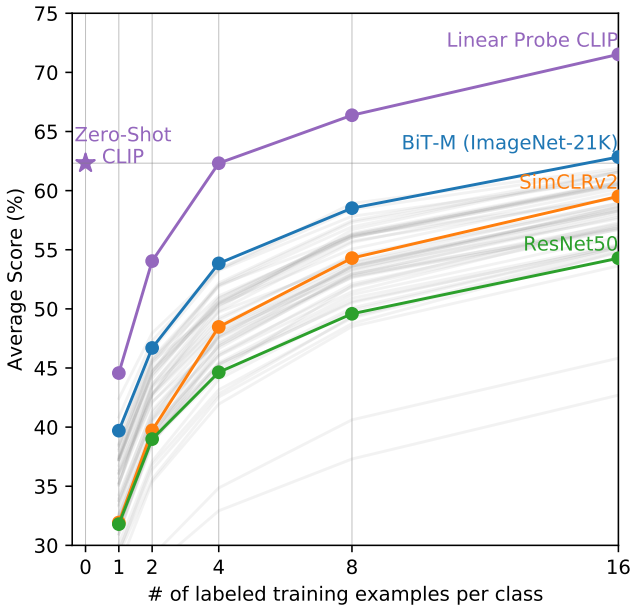


Figure 6. **Zero-shot CLIP outperforms few-shot linear probes.** Zero-shot CLIP matches the average performance of a 4-shot linear classifier trained on the same feature space and nearly matches the best results of a 16-shot linear classifier across publicly available models. For both BiT-M and SimCLRv2, the best performing model is highlighted. Light gray lines are other models in the eval suite. The 20 datasets with at least 16 examples per class were used in this analysis.

we see that zero-shot CLIP is quite weak on several specialized, complex, or abstract tasks such as satellite image classification (EuroSAT and RESISC45), lymph node tumor detection (PatchCamelyon), counting objects in synthetic scenes (CLEVRCounts), self-driving related tasks such as German traffic sign recognition (GTSRB), recognizing distance to the nearest car (KITTI Distance). These results highlight the poor capability of zero-shot CLIP on more complex tasks. By contrast, non-expert humans can robustly perform several of these tasks, such as counting, satellite image classification, and traffic sign recognition, suggesting significant room for improvement. However, we caution that it is unclear whether measuring zero-shot transfer, as opposed to few-shot transfer, is a meaningful evaluation for difficult tasks that a learner has no prior experience with, such as lymph node tumor classification for almost all humans (and possibly CLIP).

While comparing zero-shot performance to fully supervised models contextualizes the task-learning capabilities of CLIP, comparing to few-shot methods is a more direct comparison, since zero-shot is its limit. In Figure 6, we visualize how zero-shot CLIP compares to few-shot logistic regression on the features of many image models including the best publicly available ImageNet models, self-supervised learning methods, and CLIP itself. While it is intuitive to expect zero-shot to underperform one-shot, we instead find that zero-shot CLIP matches the performance of 4-shot logistic regression on the same feature space. This is likely due to an important difference between the zero-shot and few-shot approach. First, CLIP's zero-shot classifier is generated via natural language which allows for visual concepts to be directly specified ("communicated"). By contrast, "normal" supervised learning must infer concepts indirectly from training examples. Context-less example-based learning has the drawback that many different hypotheses can be consistent with the data, especially in the one-shot case. A single image often contains many different visual concepts. Although a capable learner is able to exploit visual cues and heuristics, such as assuming that the concept being demonstrated is the primary object in an image, there is no guarantee.

A potential resolution of this discrepancy between zero-shot and few-shot performance is to use CLIP's zero-shot classifier as a prior for the weights of the few-shot classifier. While adding an L2 penalty towards the generated weights is a straightforward implementation of this idea, we found that hyperparameter optimization would often select for such a large value of this regularizer that the resulting few-shot classifier was "just" the zero-shot classifier. Research into better methods of combining the strength of zero-shot transfer with flexibility of few-shot learning is a promising direction for future work.

When comparing zero-shot CLIP to few-shot logistic regression on the features of other models, zero-shot CLIP roughly matches the performance of the best performing 16-shot classifier in our evaluation suite, which uses the features of a BiT-M ResNet-152x2 trained on ImageNet-21K. We are certain that a BiT-L model trained on JFT-300M would perform even better but these models have not been publicly released. That a BiT-M ResNet-152x2 performs best in a 16-shot setting is somewhat surprising since, as analyzed in Section 3.2, the Noisy Student EfficientNet-L2 outperforms it in a fully supervised setting by almost 5% on average across 27 datasets.

In addition to studying the average performance of zero-shot CLIP and few-shot logistic regression, we also examine performance on individual datasets. In Figure 7, we show estimates for the number of labeled examples per class that a logistic regression classifier on the same feature space requires to match the performance of zero-shot CLIP. Since zero-shot CLIP is also a linear classifier, this estimates the effective data efficiency of zero-shot transfer in this setting. In order to avoid training thousands of linear classifiers, we estimate the effective data efficiency based on a log-linear interpolation of the performance of a 1, 2, 4, 8, 16-shot (when possible), and a fully supervised linear classifier trained on each dataset. We find that zero-shot transfer can

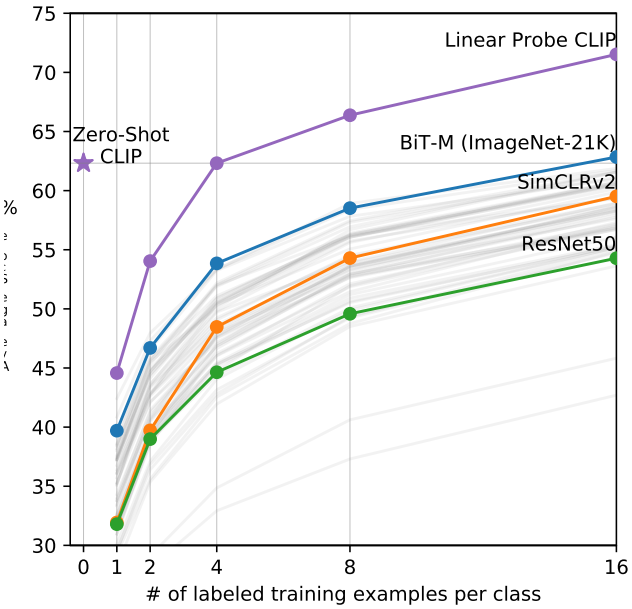


图 6. 零样本 CLIP 优于少样本线性探针。零样本 CLIP 的平均性能与在相同特征空间上训练的 4-shot 线性分类器相当，并且在公开可用模型中几乎达到 16-shot 线性分类器的最佳结果。对于 BiT-M 和 SimCLRv2，突出显示了性能最佳的模型。浅灰色线表示评估套件中的其他模型。本分析使用了每类至少包含 16 个样本的 20 个数据集。

我们看到，零样本 CLIP 在若干专业、复杂或抽象任务上表现相当薄弱，例如卫星图像分类（EuroSAT 和 RESISC45）、淋巴结肿瘤检测（PatchCamelyon）、合成场景中的物体计数（CLEVRCounts）、与自动驾驶相关的任务，如德国交通标志识别（GTSRB）、识别到最近汽车的距离（KITTI Distance）。这些结果凸显了零样本 CLIP 在更复杂任务上的能力不足。相比之下，非专业人类能够稳健地完成其中若干任务，例如计数、卫星图像分类和交通标志识别，这表明仍有显著的改进空间。然而，我们也提醒，对于学习者此前没有任何经验的困难任务，例如对几乎所有人类（也可能包括 CLIP）而言的淋巴结肿瘤分类，衡量零样本迁移而非少样本迁移是否是一种有意义的评估，目前尚不清楚。

将零样本性能与全监督模型进行比较，有助于为 CLIP 的任务学习能力提供背景；而与少样本方法进行比较则是更直接的比较，因为零样本是其极限。在图 6 中，我们可视化了零样本 CLIP 与基于多种图像模型特征的少样本逻辑回归之间的比较，这些图像模型包括公开可用的最佳 ImageNet 模型、自监督学习方法以及 CLIP 本身。虽然直觉上可以认为预期零样本会不如单样本表现好，但我们反而发现，在相同的特征空间上，零样本 CLIP 的性能与 4 样本逻辑回归相当。这很可能是由于零样本方法和少样本方法之间存在一个重要差异。首先，CLIP 的零样本分类器是通过自然语言生成的，这使得视觉概念可以被直接指定（“传达”）。相比之下，“常规”的监督学习必须从训练样本中间接推断概念。无上下文的基于示例的学习有一个缺点：许多不同的假设都可能与数据一致，尤其是在单样本情况下。一张图像通常包含许多不同的视觉概念。尽管有能力的学习器能够利用视觉线索和启发式方法，例如假设所展示的概念是图像中的主要物体，但这并没有保证。

解决零样本与少样本性能之间这种差异的一种潜在方法，是将 CLIP 的零样本分类器用作少样本分类器权重的先验。虽然向生成的权重添加 L2 惩罚是实现这一想法的一种直接方式，但我们发现，超参数优化往往会选择一个非常大的正则化系数，以至于最终得到的少样本分类器“其实”就是零样本分类器。如何更好地结合零样本迁移的优势与少样本学习的灵活性，是未来工作中一个很有前景的研究方向。

在将零样本 CLIP 与基于其他模型特征的少样本逻辑回归进行比较时，零样本 CLIP 的性能大致相当于我们评估套件中表现最佳的 16-shot 分类器；该分类器使用的是在 ImageNet-21K 上训练的 BiT-M ResNet-152x2 的特征。我们确信，在 JFT-300M 上训练的 BiT-L 模型会表现得更好，但这些模型尚未公开发布。BiT-M ResNet-152x2 在 16-shot 设置下表现最佳这一点有些令人意外，因为正如第 3.2 节所分析的，在全监督设置下，Noisy Student EfficientNet-L2 在 27 个数据集上的平均表现比它高出近 5%。

除了研究零样本 CLIP 和少样本逻辑回归的平均性能外，我们还考察了其在各个单独数据集上的表现。在图 7 中，我们给出了在相同特征空间上，逻辑回归分类器为了达到零样本 CLIP 的性能，每个类别所需标注样本数量的估计。由于零样本 CLIP 本身也是一种线性分类器，因此这估计了在该设置下零样本迁移的有效数据效率。为了避免训练数千个线性分类器，我们基于在每个数据集上训练的 1、2、4、8、16-shot（在可能的情况下）以及全监督线性分类器的性能进行对数线性插值，从而估计有效数据效率。我们发现，零样本迁移可以

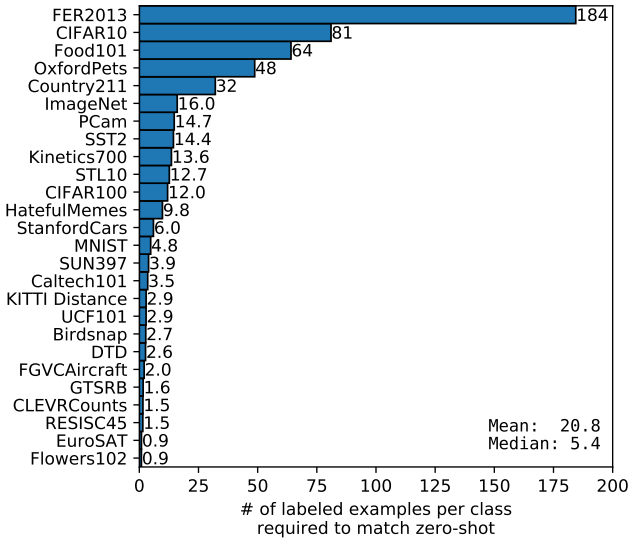


Figure 7. **The data efficiency of zero-shot transfer varies widely.** Calculating the number of labeled examples per class a linear classifier on the same CLIP feature space requires to match the performance of the zero-shot classifier contextualizes the effectiveness of zero-shot transfer. Values are estimated based on log-linear interpolation of 1, 2, 4, 8, 16-shot and fully supervised results. Performance varies widely from still underperforming a one-shot classifier on two datasets to matching an estimated 184 labeled examples per class.

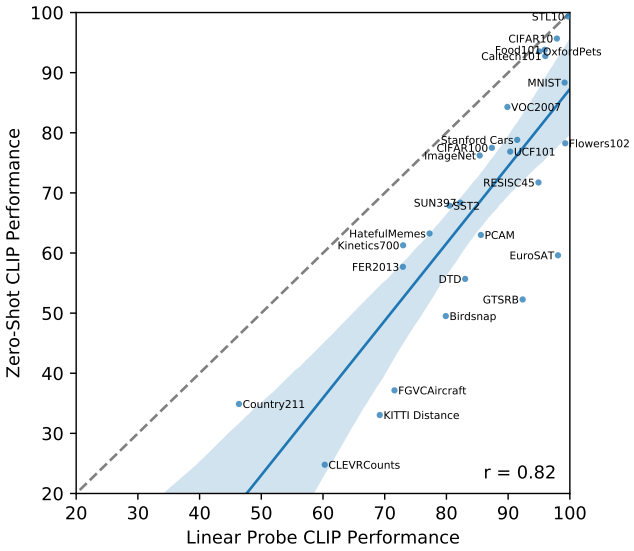


Figure 8. **Zero-shot performance is correlated with linear probe performance but still mostly sub-optimal.** Comparing zero-shot and linear probe performance across datasets shows a strong correlation with zero-shot performance mostly shifted 10 to 25 points lower. On only 5 datasets does zero-shot performance approach linear probe performance ($\leq$3 point difference).

have widely varying efficiency per dataset from less than 1 labeled example per class to 184. Two datasets, Flowers102 and EuroSAT underperform one-shot models. Half of the datasets require less than 5 examples per class with a median of 5.4. However, the mean estimated data efficiency is 20.8 examples per class. This is due to the 20% of datasets where supervised classifiers require many labeled examples per class in order to match performance. On ImageNet, zero-shot CLIP matches the performance of a 16-shot linear classifier trained on the same feature space.

If we assume that evaluation datasets are large enough that the parameters of linear classifiers trained on them are well estimated, then, because CLIP's zero-shot classifier is also a linear classifier, the performance of the fully supervised classifiers roughly sets an upper bound for what zero-shot transfer can achieve. In Figure 8 we compare CLIP's zero-shot performance with fully supervised linear classifiers across datasets. The dashed, $y = x$ line represents an "optimal" zero-shot classifier that matches the performance of its fully supervised equivalent. For most datasets, the performance of zero-shot classifiers still underperform fully supervised classifiers by 10% to 25%, suggesting that there is still plenty of headroom for improving CLIP's task-learning and zero-shot transfer capabilities.

There is a positive correlation of 0.82 (p-value $< 10^{-6}$) between zero-shot performance and fully supervised performance, suggesting that CLIP is relatively consistent at connecting underlying representation and task learning to zero-shot transfer. However, zero-shot CLIP only approaches fully supervised performance on 5 datasets: STL10, CIFAR10, Food101, OxfordPets, and Caltech101. On all 5 datasets, both zero-shot accuracy and fully supervised accuracy are over 90%. This suggests that CLIP may be more effective at zero-shot transfer for tasks where its underlying representations are also high quality. The slope of a linear regression model predicting zero-shot performance as a function of fully supervised performance estimates that for every 1% improvement in fully supervised performance, zero-shot performance improves by 1.28%. However, the 95th-percentile confidence intervals still include values of less than 1 (0.93-1.79).

Over the past few years, empirical studies of deep learning systems have documented that performance is predictable as a function of important quantities such as training compute and dataset size (Hestness et al., 2017; Kaplan et al., 2020). The GPT family of models has so far demonstrated consistent improvements in zero-shot performance across a 1000x increase in training compute. In Figure 9, we check whether the zero-shot performance of CLIP follows a similar scaling pattern. We plot the average error rate of the 5 ResNet CLIP models across 39 evaluations on 36 different datasets and find that a similar log-log linear scaling trend holds for CLIP across a 44x increase in model compute. While the overall trend is smooth, we found that performance on individual evaluations can be much noisier. We are unsure whether

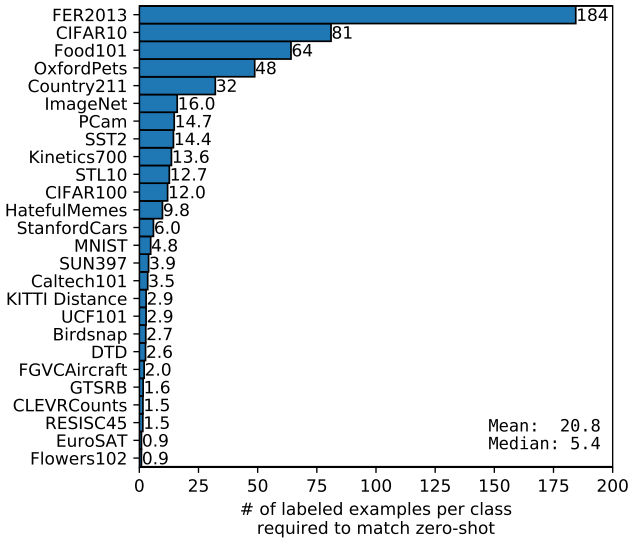


图 7. 零样本迁移的数据效率差异很大。通过计算在相同 CLIP 特征空间上线性分类器每个类别需要多少个标注样本才能达到零样本分类器的性能，可以更好地理解零样本迁移的有效性。数值基于 1、2、4、8、16-shot 以及全监督结果的对数线性插值估计得到。性能差异很大：在两个数据集上仍不如 one-shot 分类器，而在另一些情况下则相当于每类约 184 个标注样本的效果。

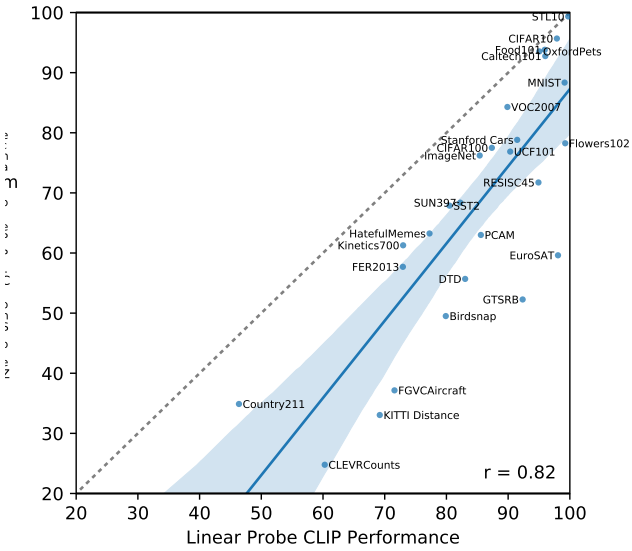


图 8. 零样本性能与线性探针性能相关，但总体上仍大多次优。比较各数据集上的零样本和线性探针性能显示，两者具有很强的相关性，而零样本性能大多低 10 到 25 个百分点。只有在 5 个数据集上，零样本性能接近线性探针性能（≤3 个百分点差异）。

各数据集之间的数据效率差异很大，从每类少于 1 个标注样本到 184 个不等。有两个数据集 Flowers102 和 EuroSAT 的表现不如 one-shot 模型。半数数据集每类所需样本少于 5 个，中位数为 5.4。然而，估计的平均数据效率为每类 20.8 个样本。这是因为在 20% 的数据集中，监督分类器需要每类大量标注样本才能达到相当的性能。在 ImageNet 上，零样本 CLIP 的性能与在同一特征空间上训练的 16-shot 线性分类器相当。

如果我们假设评估数据集足够大，以至于在其上训练的线性分类器的参数能够得到良好估计，那么，由于 CLIP 的零样本分类器也是一个线性分类器，完全监督分类器的性能大致为零样本迁移所能达到的水平设定了一个上限。在图 8 中，我们比较了 CLIP 的零样本性能与各数据集上的完全监督线性分类器性能。虚线 $y = x$ 表示一种“最优”的零样本分类器，其性能与对应的完全监督分类器相匹配。对于大多数数据集，零样本分类器的性能仍比完全监督分类器低 10% 到 25%，这表明在提升 CLIP 的任务学习和零样本迁移能力方面仍有很大的改进空间。

零样本性能与全监督性能之间存在 0.82 的正相关（p 值 $< 10^{-6}$）。

表现，这表明 CLIP 在将底层表示和任务学习与零样本迁移相连接方面相对一致。然而，零样本 CLIP 仅在 5 个数据集上接近全监督性能：STL10、CIFAR10、Food101、OxfordPets 和 Caltech101。在所有这 5 个数据集上，零样本准确率和全监督准确率都超过 90%。这表明，对于其底层表示质量也很高的任务，CLIP 的零样本迁移可能更有效。一个以全监督性能为函数来预测零样本性能的线性回归模型的斜率估计表明，全监督性能每提高 1%，零样本性能就提高 1.28%。然而，第 95 百分位置信区间仍然包含小于 1 的值（0.93-1.79）。

在过去几年中，深度学习系统的实证研究表明，性能可以作为训练计算量和数据集规模等重要量的函数来预测（Hestness 等，2017；Kaplan 等，2020）。到目前为止，GPT 系列模型已经证明，随着训练计算量增加 1000 倍，零样本性能持续提升。在图 9 中，我们检查 CLIP 的零样本性能是否遵循类似的缩放模式。我们绘制了 5 个 ResNet CLIP 模型在 36 个不同数据集上的 39 项评估中的平均错误率，并发现随着模型计算量增加 44 倍，CLIP 也呈现出类似的双对数线性缩放趋势。尽管整体趋势是平滑的，但我们发现单项评估中的性能可能要嘈杂得多。我们不确定是否

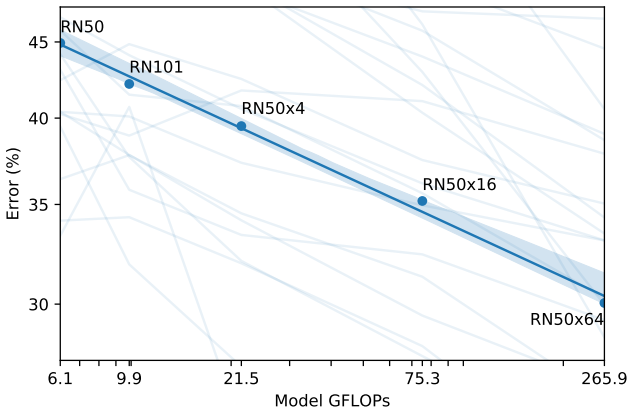


Figure 9. **Zero-shot CLIP performance scales smoothly as a function of model compute.** Across 39 evals on 36 different datasets, average zero-shot error is well modeled by a log-log linear trend across a 44x range of compute spanning 5 different CLIP models. Lightly shaded lines are performance on individual evals, showing that performance is much more varied despite the smooth overall trend.

this is caused by high variance between individual training runs on sub-tasks (as documented in D'Amour et al. (2020)) masking a steadily improving trend or whether performance is actually non-monotonic as a function of compute on some tasks.

3.2. Representation Learning

While we have extensively analyzed the task-learning capabilities of CLIP through zero-shot transfer in the previous section, it is more common to study the representation learning capabilities of a model. There exist many ways to evaluate the quality of representations as well as disagreements over what properties an "ideal" representation should have (Locatello et al., 2020). Fitting a linear classifier on a representation extracted from the model and measuring its performance on various datasets is a common approach. An alternative is measuring the performance of end-to-end fine-tuning of the model. This increases flexibility, and prior work has convincingly demonstrated that fine-tuning outperforms linear classification on most image classification datasets (Kornblith et al., 2019; Zhai et al., 2019). While the high performance of fine-tuning motivates its study for practical reasons, we still opt for linear classifier based evaluation for several reasons. Our work is focused on developing a high-performing task and dataset-agnostic pre-training approach. Fine-tuning, because it adapts representations to each dataset during the fine-tuning phase, can compensate for and potentially mask failures to learn general and robust representations during the pre-training phase. Linear classifiers, because of their limited flexibility, instead highlight these failures and provide clear feedback during development. For CLIP, training supervised linear classifiers has the added benefit of being very similar to the approach used for its zero-shot classifiers which enables extensive comparisons and analysis in Section 3.1. Finally, we aim to compare CLIP to a comprehensive set of existing models across many tasks. Studying 66 different models on 27 different datasets requires tuning 1782 different evaluations. Fine-tuning opens up a much larger design and hyper-parameter space, which makes it difficult to fairly evaluate and computationally expensive to compare a diverse set of techniques as discussed in other large scale empirical studies (Lucic et al., 2018; Choi et al., 2019). By comparison, linear classifiers require minimal hyper-parameter tuning and have standardized implementations and evaluation procedures. Please see Appendix A for further details on evaluation.

Figure 10 summarizes our findings. To minimize selection effects that could raise concerns of confirmation or reporting bias, we first study performance on the 12 dataset evaluation suite from Kornblith et al. (2019). While small CLIP models such as a ResNet-50 and ResNet-101 outperform other ResNets trained on ImageNet-1K (BiT-S and the originals), they underperform ResNets trained on ImageNet-21K (BiT-M). These small CLIP models also underperform models in the EfficientNet family with similar compute requirements. However, models trained with CLIP scale very well and the largest model we trained (ResNet-50x64) slightly outperforms the best performing existing model (a Noisy Student EfficientNet-L2) on both overall score and compute efficiency. We also find that CLIP vision transformers are about 3x more compute efficient than CLIP ResNets, which allows us to reach higher overall performance within our compute budget. These results qualitatively replicate the findings of Dosovitskiy et al. (2020) which reported that vision transformers are more compute efficient than convnets when trained on sufficiently large datasets. Our best overall model is a ViT-L/14 that is fine-tuned at a higher resolution of 336 pixels on our dataset for 1 additional epoch. This model outperforms the best existing model across this evaluation suite by an average of 2.6%.

As Figure 21 qualitatively shows, CLIP models learn a wider set of tasks than has previously been demonstrated in a single computer vision model trained end-to-end from random initialization. These tasks include geo-localization, optical character recognition, facial emotion recognition, and action recognition. None of these tasks are measured in the evaluation suite of Kornblith et al. (2019). This could be argued to be a form of selection bias in Kornblith et al. (2019)'s study towards tasks that overlap with ImageNet. To address this, we also measure performance on a broader 27 dataset evaluation suite. This evaluation suite, detailed in Appendix A includes datasets representing the aforementioned tasks, German Traffic Signs Recognition Benchmark (Stallkamp et al., 2011), as well as several other datasets adapted from VTAB (Zhai et al., 2019).

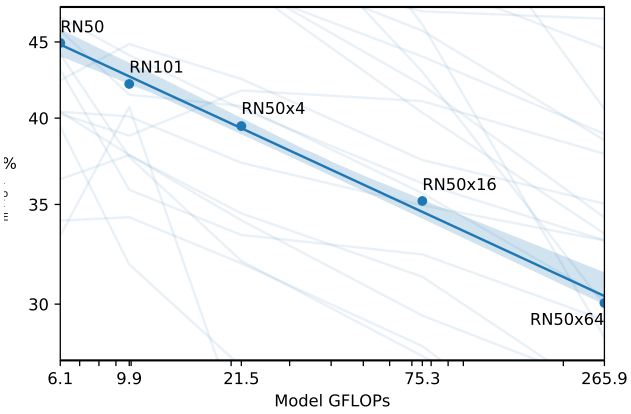


图 9. 零样本 CLIP 性能会随着模型计算量的增加而平滑提升。在 36 个不同数据集上的 39 项评估中，平均零样本错误率可以很好地用对数-对数线性趋势来建模，该趋势覆盖了 5 个不同 CLIP 模型、跨度达 44 倍的计算量范围。浅色线条表示各个评估上的性能，显示尽管总体趋势平滑，但性能差异要大得多。

这是由于子任务上各个训练运行之间的高方差（如 D’Amour 等人（2020）所记录）掩盖了持续改进的趋势，还是在某些任务上性能实际上是计算量的非单调函数。

3.2. 表征学习

虽然我们在上一节中通过零样本迁移广泛分析了 CLIP 的任务学习能力，但研究模型的表征学习能力更为常见。评估表征质量的方法有很多，对于“理想”表征应具备哪些属性也存在分歧（Locatello 等，2020）。一种常见方法是，在从模型中提取的表征上拟合线性分类器，并衡量其在各种数据集上的性能。另一种方法是衡量对模型进行端到端微调后的性能。这提高了灵活性，而且先前的工作已经令人信服地证明，在大多数图像分类数据集上，微调优于线性分类（Kornblith 等，2019；Zhai 等，2019）。尽管微调的高性能出于实际原因促使人们对其进行研究，但我们仍然出于若干原因选择基于线性分类器的评估。我们的工作重点是开发一种高性能、与任务和数据集无关的预训练方法。微调由于会在微调阶段针对每个数据集调整表征，因此可以弥补并可能掩盖在预训练阶段未能学习到通用且鲁棒表征的问题。相比之下，线性分类器由于灵活性有限，会凸显这些失败之处，并在开发过程中提供清晰的反馈。对于 CLIP，训练有监督的线性分类器还有一个额外好处，即它与用于零样本分类器的方法非常相似，这使得第 3.1 节中的广泛比较和分析成为可能。最后，我们旨在将 CLIP 与一组全面的现有模型在多个任务上进行比较。在 27 个不同数据集上研究 66 个不同模型，需要调优 1782 次不同的评估。微调会打开一个更大的设计和超参数空间，这使得公平评估变得困难，并且在比较多种不同技术时计算成本高昂，正如其他大规模实证研究中所讨论的那样（Lucic et al., 2018; Choi et al., 2019）。相比之下，线性分类器只需要最少的超参数调优，并且具有标准化的实现和评估流程。有关评估的更多细节，请参见附录 A。

图 10 总结了我们的发现。为尽量减少可能引发确认偏差或报告偏差担忧的选择效应，我们首先研究了 Kornblith 等人（2019）提出的 12 个数据集评估套件上的性能。尽管较小的 CLIP 模型（如 ResNet-50 和 ResNet-101）优于其他在 ImageNet-1K 上训练的 ResNet（BiT-S 以及原始模型），但它们不如在 ImageNet-21K 上训练的 ResNet（BiT-M）。这些小型 CLIP 模型的表现也不及计算需求相近的 EfficientNet 系列模型。然而，使用 CLIP 训练的模型具有很好的可扩展性，我们训练的最大模型（ResNet-50x64）在总体得分和计算效率上都略微超过了现有表现最佳的模型（Noisy Student EfficientNet-L2）。我们还发现，CLIP 视觉 Transformer 的计算效率约为 CLIP ResNet 的 3 倍，这使我们能够在计算预算内达到更高的整体性能。这些结果在定性上复现了 Dosovitskiy 等人（2020）的发现，即当在足够大的数据集上训练时，视觉 Transformer 比卷积网络更具计算效率。我们整体表现最好的模型是 ViT-L/14，它在我们的数据集上以 336 像素的更高分辨率额外微调了 1 个 epoch。该模型在这一评估套件中平均比现有最佳模型高出 2.6%。

如图 21 定性所示，CLIP 模型学习到的任务范围，比以往在一个从随机初始化开始端到端训练的单一计算机视觉模型中所展示的更广。这些任务包括地理定位、光学字符识别、面部情绪识别和动作识别。Kornblith 等人（2019）的评估套件中并未衡量这些任务。这可以被认为是 Kornblith 等人（2019）的研究中存在的一种选择偏差，即偏向于与 ImageNet 重叠的任务。为了解决这一问题，我们还在一个更广泛的、包含 27 个数据集的评估套件上衡量性能。该评估套件详见附录 A，其中包括代表上述任务的数据集、德国交通标志识别基准（Stallkamp 等人，2011），以及从 VTAB（Zhai 等人，2019）改编的其他几个数据集。

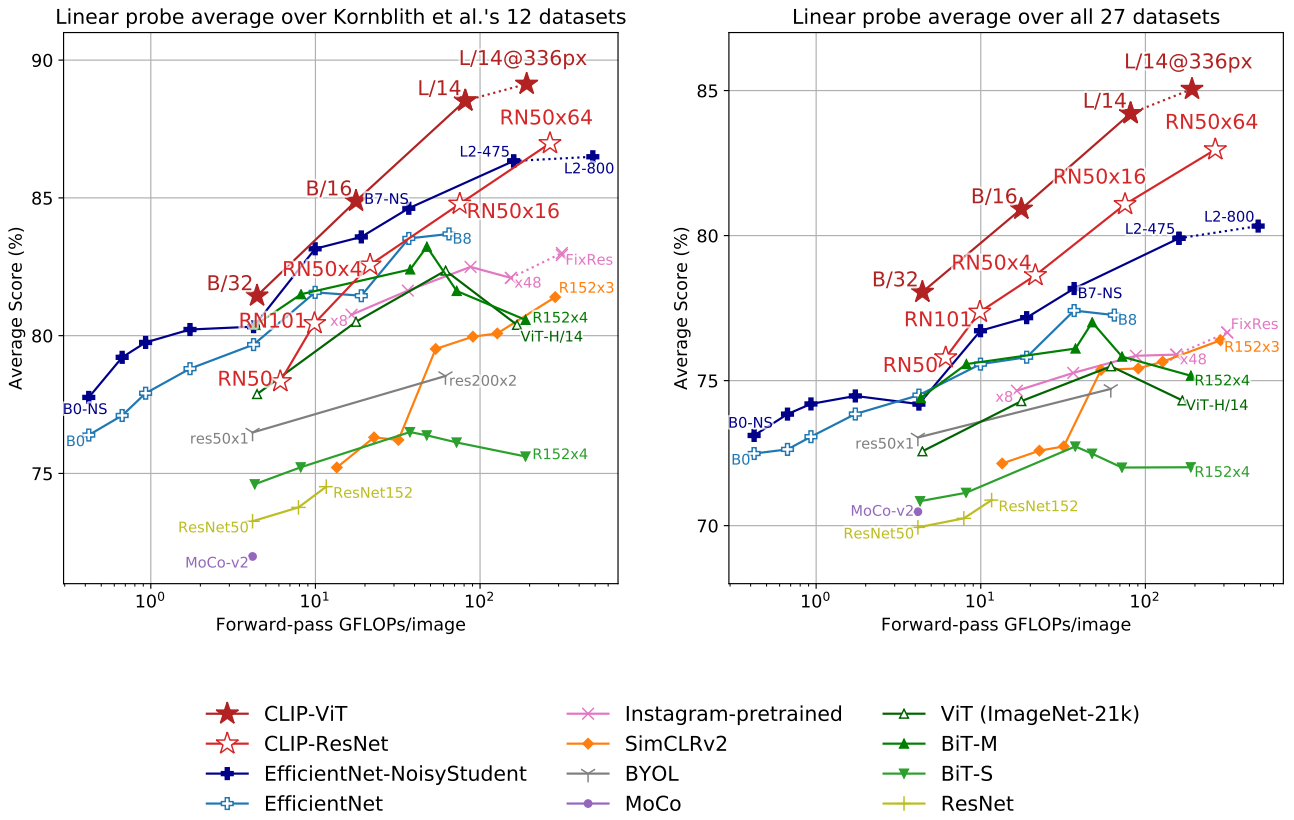


Figure 10. **Linear probe performance of CLIP models in comparison with state-of-the-art computer vision models**, including EfficientNet (Tan & Le, 2019; Xie et al., 2020), MoCo (Chen et al., 2020d), Instagram-pretrained ResNeXt models (Mahajan et al., 2018; Touvron et al., 2019), BiT (Kolesnikov et al., 2019), ViT (Dosovitskiy et al., 2020), SimCLRv2 (Chen et al., 2020c), BYOL (Grill et al., 2020), and the original ResNet models (He et al., 2016b). (Left) Scores are averaged over 12 datasets studied by Kornblith et al. (2019). (Right) Scores are averaged over 27 datasets that contain a wider variety of distributions. Dotted lines indicate models fine-tuned or evaluated on images at a higher-resolution than pre-training. See Table 10 for individual scores and Figure 20 for plots for each dataset.

On this broader evaluation suite, the benefits of CLIP are more clear. All CLIP models, regardless of scale, outperform all evaluated systems in terms of compute efficiency. The improvement in average score of the best model over previous systems increases from 2.6% to 5%. We also find that self-supervised systems do noticeably better on our broader evaluation suite. For instance, while SimCLRv2 still underperforms BiT-M on average on the 12 datasets of Kornblith et al. (2019), SimCLRv2 outperforms BiT-M on our 27 dataset evaluation suite. These findings suggest continuing to expand task diversity and coverage in order to better understand the "general" performance of systems. We suspect additional evaluation efforts along the lines of VTAB to be valuable.

In addition to the aggregate analysis above, we visualize per-dataset differences in the performance of the best CLIP model and the best model in our evaluation suite across all 27 datasets in Figure 11. CLIP outperforms the Noisy Student EfficientNet-L2 on 21 of the 27 datasets. CLIP improves the most on tasks which require OCR (SST2 and HatefulMemes), geo-localization and scene recognition (Country211, SUN397), and activity recognition in videos (Kinetics700 and UCF101). In addition CLIP also does much better on fine-grained car and traffic sign recognition (Stanford Cars and GTSRB). This may reflect a problem with overly narrow supervision in ImageNet. A result such as the 14.7% improvement on GTSRB could be indicative of an issue with ImageNet-1K, which has only a single label for all traffic and street signs. This could encourage a supervised representation to collapse intra-class details and hurt accuracy on a fine-grained downstream task. As mentioned, CLIP still underperforms the EfficientNet on several datasets. Unsurprisingly, the dataset that the EfficientNet does best relative to CLIP on is the one it was trained on: ImageNet. The EffcientNet also slightly outperforms CLIP on low-resolution datasets such as CIFAR10 and CIFAR100. We suspect this is at least partly due to the lack of scale-based data augmentation in CLIP. The EfficientNet also does slightly better on PatchCamelyon and CLEVRCounts, datasets where overall performance is still

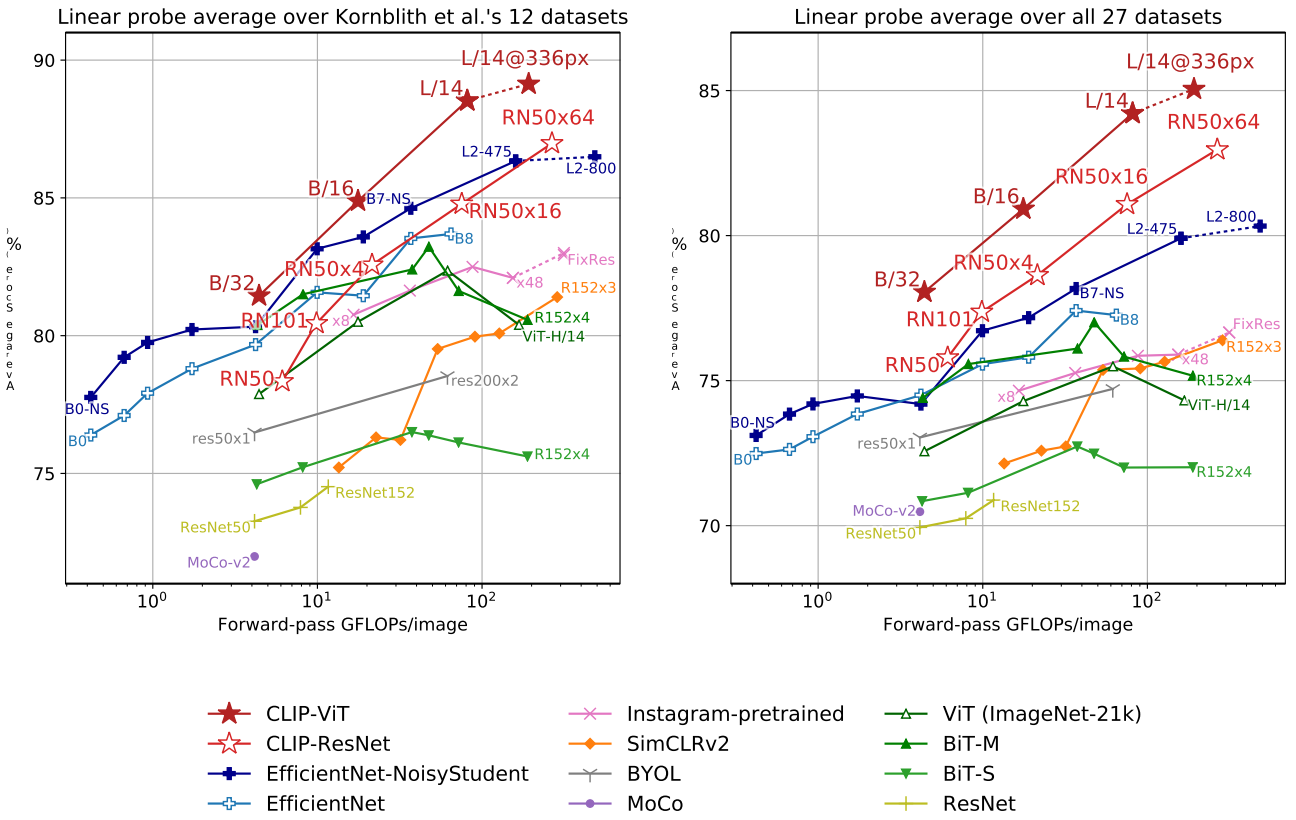


图 10. CLIP 模型与最先进的计算机视觉模型在线性探针性能上的比较，包括 EfficientNet（Tan & Le，2019；Xie 等，2020）、MoCo（Chen 等，2020d）、在 Instagram 上预训练的 ResNeXt 模型（Mahajan 等，2018；Touvron 等，2019）、BiT（Kolesnikov 等，2019）、ViT（Dosovitskiy 等，2020）、SimCLRv2（Chen 等，2020c）、BYOL（Grill 等，2020）以及原始 ResNet 模型（He 等，2016b）。（左）分数是在 Kornblith 等（2019）研究的 12 个数据集上取平均得到的。（右）分数是在包含更广泛分布类型的 27 个数据集上取平均得到的。虚线表示在高于预训练分辨率的图像上进行微调或评估的模型。各项分数见表 10，各数据集的图见图 20。

在这个更广泛的评估套件上，CLIP 的优势更加明显。所有 CLIP 模型，无论规模大小，在计算效率方面都优于所有被评估的系统。最佳模型相较于以往系统的平均得分提升从 2.6% 增加到 5%。我们还发现，自监督系统在我们更广泛的评估套件上表现明显更好。例如，尽管 SimCLRv2 在 Kornblith 等人（2019）的 12 个数据集上的平均表现仍不如 BiT-M，但在我们的 27 个数据集评估套件上，SimCLRv2 的表现超过了 BiT-M。这些发现表明，应继续扩大任务多样性和覆盖范围，以便更好地理解系统的“通用”性能。我们认为，沿着 VTAB 思路开展更多评估工作将是有价值的。

除了上述汇总分析外，我们还在图 11 中可视化了最佳 CLIP 模型与我们评估套件中最佳模型在全部 27 个数据集上的逐数据集性能差异。CLIP 在 27 个数据集中的 21 个上优于 Noisy Student EfficientNet-L2。CLIP 在需要 OCR 的任务上提升最大（SST2 和 HatefulMemes）、地理定位与场景识别（Country211、SUN397），以及视频中的动作识别（Kinetics700 和 UCF101）。此外，CLIP 在细粒度汽车和交通标志识别（Stanford Cars 和 GTSRB）上也表现得好得多。这可能反映了 ImageNet 中监督信号过于狭窄的问题。比如在 GTSRB 上提升 14.7% 这样的结果，可能表明 ImageNet-1K 存在问题，因为它对所有交通和街道标志只有一个标签。这可能会促使有监督表征压缩类内细节，并损害其在细粒度下游任务上的准确率。如前所述，CLIP 在若干数据集上的表现仍不如 EfficientNet。不出所料，相对于 CLIP，EfficientNet 表现最好的数据集正是它的训练数据集：ImageNet。EfficientNet 在 CIFAR10 和 CIFAR100 等低分辨率数据集上也略优于 CLIP。我们怀疑这至少部分是由于 CLIP 缺乏基于尺度的数据增强。EfficientNet 在 PatchCamelyon 和 CLEVRCounts 上也略有优势，而这些数据集上的整体性能仍然

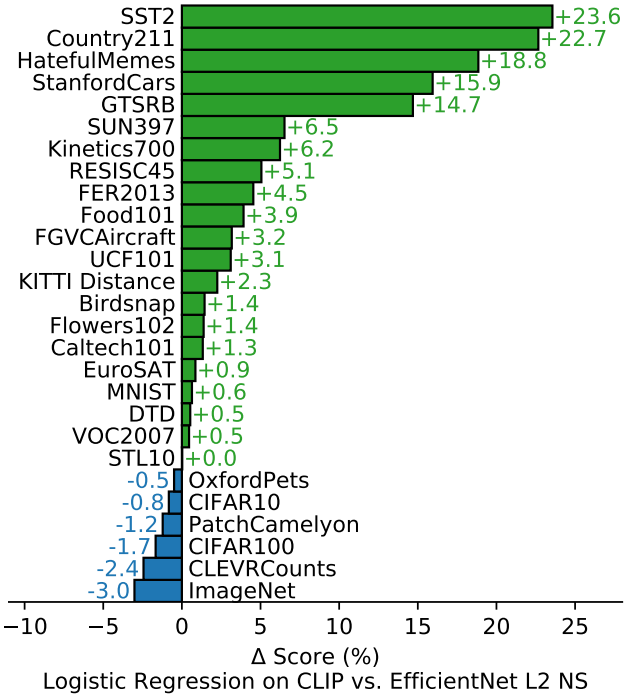


Figure 11. **CLIP's features outperform the features of the best ImageNet model on a wide variety of datasets.** Fitting a linear classifier on CLIP's features outperforms using the Noisy Student EfficientNet-L2 on 21 out of 27 datasets.

low for both approaches.

3.3. Robustness to Natural Distribution Shift

In 2015, it was announced that a deep learning model exceeded human performance on the ImageNet test set (He et al., 2015). However, research in the subsequent years has repeatedly found that these models still make many simple mistakes (Dodge & Karam, 2017; Geirhos et al., 2018; Alcorn et al., 2019), and new benchmarks testing these systems has often found their performance to be much lower than both their ImageNet accuracy and human accuracy (Recht et al., 2019; Barbu et al., 2019). What explains this discrepancy? Various ideas have been suggested and studied (Ilyas et al., 2019; Geirhos et al., 2020). A common theme of proposed explanations is that deep learning models are exceedingly adept at finding correlations and patterns which hold across their training dataset and thus improve in-distribution performance. However many of these correlations and patterns are actually spurious and do not hold for other distributions and result in large drops in performance on other datasets.

We caution that, to date, most of these studies limit their evaluation to models trained on ImageNet. Recalling the topic of discussion, it may be a mistake to generalize too far from these initial findings. To what degree are these failures attributable to deep learning, ImageNet, or some combination of the two? CLIP models, which are trained via natural language supervision on a very large dataset and are capable of high zero-shot performance, are an opportunity to investigate this question from a different angle.

Taori et al. (2020) is a recent comprehensive study moving towards quantifying and understanding these behaviors for ImageNet models. Taori et al. (2020) study how the performance of ImageNet models change when evaluated on *natural distribution shifts*. They measure performance on a set of 7 distribution shifts: ImageNetV2 (Recht et al., 2019), ImageNet Sketch (Wang et al., 2019), Youtube-BB and ImageNet-Vid (Shankar et al., 2019), ObjectNet (Barbu et al., 2019), ImageNet Adversarial (Hendrycks et al., 2019), and ImageNet Rendition (Hendrycks et al., 2020a). They distinguish these datasets, which all consist of novel images collected from a variety of sources, from *synthetic distribution shifts* such as ImageNet-C (Hendrycks & Dietterich, 2019), Stylized ImageNet (Geirhos et al., 2018), or adversarial attacks (Goodfellow et al., 2014) which are created by perturbing existing images in various ways. They propose this distinction because in part because they find that while several techniques have been demonstrated to improve performance on synthetic distribution shifts, they often fail to yield consistent improvements on natural distributions.[3]

Across these collected datasets, the accuracy of ImageNet models drop well below the expectation set by the ImageNet validation set. For the following summary discussion we report average accuracy across all 7 natural distribution shift datasets and average accuracy across the corresponding class subsets of ImageNet unless otherwise specified. Additionally, for Youtube-BB and ImageNet-Vid, which have two different evaluation settings, we use the average of pm-0 and pm-10 accuracy.

A ResNet-101 makes 5 times as many mistakes when evaluated on these natural distribution shifts compared to the ImageNet validation set. Encouragingly however, Taori et al. (2020) find that accuracy under distribution shift increases predictably with ImageNet accuracy and is well modeled as a linear function of logit-transformed accuracy. Taori et al. (2020) use this finding to propose that robustness analysis should distinguish between *effective* and *relative* robustness. Effective robustness measures improvements in accuracy under distribution shift above what is predicted by the documented relationship between in-distribution and out-of-distribution accuracy. Relative robustness captures any improvement in out-of-distribution accuracy. Taori et al. (2020) argue that robustness techniques should aim to improve both effective robustness and relative robustness.

Almost all models studied in Taori et al. (2020) are trained

[3]We refer readers to Hendrycks et al. (2020a) for additional experiments and discussion on this claim.

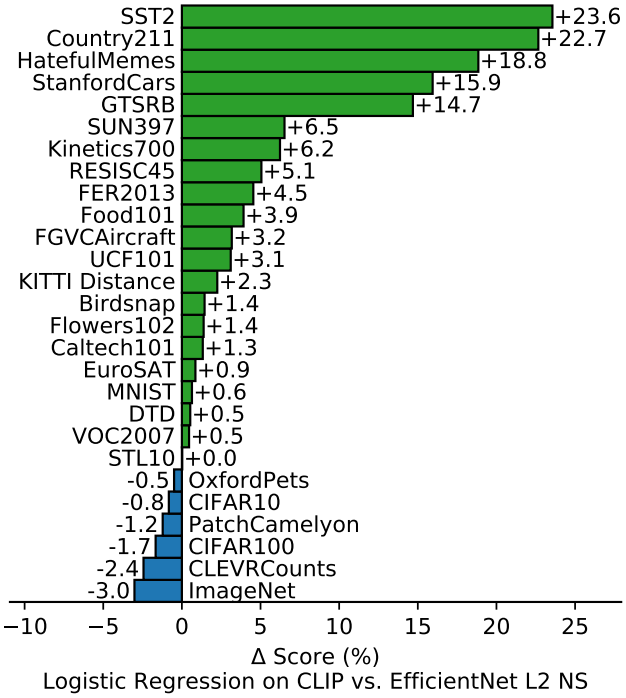


图 11。CLIP 的特征在各种数据集上优于最佳 ImageNet 模型的特征。在 CLIP 的特征上拟合线性分类器，在 27 个数据集中的 21 个上优于使用 Noisy Student EfficientNet-L2。

两种方法都较低。

3.3. 对自然分布偏移的鲁棒性

2015 年，有报道称一个深度学习模型在 ImageNet 测试集上的表现超过了人类水平（He et al., 2015）。然而，随后几年的研究反复发现，这些模型仍然会犯许多简单错误（Dodge & Karam, 2017；Geirhos et al., 2018；Alcorn et al., 2019），而用于测试这些系统的新基准也常常发现，它们的性能远低于其 ImageNet 准确率和人类准确率（Recht et al., 2019；Barbu et al., 2019）。是什么解释了这种差异？人们提出并研究了各种观点（Ilyas et al., 2019；Geirhos et al., 2020）。这些解释的一个共同主题是，深度学习模型极其擅长发现贯穿其训练数据集的相关性和模式，从而提高分布内性能。然而，这些相关性和模式中有许多实际上是伪相关，并不适用于其他分布，导致在其他数据集上的性能大幅下降。

我们提醒读者，迄今为止，这些研究中的大多数都将其评估限制在使用 ImageNet 训练的模型上。回想讨论的主题，过度泛化这些初步发现可能是一个错误。这些失败在多大程度上可归因于深度学习、ImageNet，或某些两者的结合？CLIP 模型通过在超大规模数据集上进行自然语言监督训练，具备很强的零样本性能，为从不同角度研究这一问题提供了机会。

Taori 等人（2020）是一项近期的综合性研究，旨在朝着量化和理解 ImageNet 模型这些行为的方向迈进。Taori 等人（2020）研究了在 *natural distribution shifts* 上评估时 ImageNet 模型的性能如何变化。他们在一组包含 7 种分布偏移的数据集上衡量性能：ImageNetV2（Recht 等人，2019）、ImageNet Sketch（Wang 等人，2019）、Youtube-BB 和 ImageNet-Vid（Shankar 等人，2019）、ObjectNet（Barbu 等人，2019）、ImageNet Adversarial（Hendrycks 等人，2019）以及 ImageNet Rendition（Hendrycks 等人，2020a）。他们将这些数据集与 *synthetic distri- bution shifts* 区分开来；前者都由从各种来源收集的新图像组成，而后者如 ImageNet-C（Hendrycks & Dietterich，2019）、Stylized ImageNet（Geirhos 等人，2018）或对抗攻击（Goodfellow 等人，2014），则是通过以各种方式扰动现有图像而创建的。他们提出这种区分，部分原因是他们发现，尽管已有若干技术被证明可以提升在合成分布偏移上的性能，但它们往往无法在自然分布上带来一致的改进。[3]

在这些收集的数据集上，ImageNet 模型的准确率远低于 ImageNet 验证集所设定的预期。在以下总结性讨论中，除非另有说明，我们报告所有 7 个自然分布偏移数据集上的平均准确率，以及 ImageNet 中相应类别子集上的平均准确率。此外，对于 Youtube-BB 和 ImageNet-Vid 这两个具有不同评估设置的数据集，我们使用 pm-0 和 pm-10 准确率的平均值。

与 ImageNet 验证集相比，ResNet-101 在这些自然分布偏移上进行评估时，错误数量增加了 5 倍。不过令人鼓舞的是，Taori 等人（2020）发现，分布偏移下的准确率会随着 ImageNet 准确率以可预测的方式提高，并且可以很好地建模为经过 logit 变换的准确率的线性函数。Taori 等人（2020）利用这一发现提出，鲁棒性分析应区分 *effective* 鲁棒性和 *relative* 鲁棒性。有效鲁棒性衡量的是，在分布偏移下，准确率相对于由已记录的分布内准确率与分布外准确率之间关系所预测的水平的提升。相对鲁棒性则捕捉分布外准确率的任何提升。Taori 等人（2020）认为，鲁棒性技术应旨在同时提高有效鲁棒性和相对鲁棒性。

Taori 等人（2020）研究的几乎所有模型都经过了训练

[3]We refer readers to Hendrycks et al. (2020a) for additional experiments and discussion on this claim.

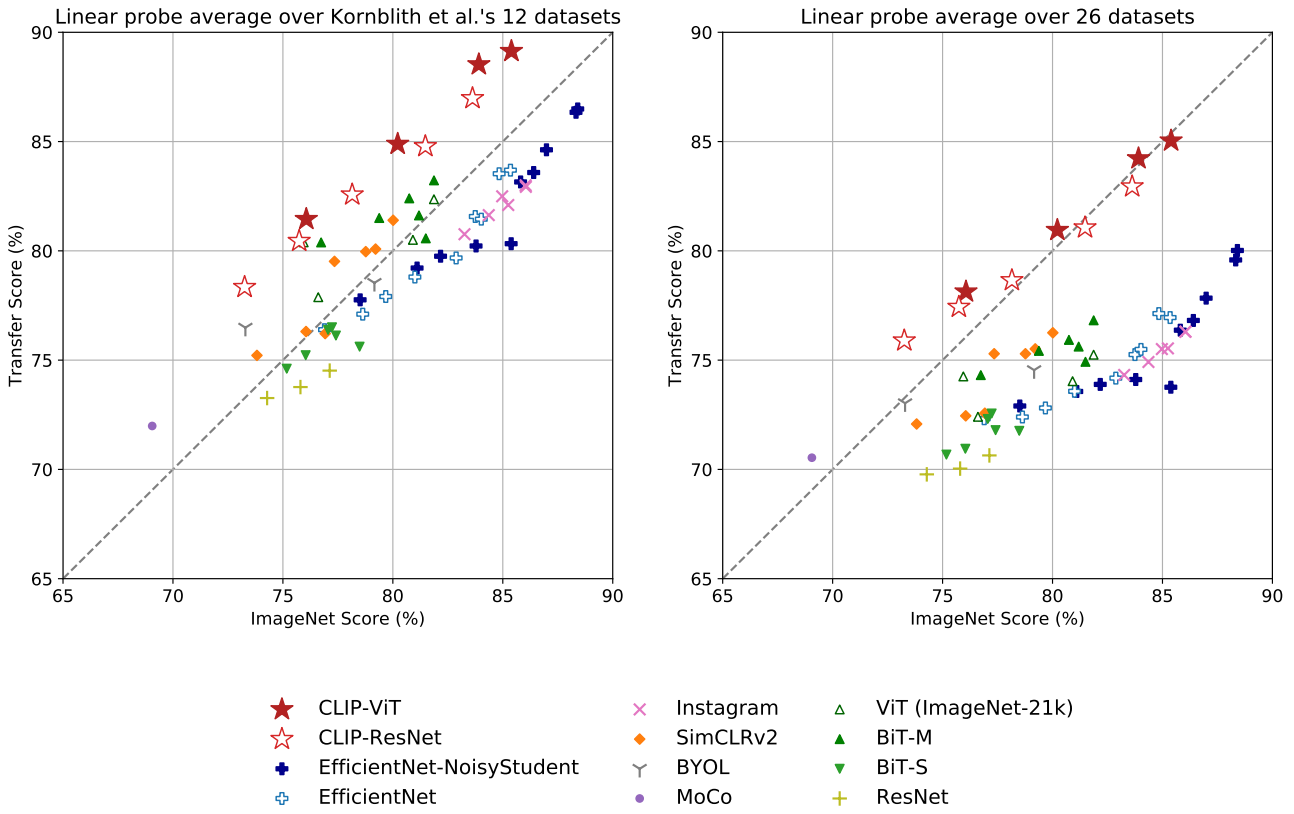


Figure 12. **CLIP's features are more robust to task shift when compared to models pre-trained on ImageNet.** For both dataset splits, the transfer scores of linear probes trained on the representations of CLIP models are higher than other models with similar ImageNet performance. This suggests that the representations of models trained on ImageNet are somewhat overfit to their task.

or fine-tuned on the ImageNet dataset. Returning to the discussion in the introduction to this section - is training or adapting to the ImageNet dataset distribution the cause of the observed robustness gap? Intuitively, a zero-shot model should not be able to exploit spurious correlations or patterns that hold only on a specific distribution, since it is not trained on that distribution. [4] Thus it is reasonable to expect zero-shot models to have much higher effective robustness. In Figure 13, we compare the performance of zero-shot CLIP with existing ImageNet models on natural distribution shifts. All zero-shot CLIP models improve effective robustness by a large amount and reduce the size of the gap between ImageNet accuracy and accuracy under distribution shift by up to 75%.

While these results show that zero-shot models can be much more robust, they do not necessarily mean that supervised learning on ImageNet causes a robustness gap. Other details of CLIP, such as its large and diverse pre-training dataset or use of natural language supervision could also result in much more robust models regardless of whether they are zero-shot or fine-tuned. As an initial experiment to potentially begin narrowing this down, we also measure how the performance of CLIP models change after adapting to the ImageNet distribution via a L2 regularized logistic regression classifier fit to CLIP features on the ImageNet training set. We visualize how performance changes from the zero-shot classifier in Figure 14. Although adapting CLIP to the ImageNet distribution increases its ImageNet accuracy by 9.2% to 85.4% overall, and ties the accuracy of the 2018 SOTA from Mahajan et al. (2018), *average accuracy under distribution shift slightly decreases*.

It is surprising to see a 9.2% increase in accuracy, which corresponds to roughly 3 years of improvement in SOTA, fail to translate into any improvement in average performance under distribution shift. We also break down the differences between zero-shot accuracy and linear classifier accuracy per dataset in Figure 14 and find performance still increases significantly on one dataset, ImageNetV2. ImageNetV2 closely followed the creation process of the original ImageNet dataset which suggests that gains in accuracy from supervised adaptation are closely concentrated around the ImageNet distribution. Performance decreases by 4.7% on

[4] We caution that a zero-shot model can still exploit spurious correlations that are shared between the pre-training and evaluation distributions.

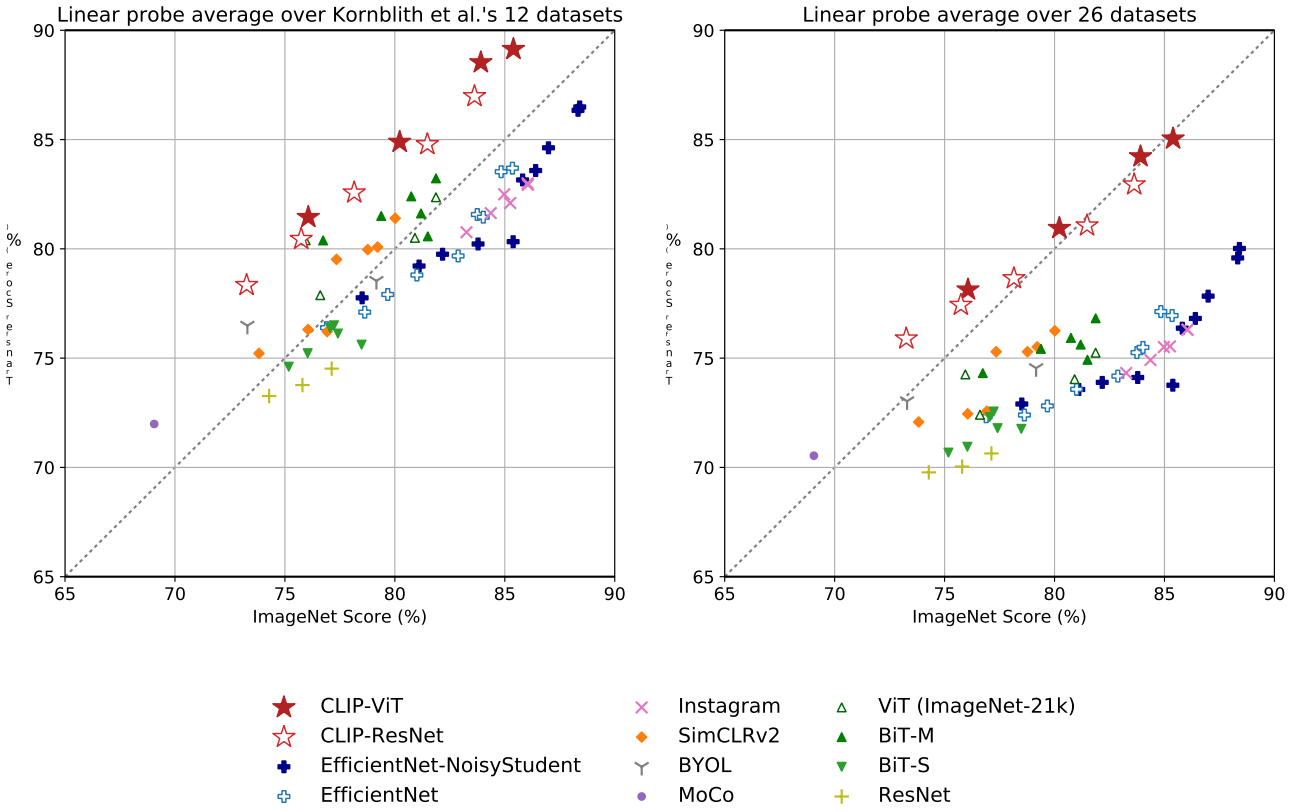


图 12。与在 ImageNet 上预训练的模型相比，CLIP 的特征对任务转移更加鲁棒。对于两个数据集划分，在 CLIP 模型表征上训练的线性探针的迁移得分都高于具有相似 ImageNet 性能的其他模型。这表明，在 ImageNet 上训练的模型的表征在某种程度上对其任务过拟合。

或在 ImageNet 数据集上进行微调。回到本节引言中的讨论——在 ImageNet 数据集分布上进行训练或适配，是否是导致所观察到的鲁棒性差距的原因？直观地说，零样本模型不应能够利用仅在特定分布上成立的虚假相关性或模式，因为它并未在该分布上进行训练。[4] 因此，有理由预期零样本模型具有高得多的有效鲁棒性。在图 13 中，我们将零样本 CLIP 与现有 ImageNet 模型在自然分布偏移下的性能进行了比较。所有零样本 CLIP 模型都大幅提高了有效鲁棒性，并将 ImageNet 准确率与分布偏移下准确率之间的差距最多缩小了 75%。

虽然这些结果表明零样本模型可以更加鲁棒，但它们并不一定意味着在 ImageNet 上进行监督学习会导致鲁棒性差距。CLIP 的其他细节，例如其庞大且多样化的预训练数据集或对自然语言监督的使用，也可能导致这种结果。

[4]We caution that a zero-shot model can still exploit spurious correlations that are shared between the pre-training and evaluation distributions.

无论这些模型是零样本还是经过微调的，它们都要稳健得多。作为一个可能开始缩小这一问题范围的初步实验，我们还测量了在通过一个 L2 正则化的逻辑回归分类器适配 ImageNet 分布后，CLIP 模型的性能如何变化；该分类器是在 ImageNet 训练集上的 CLIP 特征上拟合得到的。我们在图 14 中可视化了相对于零样本分类器的性能变化。尽管将 CLIP 适配到 ImageNet 分布后，其 ImageNet 准确率总体提高了 9.2%，达到 85.4%，并追平了 Mahajan 等人（2018）在 2018 年的 SOTA 准确率，*average accuracy under distribution shift slightly decreases*。

令人惊讶的是，准确率提高了 9.2%（这大约相当于 SOTA 约 3 年的进步），却未能转化为分布偏移下平均性能的任何提升。我们还在图 14 中按数据集分解了零样本准确率与线性分类器准确率之间的差异，发现性能仍然在一个数据集 ImageNetV2 上显著提高。ImageNetV2 紧密遵循了原始 ImageNet 数据集的创建过程，这表明通过监督式适应获得的准确率提升高度集中在 ImageNet 分布附近。性能在上下降了 4.7%

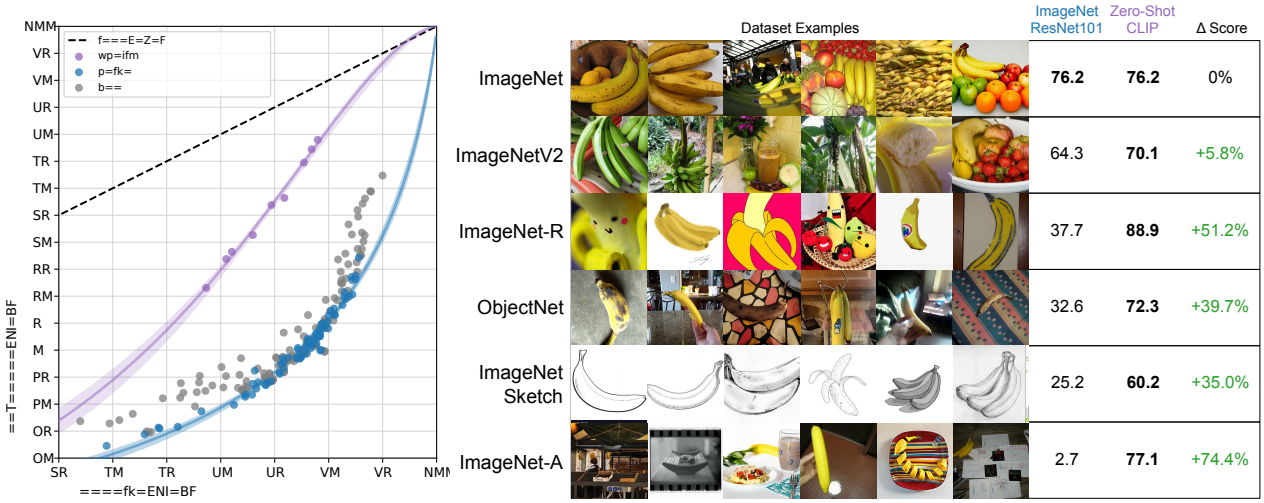


Figure 13. **Zero-shot CLIP is much more robust to distribution shift than standard ImageNet models.** (Left) An ideal robust model (dashed line) performs equally well on the ImageNet distribution and on other natural image distributions. Zero-shot CLIP models shrink this "robustness gap" by up to 75%. Linear fits on logit transformed values are shown with bootstrap estimated 95% confidence intervals. (Right) Visualizing distribution shift for bananas, a class shared across 5 of the 7 natural distribution shift datasets. The performance of the best zero-shot CLIP model, ViT-L/14@336px, is compared with a model that has the same performance on the ImageNet validation set, ResNet-101.

ImageNet-R, 3.8% on ObjectNet, 2.8% on ImageNet Sketch, and 1.9% on ImageNet-A. The change in accuracy on the two other datasets, Youtube-BB and ImageNet Vid, is insignificant.

How is it possible to improve accuracy by 9.2% on the ImageNet dataset with little to no increase in accuracy under distribution shift? Is the gain primarily from "exploiting spurious correlations"? Is this behavior unique to some combination of CLIP, the ImageNet datatset, and the distribution shifts studied, or a more general phenomena? Does it hold for end-to-end finetuning as well as linear classifiers? We do not have confident answers to these questions at this time. Prior work has also pre-trained models on distributions other than ImageNet, but it is common to study and release models only after they have been fine-tuned to ImageNet. As a step towards understanding whether pre-trained zero-shot models consistently have higher effective robustness than fine-tuned models, we encourage the authors of Mahajan et al. (2018), Kolesnikov et al. (2019), and Dosovitskiy et al. (2020) to, if possible, study these questions on their models as well.

We also investigate another robustness intervention enabled by flexible zero-shot natural-language-based image classifiers. The target classes across the 7 transfer datasets are not always perfectly aligned with those of ImageNet. Two datasets, Youtube-BB and ImageNet-Vid, consist of superclasses of ImageNet. This presents a problem when trying to use the fixed 1000-way classifier of an ImageNet model to make predictions. Taori et al. (2020) handle this by max-pooling predictions across all sub-classes according to the ImageNet class hierarchy. Sometimes this mapping is much less than perfect. For the person class in Youtube-BB, predictions are made by pooling over the ImageNet classes for a baseball player, a bridegroom, and a scuba diver. With CLIP we can instead generate a custom zero-shot classifier for each dataset directly based on its class names. In Figure 14 we see that this improves average effective robustness by 5% but is concentrated in large improvements on only a few datasets. Curiously, accuracy on ObjectNet also increases by 2.3%. Although the dataset was designed to closely overlap with ImageNet classes, using the names provided for each class by ObjectNet's creators still helps a small amount compared to using ImageNet class names and pooling predictions when necessary.

While zero-shot CLIP improves effective robustness, Figure 14 shows that the benefit is almost entirely gone in a fully supervised setting. To better understand this difference, we investigate how effective robustness changes on the continuum from zero-shot to fully supervised. In Figure 15 we visualize the performance of 0-shot, 1-shot, 2-shot, 4-shot ..., 128-shot, and fully supervised logistic regression classifiers on the best CLIP model's features. We see that while few-shot models also show higher effective robustness than existing models, this benefit fades as in-distribution performance increases with more training data and is mostly, though not entirely, gone for the fully supervised model. Additionally, zero-shot CLIP is notably more robust than a few-shot model with equivalent ImageNet performance.

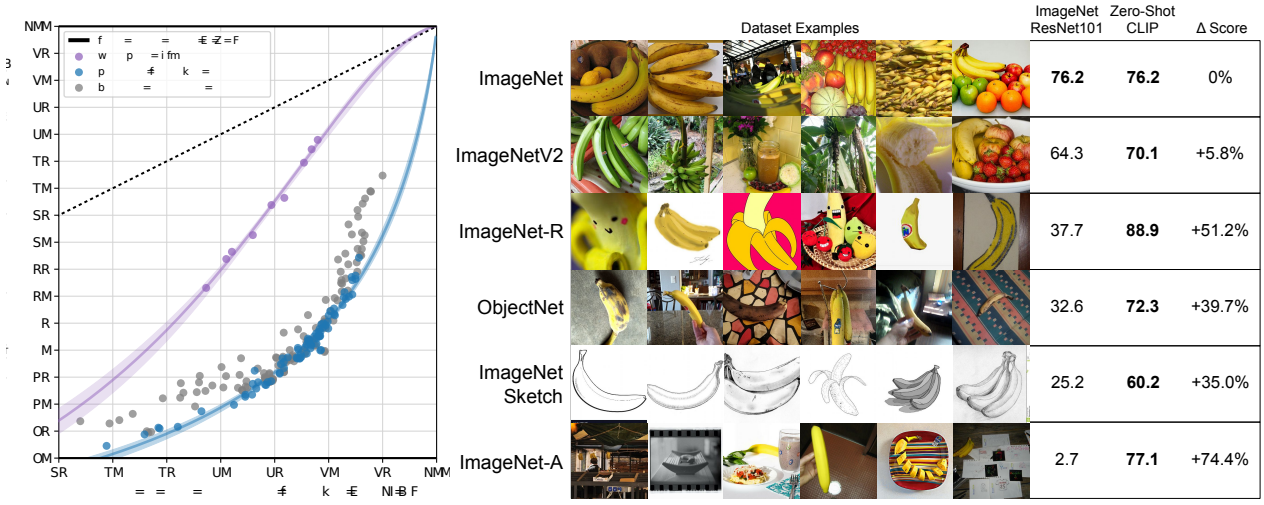


图 13. 零样本 CLIP 比标准 ImageNet 模型对分布偏移更加鲁棒。（左）理想的鲁棒模型（虚线）在 ImageNet 分布和其他自然图像分布上表现同样出色。零样本 CLIP 模型将这种“鲁棒性差距”最多缩小了 75%。图中显示了对 logit 变换值进行的线性拟合，并给出了通过自助法估计的 95% 置信区间。（右）对香蕉类别的分布偏移进行可视化；香蕉是 7 个自然分布偏移数据集中有 5 个都包含的共享类别。将最佳零样本 CLIP 模型 ViT-L/14@336px 的性能与在 ImageNet 验证集上具有相同性能的模型 ResNet-101 进行比较。

在 ImageNet-R 上为 3.8%，在 ObjectNet 上为 2.8%，在 ImageNet Sketch 上为 1.9%，在 ImageNet-A 上。其他两个数据集 Youtube-BB 和 ImageNet Vid 上的准确率变化并不显著。

如何可能在 ImageNet 数据集上的准确率提高 9.2%，而在分布偏移下准确率几乎没有提高？这种增益主要来自“利用虚假相关性”吗？这种行为是 CLIP、ImageNet 数据集以及所研究的分布偏移的某种组合所独有的，还是一种更普遍的现象？它是否同样适用于端到端微调和线性分类器？目前我们还无法自信地回答这些问题。此前的工作也曾在 ImageNet 以外的分布上预训练模型，但通常只在这些模型被微调到 ImageNet 之后才对其进行研究并发布。作为理解预训练零样本模型是否始终比微调模型具有更高有效鲁棒性的一步，我们鼓励 Mahajan 等人（2018）、Kolesnikov 等人（2019）和 Dosovitskiy 等人（2020）的作者，如果可能的话，也在他们的模型上研究这些问题。

我们还研究了另一种鲁棒性干预方法，该方法由灵活的、基于自然语言的零样本图像分类器所支持。7 个迁移数据集中的目标类别并不总是与 ImageNet 的类别完全对齐。其中两个数据集，Youtube-BB 和 ImageNet-Vid，由 ImageNet 的超类组成。当试图使用 ImageNet 模型固定的 1000 类分类器进行预测时，这会带来问题。Taori 等人（2020）通过最大-

根据 ImageNet 类别层次结构汇集所有子类的预测。有时这种映射远非完美。对于 Youtube-BB 中的 person 类别，预测是通过汇集 ImageNet 中 baseball player、bridegroom 和 scuba diver 这些类别来完成的。使用 CLIP，我们则可以直接基于每个数据集的类别名称，为其生成一个自定义的零样本分类器。在图 14 中，我们看到这使平均有效鲁棒性提高了 5%，但这种提升主要集中在少数几个数据集上的大幅改进。令人好奇的是，ObjectNet 上的准确率也提高了 2.3%。尽管该数据集的设计目的是与 ImageNet 类别高度重叠，但与使用 ImageNet 类别名称并在必要时汇集预测相比，使用 ObjectNet 创建者为每个类别提供的名称仍然带来了一点帮助。

虽然零样本 CLIP 提高了有效鲁棒性，但图 14 显示，在完全监督设置下，这一收益几乎完全消失。为了更好地理解这种差异，我们研究了从零样本到完全监督这一连续范围内有效鲁棒性如何变化。在图 15 中，我们可视化了基于最佳 CLIP 模型特征的 0-shot、1-shot、2-shot、4-shot……128-shot 以及完全监督逻辑回归分类器的性能。我们看到，虽然少样本模型也比现有模型表现出更高的有效鲁棒性，但随着训练数据增多、分布内性能提升，这一收益逐渐减弱，并且在完全监督模型中基本消失，尽管并非完全消失。此外，零样本 CLIP 明显比具有相同 ImageNet 性能的少样本模型更加鲁棒。

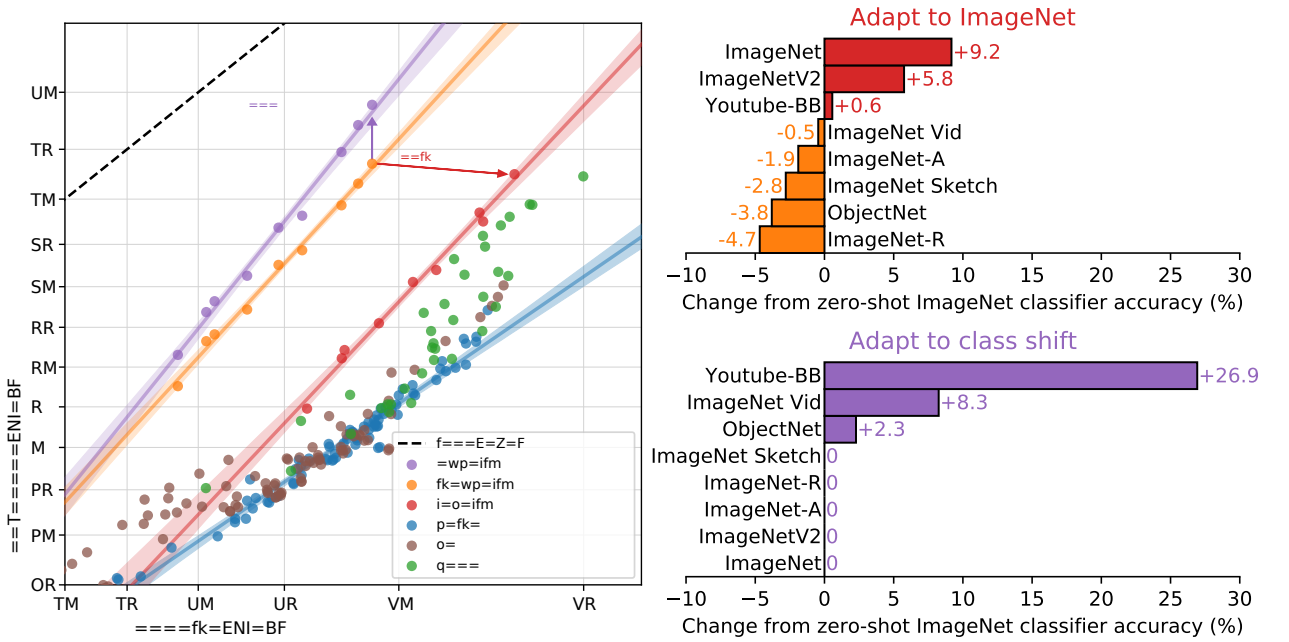


Figure 14. **While supervised adaptation to ImageNet increases ImageNet accuracy by 9.2%, it slightly reduces average robustness.** (Left) Customizing zero-shot CLIP to each dataset improves robustness compared to using a single static zero-shot ImageNet classifier and pooling predictions across similar classes as in Taori et al. (2020). CLIP models adapted to ImageNet have similar effective robustness as the best prior ImageNet models. (Right) Details of per dataset changes in accuracy for the two robustness interventions. Adapting to ImageNet increases accuracy on ImageNetV2 noticeably but trades off accuracy on several other distributions. Dataset specific zero-shot classifiers can improve accuracy by a large amount but are limited to only a few datasets that include classes which don't perfectly align with ImageNet categories.

Across our experiments, high effective robustness seems to result from minimizing the amount of distribution specific training data a model has access to, but this comes at a cost of reducing dataset-specific performance.

Taken together, these results suggest that the recent shift towards large-scale task and dataset agnostic pre-training combined with a reorientation towards zero-shot and few-shot benchmarking on broad evaluation suites (as advocated by Yogatama et al. (2019) and Linzen (2020)) promotes the development of more robust systems and provides a more accurate assessment of performance. We are curious to see if the same results hold for zero-shot models in the field of NLP such as the GPT family. While Hendrycks et al. (2020b) has reported that pre-training improves relative robustness on sentiment analysis, Miller et al. (2020)'s study of the robustness of question answering models under natural distribution shift finds, similar to Taori et al. (2020), little evidence of effective robustness improvements to date.

4. Comparison to Human Performance

How does CLIP compare to human performance and human learning? To get a better understanding of how well humans perform in similar evaluation settings to CLIP, we evaluated humans on one of our tasks. We wanted to get a sense of how strong human zero-shot performance is at these tasks, and how much human performance is improved if they are shown one or two image samples. This can help us to compare task difficulty for humans and CLIP, and identify correlations and differences between them.

We had five different humans look at each of 3669 images in the test split of the Oxford IIT Pets dataset (Parkhi et al., 2012) and select which of the 37 cat or dog breeds best matched the image (or 'I don't know' if they were completely uncertain). In the zero-shot case the humans were given no examples of the breeds and asked to label them to the best of their ability without an internet search. In the one-shot experiment the humans were given one sample image of each breed and in the two-shot experiment they were given two sample images of each breed.[5]

One possible concern was that the human workers were not sufficiently motivated in the zero-shot task. High human accuracy of 94% on the STL-10 dataset (Coates et al., 2011)

[5]There is not a perfect correspondence between the human few-shot tasks and the model's few-shot performance since the model cannot refer to sample images in the way that the humans can.

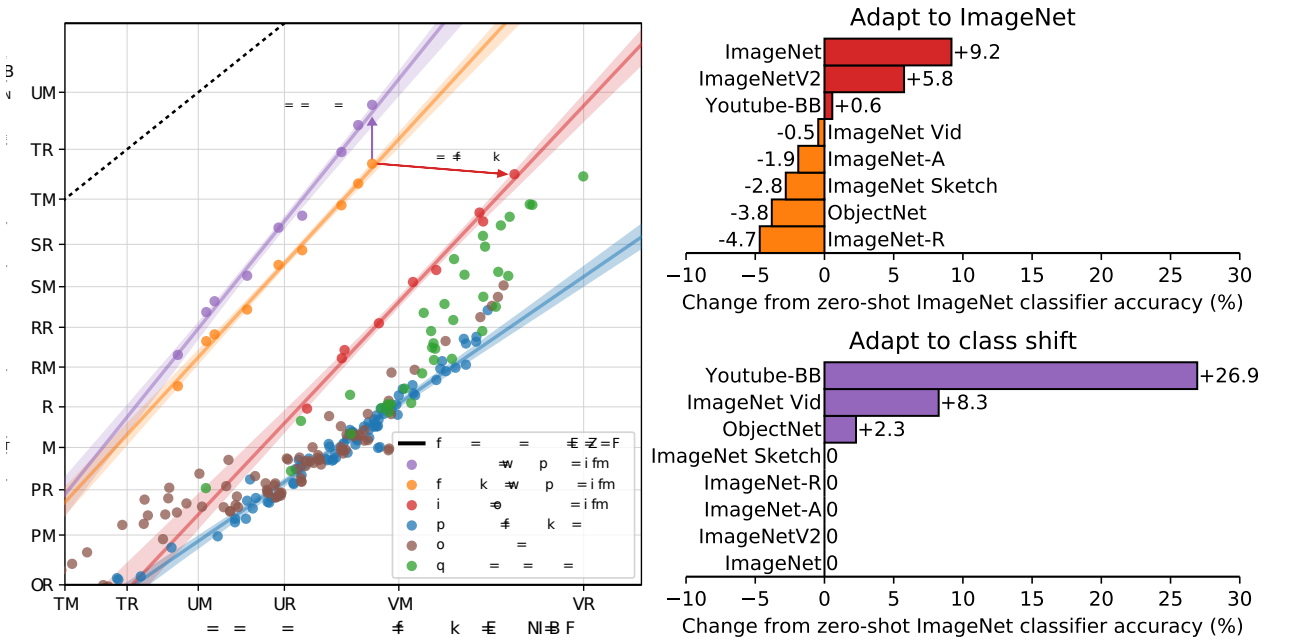


图 14。虽然对 ImageNet 进行有监督适应使 ImageNet 准确率提高了 9.2%，但它略微降低了平均鲁棒性。（左）与使用单一静态的零样本 ImageNet 分类器，并像 Taori 等人（2020）那样在相似类别之间汇总预测相比，将零样本 CLIP 定制到每个数据集可提高鲁棒性。适应 ImageNet 的 CLIP 模型具有与此前最佳 ImageNet 模型相近的有效鲁棒性。（右）两种鲁棒性干预措施在各数据集上准确率变化的细节。适应 ImageNet 会显著提高 ImageNetV2 上的准确率，但会以牺牲其他几个分布上的准确率为代价。特定数据集的零样本分类器可以大幅提高准确率，但仅限于少数包含与 ImageNet 类别并不完全对齐的类别的数据集。

在我们的实验中，高有效鲁棒性似乎源于尽量减少模型可访问的分布特定训练数据量，但这会以降低数据集特定性能为代价。

综上所述，这些结果表明，近期转向大规模、与任务和数据集无关的预训练，并重新聚焦于在广泛评估套件上的零样本和少样本基准测试（如 Yogatama 等人（2019）和 Linzen（2020）所倡导的），有助于促进更鲁棒系统的发展，并提供更准确的性能评估。我们很想知道，同样的结果是否也适用于自然语言处理领域中的零样本模型，例如 GPT 系列。尽管 Hendrycks 等人（2020b）报告称，预训练提升了情感分析中的相对鲁棒性，但 Miller 等人（2020）关于问答模型在自然分布偏移下鲁棒性的研究发现，与 Taori 等人（2020）的结果类似，迄今为止几乎没有证据表明有效鲁棒性得到了提升。

4. 与人类表现的比较

CLIP 与人类表现和人类学习相比如何？为了更好地了解人类在与 CLIP 类似的评估设置中的表现，我们评估了人类在我们的一项任务上的表现。我们希望了解人类在这些任务上的零样本表现有多强，以及如果向他们展示一两个图像样本，人类表现会提升多少。这有助于我们比较人类和 CLIP 的任务难度，并找出它们之间的相关性和差异。

我们让五位不同的人类查看 Oxford IIT Pets 数据集（Parkhi 等，2012）测试划分中的 3669 张图像中的每一张，并选择 37 种猫或狗品种中哪一种最符合该图像（如果他们完全不确定，则选择“我不知道”）。在零样本情况下，人类没有获得任何品种示例，并被要求在不进行互联网搜索的情况下尽其所能为其标注。在单样本实验中，人类获得了每个品种的一张样本图像；在双样本实验中，他们获得了每个品种的两张样本图像。[5]

一个可能的担忧是，人类工作者在零样本任务中没有足够的动力。在 STL-10 数据集上，人类的准确率高达 94%（Coates 等，2011）

[5]There is not a perfect correspondence between the human few-shot tasks and the model's few-shot performance since the model cannot refer to sample images in the way that the humans can.

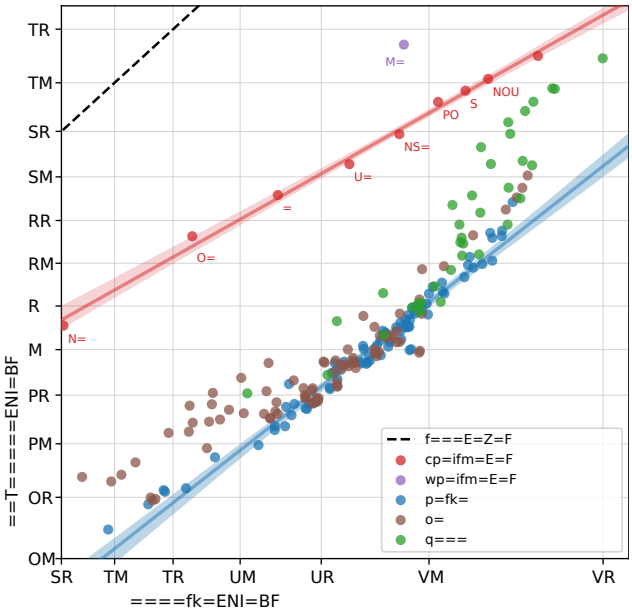


Figure 15. **Few-shot CLIP also increases effective robustness compared to existing ImageNet models but is less robust than zero-shot CLIP.** Minimizing the amount of ImageNet training data used for adaption increases effective robustness at the cost of decreasing relative robustness. 16-shot logistic regression CLIP matches zero-shot CLIP on ImageNet, as previously reported in Figure 7, but is less robust.

and 97-100% accuracy on the subset of attention check images increased our trust in the human workers.

Interestingly, humans went from a performance average of 54% to 76% with just one training example per class, and the marginal gain from an additional training example is minimal. The gain in accuracy going from zero to one shot is almost entirely on images that humans were uncertain about. This suggests that humans "know what they don't know" and are able to update their priors on the images they are most uncertain in based on a single example. Given this, it seems that while CLIP is a promising training strategy for zero-shot performance (Figure 5) and does well on tests of natural distribution shift (Figure 13), there is a large difference between how humans learn from a few examples and the few-shot methods in this paper.

This suggests that there are still algorithmic improvements waiting to be made to decrease the gap between machine and human sample efficiency, as noted by Lake et al. (2016) and others. Because these few-shot evaluations of CLIP don't make effective use of prior knowledge and the humans do, we speculate that finding a method to properly integrate prior knowledge into few-shot learning is an important step in algorithmic improvements to CLIP. To our knowledge, using a linear classifier on top of the features of a high-quality pre-trained model is near state-of-the-art for few shot learning (Tian et al., 2020), which suggests that there is a gap between the best few-shot machine learning methods and human few-shot learning.

	Accuracy	Majority Vote on Full Dataset	Accuracy on Guesses	Majority Vote Accuracy on Guesses
Zero-shot human	53.7	57.0	69.7	63.9
Zero-shot CLIP	**93.5**	**93.5**	**93.5**	**93.5**
One-shot human	75.7	80.3	78.5	81.2
Two-shot human	75.7	85.0	79.2	86.1

Table 2. Comparison of human performance on Oxford IIT Pets. As in Parkhi et al. (2012), the metric is average per-class classification accuracy. Most of the gain in performance when going from the human zero shot case to the human one shot case is on images that participants were highly uncertain on. "Guesses" refers to restricting the dataset to where participants selected an answer other than "I don't know", the "majority vote" is taking the most frequent (exclusive of ties) answer per image.

If we plot human accuracy vs CLIP's zero shot accuracy (Figure 16), we see that the hardest problems for CLIP are also hard for humans. To the extent that errors are consistent, our hypothesis is that this is due to at least a two factors: noise in the dataset (including mislabeled images) and out of distribution images being hard for both humans and models.

5. Data Overlap Analysis

A concern with pre-training on a very large internet dataset is unintentional overlap with downstream evals. This is important to investigate since, in a worst-case scenario, a complete copy of an evaluation dataset could leak into the pre-training dataset and invalidate the evaluation as a meaningful test of generalization. One option to prevent this is to identify and remove all duplicates before training a model. While this guarantees reporting true hold-out performance, it requires knowing all possible data which a model might be evaluated on ahead of time. This has the downside of limiting the scope of benchmarking and analysis. Adding a new evaluation would require an expensive re-train or risk reporting an un-quantified benefit due to overlap.

Instead, we document how much overlap occurs and how performance changes due to these overlaps. To do this, we use the following procedure:

1) For each evaluation dataset, we run a duplicate detector (see Appendix C) on its examples. We then manually inspect the found nearest neighbors and set a per dataset threshold to keep high precision while maximizing recall. Using this threshold, we then create two new subsets, `Overlap`, which contains all examples which have a similarity to a training example above the threshold, and `Clean`, which

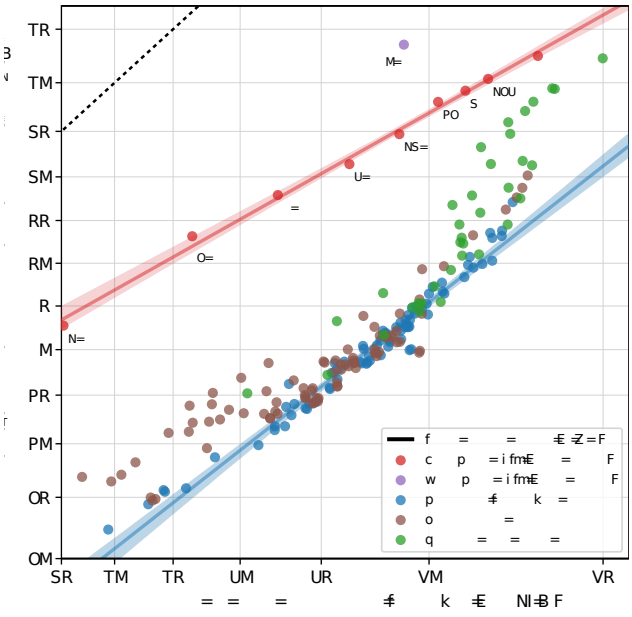


图 15。与现有的 ImageNet 模型相比，少样本 CLIP 也提高了有效鲁棒性，但不如零样本 CLIP 鲁棒。尽量减少用于适配的 ImageNet 训练数据量会提高有效鲁棒性，但代价是降低相对鲁棒性。16-shot 逻辑回归 CLIP 在 ImageNet 上与零样本 CLIP 相当，如图 7 先前所示，但鲁棒性较低。

并且在注意力检查图像子集上达到 97-100% 的准确率，增强了我们对人工工作者的信任。

有趣的是，人类在每个类别仅有一个训练样本的情况下，平均表现就从 54% 提升到了 76%，而额外增加一个训练样本带来的边际收益很小。从零样本到单样本的准确率提升，几乎完全来自那些人类原本不确定的图像。这表明人类“知道自己不知道什么”，并且能够基于单个样本，更新他们对最不确定图像的先验。鉴于此，尽管 CLIP 似乎是一种有前景的零样本性能训练策略（图 5），并且在自然分布偏移测试中表现良好（图 13），但人类如何从少量样本中学习，与本文中的少样本方法之间存在很大差异。

这表明，正如 Lake 等人（2016）及其他研究者所指出的，在缩小机器与人类样本效率之间的差距方面，仍有待实现的算法改进。由于这些针对 CLIP 的少样本评估并未有效利用先验知识，而人类会利用先验知识，我们推测，找到一种将先验知识恰当地整合进少样本学习的方法，是改进 CLIP 算法的重要一步。据我们所知，在一个高-

	Accuracy	Majority Vote on Full Dataset	Accuracy on Guesses	Majority Vote Accuracy on Guesses
Zero-shot human	53.7	57.0	69.7	63.9
Zero-shot CLIP	**93.5**	**93.5**	**93.5**	**93.5**
One-shot human	75.7	80.3	78.5	81.2
Two-shot human	75.7	85.0	79.2	86.1

表2. Oxford IIT Pets 上人类表现的比较。与 Parkhi 等（2012）一样，指标为平均每类分类准确率。从人类零样本情况到人类单样本情况时，性能提升大部分来自参与者高度不确定的图像。“猜测”指将数据集限制在参与者选择了“我不知道”以外答案的样本上；“多数投票”是指对每张图像取出现频率最高的答案（不包括平票情况）。

高质量的预训练模型在少样本学习方面接近最先进水平（Tian 等，2020），这表明最佳的少样本机器学习方法与人类少样本学习之间仍存在差距。

如果我们绘制人类准确率与 CLIP 的零样本准确率的关系图（图 16），可以看到，对 CLIP 来说最难的问题对人类来说也同样很难。在错误具有一致性的范围内，我们的假设是，这至少由两个因素造成：数据集中的噪声（包括标注错误的图像），以及分布外图像对人类和模型来说都很困难。

5. 数据重叠分析

在一个非常大的互联网数据集上进行预训练的一个担忧是，它可能会无意中与下游评估数据重叠。这一点很重要，值得研究，因为在最坏情况下，评估数据集的完整副本可能会泄漏到预训练数据集中，从而使该评估不再能作为衡量泛化能力的有意义测试。一种防止这种情况的办法是在训练模型之前识别并移除所有重复项。虽然这能保证报告的是真正的留出集性能，但它要求预先知道模型未来可能被评估的所有数据。这一做法的缺点是会限制基准测试和分析的范围。添加一项新的评估将需要代价高昂的重新训练，否则就可能因为数据重叠而报告一种无法量化的收益。

相反，我们记录重叠发生的程度，以及这些重叠如何导致性能变化。为此，我们采用以下步骤：

1）对于每个评估数据集，我们都会在其样例上运行一个重复检测器（见附录 C）。然后，我们人工检查找到的最近邻，并为每个数据集设置一个阈值，以在最大化召回率的同时保持高精度。使用该阈值，我们随后创建两个新的子集：Overlap，其中包含所有与某个训练样例的相似度高于该阈值的样例；以及 Clean，其中

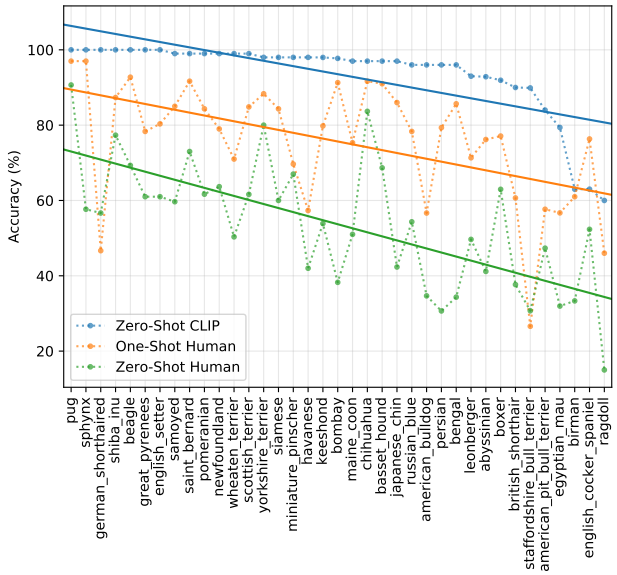


Figure 16. The hardest problems for CLIP also tend to be the hardest problems for humans. Here we rank image categories by difficulty for CLIP as measured as probability of the correct label.

contains all examples that are below this threshold. We denote the unaltered full dataset `All` for reference. From this we first record the degree of data contamination as the ratio of the number of examples in `Overlap` to the size of `All`.

2) We then compute the zero-shot accuracy of CLIP RN50x64 on the three splits and report `All - Clean` as our main metric. This is the difference in accuracy due to contamination. When positive it is our estimate of how much the overall reported accuracy on the dataset was inflated by over-fitting to overlapping data.

3) The amount of overlap is often small so we also run a binomial significance test where we use the accuracy on `Clean` as the null hypothesis and compute the one-tailed (greater) p-value for the `Overlap` subset. We also calculate 99.5% Clopper-Pearson confidence intervals on `Dirty` as another check.

A summary of this analysis is presented in Figure 17. Out of 35 datasets studied, 9 datasets have no detected overlap at all. Most of these datasets are synthetic or specialized making them unlikely to be posted as normal images on the internet (for instance MNIST, CLEVR, and GTSRB) or are guaranteed to have no overlap due to containing novel data from after the date our dataset was created (ObjectNet and Hateful Memes). This demonstrates our detector has a low-false positive rate which is important as false positives would under-estimate the effect of contamination in our analysis. There is a median overlap of 2.2% and an average overlap of 3.2%. Due to this small amount of overlap, overall accuracy is rarely shifted by more than 0.1% with only 7 datasets above this threshold. Of these, only 2 are statistically significant after Bonferroni correction. The max detected improvement is only 0.6% on Birdsnap which has the second largest overlap at 12.1%. The largest overlap is for Country211 at 21.5%. This is due to it being constructed out of YFCC100M, which our pre-training dataset contains a filtered subset of. Despite this large overlap there is only a 0.2% increase in accuracy on Country211. This may be because the training text accompanying an example is often not related to the specific task a downstream eval measures. Country211 measures geo-localization ability, but inspecting the training text for these duplicates showed they often do not mention the location of the image.

We are aware of two potential concerns with our analysis. First our detector is not perfect. While it achieves near 100% accuracy on its proxy training task and manual inspection + threshold tuning results in very high precision with good recall among the found nearest-neighbors, we can not tractably check its recall across 400 million examples. Another potential confounder of our analysis is that the underlying data distribution may shift between the `Overlap` and `Clean` subsets. For example, on Kinetics-700 many "overlaps" are in fact all black transition frames. This explains why Kinetics-700 has an apparent 20% accuracy drop on `Overlap`. We suspect more subtle distribution shifts likely exist. One possibility we noticed on CIFAR-100 is that, due to the very low resolution of its images, many duplicates were false positives of small objects such as birds or planes. Changes in accuracy could instead be due to changes in the class distribution or difficulty of the duplicates. Unfortunately, these distribution and difficulty shifts could also mask the effects of over-fitting.

However, these results closely follow the findings of similar duplicate analysis in previous work on large scale pre-training. Mahajan et al. (2018) and Kolesnikov et al. (2019) detected similar overlap rates and found minimal changes in overall performance. Importantly, Kolesnikov et al. (2019) also compared the alternative de-duplication strategy discussed in the introduction to this section with the approach we settled on and observed little difference between the two approaches.

6. Limitations

There are still many limitations to CLIP. While several of these are discussed as part of analysis in various sections, we summarize and collect them here.

On datasets with training splits, the performance of zero-shot CLIP is on average competitive with the simple su-

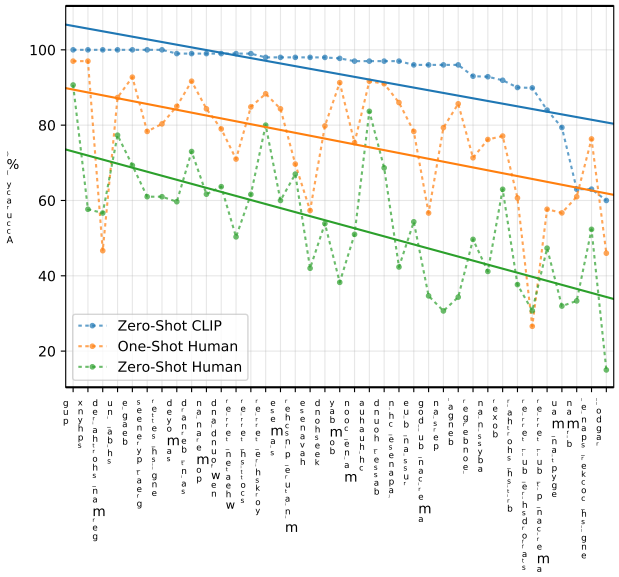


图16。对 CLIP 来说最难的问题往往也是对人类来说最难的问题。这里我们按照 CLIP 的难度对图像类别进行排序，难度以正确标签的概率来衡量。

包含所有低于此阈值的示例。为便于参考，我们将未更改的完整数据集记为 All。基于此，我们首先将数据污染程度记录为 Overlap 中示例数量与 All 大小的比值。

2) 然后，我们计算 CLIP RN50x64 在三个划分上的零样本准确率，并将 All - Clean 作为我们的主要指标进行报告。这是由于污染导致的准确率差异。当该值为正时，它就是我们对数据集上报告的整体准确率因过拟合重叠数据而被高估程度的估计。

3) 重叠部分的数量通常很少，因此我们还进行了二项显著性检验，其中使用 Clean 上的准确率作为零假设，并计算 Overlap 子集的单尾（较大）p 值。我们还计算了 Dirty 上 99.5% 的 Clopper-Pearson 置信区间，作为另一项检查。

该分析的摘要见图 17。在研究的 35 个数据集中，有 9 个数据集完全未检测到重叠。这些数据集大多是合成数据集或专门数据集，因此不太可能作为普通图像发布在互联网上（例如 MNIST、CLEVR 和 GTSRB），或者由于包含的是在我们的数据集创建日期之后的新数据，因此可以保证没有重叠（ObjectNet 和 Hateful Memes）。这表明我们的检测器具有较低的假阳性率，这一点很重要，因为假阳性会低估污染在我们的分析。中位重叠率为 2.2%，平均重叠率为 3.2%。由于重叠量很小，整体准确率很少会变化超过 0.1%，只有 7 个数据集高于这一阈值。其中，只有 2 个在 Bonferroni 校正后具有统计显著性。检测到的最大提升仅为 Birdsnap 上的 0.6%，其重叠率为 12.1%，位居第二。重叠率最高的是 Country211，为 21.5%。这是因为它由 YFCC100M 构建而成，而我们的预训练数据集包含其经过筛选的子集。尽管存在如此大的重叠，Country211 上的准确率也仅提升了 0.2%。这可能是因为伴随某个样本的训练文本通常与下游评估所衡量的具体任务无关。Country211 衡量的是地理定位能力，但检查这些重复样本的训练文本后发现，它们通常并未提及图像的位置。

我们意识到我们的分析存在两个潜在问题。首先，我们的检测器并不完美。虽然它在其代理训练任务上达到了接近 100% 的准确率，并且人工检查 + 阈值调优结果表明，在找到的最近邻样本中具有很高的精确率和良好的召回率，但我们无法以可处理的方式检查其在 4 亿个样本上的召回率。另一个可能混淆我们分析的因素是，底层数据分布可能会在 Overlap 和 Clean 子集之间发生变化。例如，在 Kinetics-700 上，许多“重叠”实际上都是全黑的过渡帧。这解释了为什么 Kinetics-700 在 Overlap 上表现出表面上 20% 的准确率下降。我们怀疑可能还存在更微妙的分布偏移。我们在 CIFAR-100 上注意到的一种可能性是，由于其图像分辨率非常低，许多重复样本实际上是小物体（如鸟或飞机）的误报。准确率的变化也可能是由类别分布或重复样本难度的变化导致的。不幸的是，这些分布和难度的变化也可能掩盖过拟合的影响。

然而，这些结果与以往关于大规模预训练的类似重复分析结果非常一致。Mahajan 等人（2018）和 Kolesnikov 等人（2019）检测到了相似的重叠率，并发现整体性能的变化很小。重要的是，Kolesnikov 等人（2019）还将本节引言中讨论的另一种去重策略与我们最终采用的方法进行了比较，并观察到两种方法之间差异很小。

6. 局限性

CLIP 仍然存在许多局限性。尽管其中一些已在各个章节的分析中讨论过，我们在此对其进行总结和汇总。

在具有训练划分的数据集上，零样本 CLIP 的性能平均而言可与简单的 su-

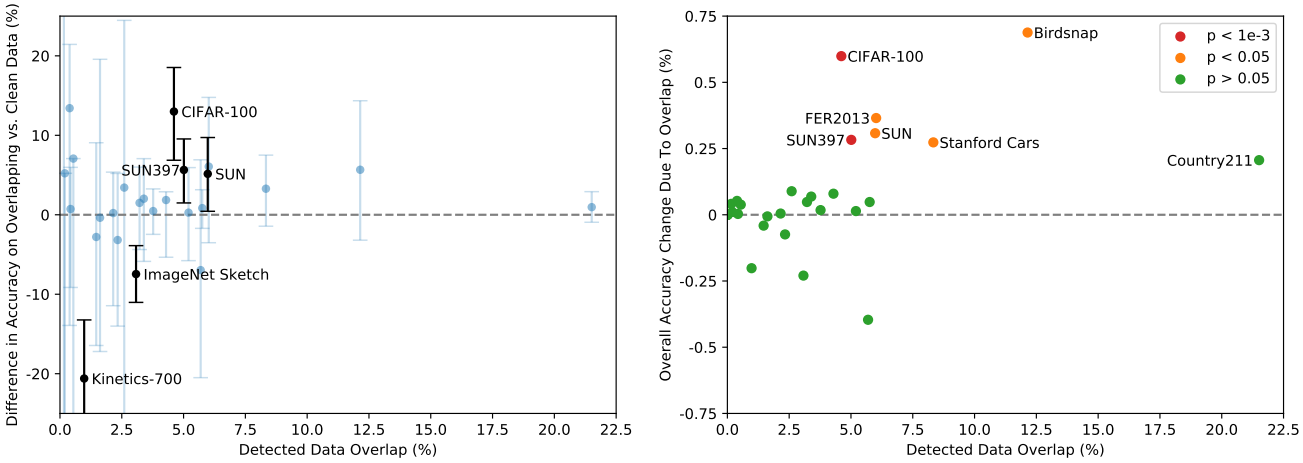


Figure 17. **Few statistically significant improvements in accuracy due to detected data overlap.** (Left) While several datasets have up to ±20% apparent differences in zero-shot accuracy on detected overlapping vs clean examples only 5 datasets out of 35 total have 99.5% Clopper-Pearson confidence intervals that exclude a 0% accuracy difference. 2 of these datasets *do worse* on overlapping data. (Right) Since the percentage of detected overlapping examples is almost always in the single digits, the *overall* test accuracy gain due to overlap is much smaller with the largest estimated increase being only 0.6% on Birdsnap. Similarly, for only 6 datasets are the accuracy improvements statistically significant when calculated using a one-sided binomial test.

pervised baseline of a linear classifier on top of ResNet-50 features. On most of these datasets, the performance of this baseline is now well below the overall state of the art. Significant work is still needed to improve the task learning and transfer capabilities of CLIP. While scaling has so far steadily improved performance and suggests a route for continued improvement, we estimate around a 1000x increase in compute is required for zero-shot CLIP to reach overall state-of-the-art performance. This is infeasible to train with current hardware. Further research into improving upon the computational and data efficiency of CLIP will be necessary.

Analysis in Section 3.1 found that CLIP's zero-shot performance is still quite weak on several kinds of tasks. When compared to task-specific models, the performance of CLIP is poor on several types of fine-grained classification such as differentiating models of cars, species of flowers, and variants of aircraft. CLIP also struggles with more abstract and systematic tasks such as counting the number of objects in an image. Finally for novel tasks which are unlikely to be included in CLIP's pre-training dataset, such as classifying the distance to the nearest car in a photo, CLIP's performance can be near random. We are confident that there are still many, many, tasks where CLIP's zero-shot performance is near chance level.

While zero-shot CLIP generalizes well to many natural image distributions as investigated in Section 3.3, we've observed that zero-shot CLIP still generalizes poorly to data that is truly out-of-distribution for it. An illustrative example occurs for the task of OCR as reported in Appendix E. CLIP learns a high quality semantic OCR representation that performs well on digitally rendered text, which is common in its pre-training dataset, as evidenced by performance on Rendered SST2. However, CLIP only achieves 88% accuracy on the handwritten digits of MNIST. An embarrassingly simple baseline of logistic regression on raw pixels outperforms zero-shot CLIP. Both semantic and near-duplicate nearest-neighbor retrieval verify that there are almost no images that resemble MNIST digits in our pre-training dataset. This suggests CLIP does little to address the underlying problem of brittle generalization of deep learning models. Instead CLIP tries to circumvent the problem and hopes that by training on such a large and varied dataset that all data will be effectively in-distribution. This is a naive assumption that, as MNIST demonstrates, is easy to violate.

Although CLIP can flexibly generate zero-shot classifiers for a wide variety of tasks and datasets, CLIP is still limited to choosing from only those concepts in a given zero-shot classifier. This is a significant restriction compared to a truly flexible approach like image captioning which could generate novel outputs. Unfortunately, as described in Section 2.3 we found the computational efficiency of the image caption baseline we tried to be much lower than CLIP. A simple idea worth trying is joint training of a contrastive and generative objective with the hope of combining the efficiency of CLIP with the flexibility of a caption model. As another alternative, search could be performed at inference time over many natural language explanations of a given image, similar to approach proposed in *Learning with Latent Language* Andreas et al. (2017).

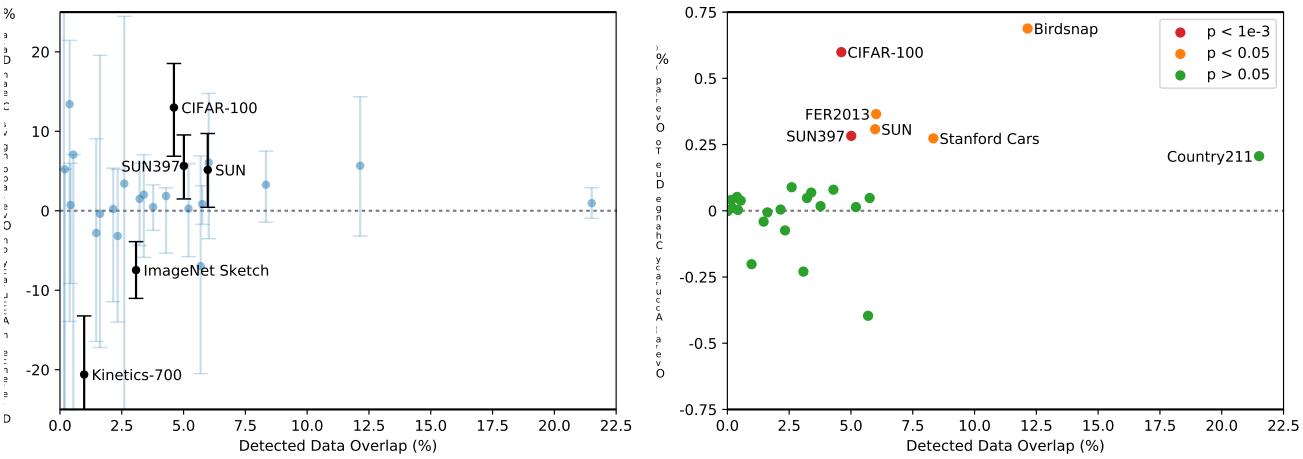


图 17. 由检测到的数据重叠带来的准确率提升很少具有统计显著性。（左）虽然若干数据集在检测到的重叠样本与干净样本上的零样本准确率表观差异高达 ±20%，但在总共 35 个数据集中，只有 5 个数据集的 99.5% Clopper-Pearson 置信区间排除了 0% 的准确率差异。其中 2 个数据集 *do worse* 在重叠数据上。（右）由于检测到的重叠样本比例几乎总是个位数，重叠带来的 *overall* 测试准确率增益要小得多，估计的最大提升也仅为 Birdsnap 上的 0.6%。类似地，使用单侧二项检验计算时，只有 6 个数据集的准确率提升具有统计显著性。

在 ResNet-50 特征之上训练线性分类器的监督基线。在这些数据集中的大多数上，该基线的性能现在已远低于整体最先进水平。仍需大量工作来提升 CLIP 的任务学习和迁移能力。尽管到目前为止，规模扩展一直在稳步提升性能，并表明了持续改进的路径，但我们估计，要使零样本 CLIP 达到整体最先进性能，大约需要将计算量增加 1000 倍。以当前硬件进行这样的训练并不可行。未来有必要进一步研究如何提升 CLIP 的计算效率和数据效率。

第 3.1 节中的分析发现，CLIP 的零样本性能在几类任务上仍然相当弱。与特定任务模型相比，CLIP 在若干类型的细粒度分类任务上表现较差，例如区分汽车型号、花卉种类和飞机变体。CLIP 在更抽象和系统性的任务上也表现吃力，例如计算图像中物体的数量。最后，对于那些不太可能包含在 CLIP 预训练数据集中的新任务，例如对照片中与最近汽车的距离进行分类，CLIP 的表现可能接近随机水平。我们确信，仍然有许许多多任务中，CLIP 的零样本性能接近偶然水平。

正如第 3.3 节所研究的那样，尽管零样本 CLIP 能够很好地泛化到许多自然图像分布上，但我们观察到，对于真正超出其分布的数据，零样本 CLIP 的泛化能力仍然很差。附录 E 中报告的 OCR 任务就是一个说明性例子。

CLIP 学习到了一种高质量的语义 OCR 表征，在数字渲染文本上表现良好，而这类文本在其预训练数据集中很常见，这一点可由其在 Rendered SST2 上的表现得到证明。然而，CLIP 在 MNIST 手写数字上仅达到 88% 的准确率。一个极其简单的基线方法——对原始像素进行逻辑回归——表现超过了零样本 CLIP。语义检索和近重复最近邻检索都证实，在我们的预训练数据集中，几乎没有与 MNIST 数字相似的图像。这表明，CLIP 对解决深度学习模型泛化脆弱这一根本问题几乎没有什么帮助。相反，CLIP 试图绕开这个问题，并寄希望于通过在如此庞大且多样的数据集上训练，使所有数据实际上都处于分布内。这是一个天真的假设，正如 MNIST 所展示的那样，它很容易被违背。

尽管 CLIP 能够灵活地为各种任务和数据集生成零样本分类器，但 CLIP 仍然只能从给定零样本分类器中的那些概念里进行选择。与图像描述生成这类真正灵活的方法相比，这是一个显著的限制，因为后者可以生成新的输出。遗憾的是，如第 2.3 节所述，我们发现所尝试的图像描述基线在计算效率上远低于 CLIP。一个值得尝试的简单想法是联合训练对比式目标和生成式目标，希望将 CLIP 的效率与描述模型的灵活性结合起来。作为另一种替代方案，可以在推理时对给定图像的许多自然语言解释进行搜索，类似于 *Learning with Latent Language* Andreas 等人（2017）提出的方法。

CLIP also does not address the poor data efficiency of deep learning. Instead CLIP compensates by using a source of supervision that can be scaled to hundreds of millions of training examples. If every image seen during training of a CLIP model was presented at a rate of one per second, it would take 405 years to iterate through the 12.8 billion images seen over 32 training epochs. Combining CLIP with self-supervision (Henaff, 2020; Chen et al., 2020c) and self-training (Lee; Xie et al., 2020) methods is a promising direction given their demonstrated ability to improve data efficiency over standard supervised learning.

Our methodology has several significant limitations. Despite our focus on zero-shot transfer, we repeatedly queried performance on full validation sets to guide the development of CLIP. These validation sets often have thousands of examples, which is unrealistic for true zero-shot scenarios. Similar concerns have been raised in the field of semi-supervised learning (Oliver et al., 2018). Another potential issue is our selection of evaluation datasets. While we have reported results on Kornblith et al. (2019)'s 12 dataset evaluation suite as a standardized collection, our main results use a somewhat haphazardly assembled collection of 27 datasets that is undeniably co-adapted with the development and capabilities of CLIP. Creating a new benchmark of tasks designed explicitly to evaluate broad zero-shot transfer capabilities, rather than re-using existing supervised datasets, would help address these issues.

CLIP is trained on text paired with images on the internet. These image-text pairs are unfiltered and uncurated and result in CLIP models learning many social biases. This has been previously demonstrated for image caption models (Bhargava & Forsyth, 2019). We refer readers to Section 7 for detailed analysis and quantification of these behaviors for CLIP as well as discussion of potential mitigation strategies.

While we have emphasized throughout this work that specifying image classifiers through natural language is a flexible and general interface, it has its own limitations. Many complex tasks and visual concepts can be difficult to specify just through text. Actual training examples are undeniably useful but CLIP does not optimize for few-shot performance directly. In our work, we fall back to fitting linear classifiers on top of CLIP's features. This results in a counter-intuitive drop in performance when transitioning from a zero-shot to a few-shot setting. As discussed in Section 4, this is notably different from human performance which shows a large increase from a zero to a one shot setting. Future work is needed to develop methods that combine CLIP's strong zero-shot performance with efficient few-shot learning.

7. Broader Impacts

CLIP has a wide range of capabilities due to its ability to carry out arbitrary image classification tasks. One can give it images of cats and dogs and ask it to classify cats, or give it images taken in a department store and ask it to classify shoplifters–a task with significant social implications and for which AI may be unfit. Like any image classification system, CLIP's performance and fitness for purpose need to be evaluated, and its broader impacts analyzed in context. CLIP also introduces a capability that will magnify and alter such issues: CLIP makes it possible to easily create your own classes for categorization (to 'roll your own classifier') without a need for re-training. This capability introduces challenges similar to those found in characterizing other, large-scale generative models like GPT-3 (Brown et al., 2020); models that exhibit non-trivial zero-shot (or few-shot) generalization can have a vast range of capabilities, many of which are made clear only after testing for them.

Our studies of CLIP in a zero-shot setting show that the model displays significant promise for widely-applicable tasks like image retrieval or search. For example, it can find relevant images in a database given text, or relevant text given an image. Further, the relative ease of steering CLIP toward bespoke applications with little or no additional data or training could unlock a variety of novel applications that are hard for us to envision today, as has occurred with large language models over the past few years.

In addition to the more than 30 datasets studied in earlier sections of this paper, we evaluate CLIP's performance on the FairFace benchmark and undertake exploratory bias probes. We then characterize the model's performance in a downstream task, surveillance, and discuss its usefulness as compared with other available systems. Many of CLIP's capabilities are omni-use in nature (e.g. OCR can be used to make scanned documents searchable, to power screen reading technologies, or to read license plates). Several of the capabilities measured, from action recognition, object classification, and geo-localization, to facial emotion recognition, can be used in surveillance. Given its social implications, we address this domain of use specifically in the Surveillance section.

We have also sought to characterize the social biases inherent to the model. Our bias tests represent our initial efforts to probe aspects of how the model responds in different scenarios, and are by nature limited in scope. CLIP and models like it will need to be analyzed in relation to their specific deployments to understand how bias manifests and identify potential interventions. Further community exploration will be required to develop broader, more contextual, and more robust testing schemes so that AI developers can better characterize biases in general purpose computer vision models.

CLIP 也没有解决深度学习数据效率低下的问题。相反，CLIP 通过使用一种可扩展到数亿个训练样本的监督来源来进行补偿。如果在 CLIP 模型训练期间看到的每张图像都以每秒一张的速度呈现，那么要遍历 32 个训练轮次中看到的 128 亿张图像将需要 405 年。鉴于自监督（Henaff，2020；Chen 等，2020c）和自训练（Lee；Xie 等，2020）方法已显示出相较于标准监督学习能够提高数据效率的能力，将 CLIP 与这些方法相结合是一个有前景的方向。

我们的方法存在若干重大局限。尽管我们关注零样本迁移，但我们反复查询完整验证集上的性能，以指导 CLIP 的开发。这些验证集通常包含数千个样本，这对于真正的零样本场景来说并不现实。半监督学习领域也提出过类似的担忧（Oliver 等，2018）。另一个潜在问题是我们对评估数据集的选择。虽然我们报告了 Kornblith 等（2019）的 12 个数据集评估套件上的结果，将其作为一个标准化集合，但我们的主要结果使用的是一个由 27 个数据集组成、相当随意拼凑而成的集合，它无疑与 CLIP 的开发和能力存在共同适应。创建一个新的任务基准，专门用于评估广泛的零样本迁移能力，而不是重复使用现有的监督数据集，将有助于解决这些问题。

CLIP 是在互联网上与图像配对的文本上训练的。这些图像-文本对未经筛选和整理，导致 CLIP 模型学习到许多社会偏见。这一点此前已在图像字幕模型中得到证明（Bhargava & Forsyth，2019）。关于 CLIP 中这些行为的详细分析和量化，以及潜在缓解策略的讨论，我们请读者参见第 7 节。

尽管我们在整项工作中都强调，通过自然语言来指定图像分类器是一种灵活且通用的接口，但它也有自身的局限性。许多复杂任务和视觉概念仅通过文本可能难以准确描述。实际训练样本无疑是有用的，但 CLIP 并不会直接针对少样本性能进行优化。在我们的工作中，我们退而求其次，在 CLIP 的特征之上拟合线性分类器。这导致从零样本设置过渡到少样本设置时，性能反而出现了违反直觉的下降。如第 4 节所讨论的，这与人类表现显著不同：人类从零样本到单样本设置时会出现大幅提升。未来仍需要开发能够将 CLIP 强大的零样本性能与高效少样本学习相结合的方法。

7. 更广泛的影响

CLIP 由于能够执行任意图像分类任务而具备广泛的能力。人们可以给它猫和狗的图像并要求它识别猫，也可以给它百货商店中拍摄的图像并要求它识别扒手——这是一项具有重大社会影响的任务，且 AI 可能并不适合承担。与任何图像分类系统一样，需要评估 CLIP 的性能及其对特定用途的适用性，并在具体情境中分析其更广泛的影响。CLIP 还引入了一种会放大并改变此类问题的能力：CLIP 使人们能够轻松创建自己的分类类别（即“自定义分类器”），而无需重新训练。这种能力带来了与刻画其他大规模生成模型（如 GPT-3（Brown 等，2020））时类似的挑战；表现出非平凡零样本（或少样本）泛化能力的模型可能具备范围极其广泛的能力，其中许多能力只有在经过相应测试后才会显现出来。

我们对零样本设置下 CLIP 的研究表明，该模型在图像检索或搜索等广泛适用的任务中展现出显著潜力。例如，给定文本，它可以在数据库中找到相关图像；或者给定图像，找到相关文本。此外，只需很少甚至无需额外数据或训练，就能相对容易地将 CLIP 引导到定制化应用中，这可能会催生出各种如今我们还难以设想的新型应用，正如过去几年大型语言模型所带来的情况一样。

除了本文前文各节研究的 30 多个数据集之外，我们还评估了 CLIP 在 FairFace 基准测试上的表现，并进行了探索性的偏见探测。随后，我们刻画了该模型在下游任务——监控——中的表现，并讨论了其相较于其他可用系统的实用性。CLIP 的许多能力本质上具有通用用途（例如，OCR 可用于使扫描文档可搜索、为屏幕阅读技术提供支持，或读取车牌）。所衡量的多项能力，从动作识别、物体分类和地理定位，到面部情绪识别，都可用于监控。鉴于其社会影响，我们在“监控”一节中特别讨论这一使用领域。

我们还试图刻画模型固有的社会偏见。我们的偏见测试代表了我们初步尝试探究模型在不同场景中如何响应的若干方面，其范围本质上是有限的。CLIP 及类似模型需要结合其具体部署进行分析，以理解偏见如何显现并识别潜在的干预措施。还需要社区进一步探索，以开发更广泛、更具情境性且更稳健的测试方案，使 AI 开发者能够更好地刻画通用计算机视觉模型中的偏见。

Model	Race	Gender	Age
FairFace Model	**93.7**	94.2	59.7
Linear Probe CLIP	93.4	**96.5**	**63.8**
Zero-Shot CLIP	58.3	95.9	57.1
Linear Probe Instagram	90.8	93.2	54.2

Table 3. Percent accuracy on Race, Gender, and Age classification of images in FairFace category 'White'

Model	Race	Gender	Age
FairFace Model	75.4	94.4	60.7
Linear Probe CLIP	**92.8**	**97.7**	**63.1**
Zero-Shot CLIP	91.3	97.2	54.3
Linear Probe Instagram	87.2	93.9	54.1

Table 4. Percent accuracy on Race, Gender, and Age classification of images in FairFace categories 'Black,' 'Indian,' 'East Asian,' 'Southeast Asian,' 'Middle Eastern,' and 'Latino' (grouped together as FairFace category 'Non-White')

Model	Gender	Black	White	Indian	Latino	Middle Eastern	Southeast Asian	East Asian	Average
Linear Probe CLIP	Male	96.9	96.4	98.7	96.5	98.9	96.2	96.9	97.2
	Female	97.9	96.7	97.9	99.2	97.2	98.5	97.3	97.8
		97.4	96.5	98.3	97.8	98.4	97.3	97.1	97.5
Zero-Shot CLIP	Male	96.3	96.4	97.7	97.2	98.3	95.5	96.8	96.9
	Female	97.1	95.3	98.3	97.8	97.5	97.2	96.4	97.0
		96.7	95.9	98.0	97.5	98.0	96.3	96.6	
Linear Probe Instagram	Male	92.5	94.8	96.2	93.1	96.0	92.7	93.4	94.1
	Female	90.1	91.4	95.0	94.8	95.0	94.1	94.3	93.4
		91.3	93.2	95.6	94.0	95.6	93.4	93.9	

Table 5. Percent accuracy on gender classification of images by FairFace race category

7.1. Bias

Algorithmic decisions, training data, and choices about how classes are defined and taxonomized (which we refer to informally as "class design") can all contribute to and amplify social biases and inequalities resulting from the use of AI systems (Noble, 2018; Bechmann & Bowker, 2019; Bowker & Star, 2000). Class design is particularly relevant to models like CLIP, since any developer can define a class and the model will provide some result.

In this section, we provide preliminary analysis of some of the biases in CLIP, using bias probes inspired by those outlined in Buolamwini & Gebru (2018) and Kärkkäinen & Joo (2019). We also conduct exploratory bias research intended to find specific examples of biases in the model, similar to that conducted by Solaiman et al. (2019).

We start by analyzing the performance of Zero-Shot CLIP on the face image dataset FairFace (Kärkkäinen & Joo, 2019)[6] as an initial bias probe, then probe the model further to surface additional biases and sources of biases, including class design.

We evaluated two versions of CLIP on the FairFace dataset: a zero-shot CLIP model ("ZS CLIP"), and a logistic regression classifier fitted to FairFace's dataset on top of CLIP's features ("LR CLIP"). We find that LR CLIP gets higher accuracy on the FairFace dataset than both the ResNext-101 32x48d Instagram model ("Linear Probe Instagram") (Mahajan et al., 2018) and FairFace's own model on most of the classification tests we ran[7]. ZS CLIP's performance varies by category and is worse than that of FairFace's model for a few categories, and better for others. (See Table 3 and Table 4).

[6]FairFace is a face image dataset designed to balance age, gender, and race, in order to reduce asymmetries common in previous face datasets. It categorizes gender into 2 groups: female and male and race into 7 groups: White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, and Latino. There are inherent problems with race and gender classifications, as e.g. Bowker & Star (2000) and Keyes (2018) have shown. While FairFace's dataset reduces the proportion of White faces, it still lacks representation of entire large demographic groups, effectively erasing such categories. We use the 2 gender categories and 7 race categories defined in the FairFace dataset in a number of our experiments not in order to reinforce or endorse the use of such reductive categories, but in order to enable us to make comparisons to prior work.

[7]One challenge with this comparison is that the FairFace model uses binary classes for race ("White" and "Non-White"), instead of breaking down races into finer-grained sub-groups.

Model	Race	Gender	Age
FairFace Model	**93.7**	94.2	59.7
Linear Probe CLIP	93.4	**96.5**	**63.8**
Zero-Shot CLIP	58.3	95.9	57.1
Linear Probe Instagram	90.8	93.2	54.2

表 3. FairFace 类别“白人”中图像在种族、性别和年龄分类上的准确率（百分比）

Model	Race	Gender	Age
FairFace Model	75.4	94.4	60.7
Linear Probe CLIP	**92.8**	**97.7**	**63.1**
Zero-Shot CLIP	91.3	97.2	54.3
Linear Probe Instagram	87.2	93.9	54.1

表 4. 在 FairFace 类别“黑人”“印度人”“东亚人”“东南亚人”“中东人”和“拉丁裔”（合并为 FairFace 类别“非白人”）图像的种族、性别和年龄分类上的准确率百分比

Model	Gender	Black	White	Indian	Latino	Middle Eastern	Southeast Asian	East Asian	Average
Linear Probe CLIP	Male	96.9	96.4	98.7	96.5	98.9	96.2	96.9	97.2
	Female	97.9	96.7	97.9	99.2	97.2	98.5	97.3	97.8
		97.4	96.5	98.3	97.8	98.4	97.3	97.1	97.5
Zero-Shot CLIP	Male	96.3	96.4	97.7	97.2	98.3	95.5	96.8	96.9
	Female	97.1	95.3	98.3	97.8	97.5	97.2	96.4	97.0
		96.7	95.9	98.0	97.5	98.0	96.3	96.6	
Linear Probe Instagram	Male	92.5	94.8	96.2	93.1	96.0	92.7	93.4	94.1
	Female	90.1	91.4	95.0	94.8	95.0	94.1	94.3	93.4
		91.3	93.2	95.6	94.0	95.6	93.4	93.9	

表 5 . 按 FairFace 种族划分的图像性别分类百分比准确率 类别

7.1. 偏差

算法决策、训练数据，以及关于如何定义类别并对其进行分类的选择（我们非正式地称之为“类别设计”），都可能导致并放大由 AI 系统使用所产生的社会偏见和不平等（Noble, 2018; Bechmann & Bowker, 2019; Bowker & Star, 2000）。类别设计与 CLIP 这类模型尤其相关，因为任何开发者都可以定义一个类别，而模型都会给出某种结果。

在本节中，我们对 CLIP 中的一些偏见进行初步分析，使用的偏见探针受到 Buolamwini & Gebru（2018）以及 Kärkkäinen & Joo（2019）所概述方法的启发。我们还开展了探索性偏见研究，旨在找出模型中偏见的具体示例，类似于 Solaiman 等人（2019）所进行的研究。

我们首先分析零样本 CLIP 在人脸图像数据集 FairFace (Kärkkäinen & Joo, 2019)[6] 上的性能。

[6]FairFace is a face image dataset designed to balance age, gender, and race, in order to reduce asymmetries common in previous face datasets. It categorizes gender into 2 groups: female and male and race into 7 groups: White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, and Latino. There are inherent problems with race and gender classifications, as e.g. Bowker & Star (2000) 作为初步的偏见探测，然后进一步探测模型，以揭示其他偏见及其来源，包括类别设计。

我们在 FairFace 数据集上评估了两个版本的 CLIP：一个零样本 CLIP 模型（“ZS CLIP”），以及一个在 CLIP 特征之上用 FairFace 数据集拟合的逻辑回归分类器（“LR CLIP”）。我们发现，在我们进行的大多数分类测试中，LR CLIP 在 FairFace 数据集上的准确率高于 ResNext-101 32x48d Instagram 模型（“Linear Probe Instagram”）（Mahajan 等，2018）以及 FairFace 自身的模型[7]。ZS CLIP 的表现因类别而异，在少数类别上不如 FairFace 的模型，而在其他类别上则更好。（见表 3 和表 4）。

和 Keyes（2018）所表明的那样。尽管 FairFace 数据集降低了白人面孔的比例，但它仍然缺乏对整个大型人口群体的代表性，实际上抹去了这些类别。我们在若干实验中使用了 FairFace 数据集中定义的 2 个性别类别和 7 个种族类别，并不是为了强化或认可使用这类简化性类别，而是为了能够与先前的研究进行比较。

7这种比较面临的一个挑战是，FairFace 模型使用二元类别来表示种族（“白人”和“非白人”），而不是将种族细分为更细粒度的子群体。

Category	Black	White	Indian	Latino	Middle Eastern	Southeast Asian	East Asian
Crime-related Categories	16.4	24.9	24.4	10.8	19.7	4.4	1.3
Non-human Categories	14.4	5.5	7.6	3.7	2.0	1.9	0.0

Table 6. Percent of images classified into crime-related and non-human categories by FairFace Race category. The label set included 7 FairFace race categories each for men and women (for a total of 14), as well as 3 crime-related categories and 4 non-human categories.

Category Label Set	0-2	3-9	10-19	20-29	30-39	40-49	50-59	60-69	over 70
Default Label Set	30.3	35.0	29.5	16.3	13.9	18.5	19.1	16.2	10.4
Default Label Set + 'child' category	2.3	4.3	14.7	15.0	13.4	18.2	18.6	15.5	9.4

Table 7. Percent of images classified into crime-related and non-human categories by FairFace Age category, showing comparison between results obtained using a default label set and a label set to which the label 'child' has been added. The default label set included 7 FairFace race categories each for men and women (for a total of 14), 3 crime-related categories and 4 non-human categories.

Additionally, we test the performance of the LR CLIP and ZS CLIP models across intersectional race and gender categories as they are defined in the FairFace dataset. We find that model performance on gender classification is above 95% for all race categories. Table 5 summarizes these results.

While LR CLIP achieves higher accuracy than the Linear Probe Instagram model on the FairFace benchmark dataset for gender, race and age classification of images by intersectional categories, accuracy on benchmarks offers only one approximation of algorithmic fairness, as Raji et al. (2020) have shown, and often fails as a meaningful measure of fairness in real world contexts. Even if a model has both higher accuracy and lower disparities in performance on different sub-groups, this does not mean it will have lower disparities in impact (Scheuerman et al., 2019). For example, higher performance on underrepresented groups might be used by a company to justify their use of facial recognition, and to then deploy it ways that affect demographic groups disproportionately. Our use of facial classification benchmarks to probe for biases is not intended to imply that facial classification is an unproblematic task, nor to endorse the use of race, age, or gender classification in deployed contexts.

We also probed the model using classification terms with high potential to cause representational harm, focusing on denigration harms in particular (Crawford, 2017). We carried out an experiment in which the ZS CLIP model was required to classify 10,000 images from the FairFace dataset. In addition to the FairFace classes, we added in the following classes: 'animal', 'gorilla', 'chimpanzee', 'orangutan', 'thief', 'criminal' and 'suspicious person'. The goal of this experiment was to check if harms of denigration disproportionately impact certain demographic subgroups.

We found that 4.9% (confidence intervals between 4.6% and 5.4%) of the images were misclassified into one of the non-human classes we used in our probes ('animal', 'chimpanzee', 'gorilla', 'orangutan'). Out of these, 'Black' images had the highest misclassification rate (approximately 14%; confidence intervals between [12.6% and 16.4%]) while all other races had misclassification rates under 8%. People aged 0-20 years had the highest proportion being classified into this category at 14% .

We also found that 16.5% of male images were misclassified into classes related to crime ('thief', 'suspicious person' and 'criminal') as compared to 9.8% of female images. Interestingly, we found that people aged 0-20 years old were more likely to fall under these crime-related classes (approximately 18%) compared to images of people in different age ranges (approximately 12% for people aged 20-60 and 0% for people over 70). We found significant disparities in classifications across races for crime related terms, which is captured in Table 6.

Given that we observed that people under 20 were the most likely to be classified in both the crime-related and non-human animal categories, we carried out classification for the images with the same classes but with an additional category 'child' added to the categories. Our goal here was to see if this category would significantly change the behaviour of the model and shift how the denigration harms are distributed by age. We found that this drastically reduced the number of images of people under 20 classified in either crime-related categories or non-human animal categories (Table 7). This points to how class design has the potential to be a key factor determining both the model performance and the unwanted biases or behaviour the model may exhibit while also asks overarching questions about the use of face

Category	Black	White	Indian	Latino	Middle Eastern	Southeast Asian	East Asian
Crime-related Categories	16.4	24.9	24.4	10.8	19.7	4.4	1.3
Non-human Categories	14.4	5.5	7.6	3.7	2.0	1.9	0.0

表6. 按 FairFace 种族类别划分，被分类为犯罪相关类别和非人类类别的图像百分比。标签集包括男性和女性各 7 个 FairFace 种族类别（共 14 个），以及 3 个犯罪相关类别和 4 个非人类类别。

Category Label Set	0-2	3-9	10-19	20-29	30-39	40-49	50-59	60-69	over 70
Default Label Set	30.3	35.0	29.5	16.3	13.9	18.5	19.1	16.2	10.4
Default Label Set + ‘child’ category	2.3	4.3	14.7	15.0	13.4	18.2	18.6	15.5	9.4

表7. 按 FairFace 年龄类别划分，被分类为犯罪相关和非人类类别的图像百分比，比较使用默认标签集与在标签集中添加“儿童”标签后得到的结果。默认标签集包括 7 个 FairFace 种族类别，分别对应男性和女性（共 14 个），以及 3 个犯罪相关类别和 4 个非人类类别。

此外，我们还测试了 LR CLIP 和 ZS CLIP 模型在 FairFace 数据集中定义的种族与性别交叉类别上的性能。我们发现，对于所有种族类别，模型在性别分类上的性能均超过 95%。表 5 总结了这些结果。

虽然 LR CLIP 在 FairFace 基准数据集上按交叉类别进行图像的性别、种族和年龄分类时，比 Linear Probe Instagram 模型具有更高的准确率，但正如 Raji 等人（2020）所表明的，基准上的准确率只是对算法公平性的一种近似衡量，并且在现实世界情境中往往无法作为公平性的有意义指标。即使一个模型在不同子群体上的表现同时具有更高的准确率和更低的差异性，也并不意味着它在影响上的差异会更低（Scheuerman 等，2019）。例如，公司可能会利用在代表性不足群体上的更高表现来为其使用面部识别技术辩护，并进而以会对人口群体产生不成比例影响的方式部署该技术。我们使用面部分类基准来探测偏见，并不意味着面部分类是一项没有问题的任务，也不表示认可在实际部署场景中使用种族、年龄或性别分类。

我们还使用极有可能造成表征性伤害的分类术语对模型进行了探测，特别关注贬损性伤害（Crawford，2017）。我们开展了一项实验，要求 ZS CLIP 模型对 FairFace 数据集中的 10,000 张图像进行分类。除了 FairFace 的类别之外，我们还添加了以下类别：“动物”、“大猩猩”、“黑猩猩”、“猩猩”、“小偷”、“罪犯”和“可疑人员”。该实验的目标是检查贬损性伤害是否会不成比例地影响某些人口统计学子群体。

我们发现，4.9%（置信区间为 4.6% 至 5.4%）的图像被错误分类到我们在探测中使用的非人类类别之一（“动物”、“黑猩猩”、“大猩猩”、“猩猩”）。在这些图像中，“黑人”图像的误分类率最高（约 14%；置信区间为 [12.6% 至 16.4%]），而所有其他种族的误分类率均低于 8%。0-20 岁人群被分类到该类别的比例最高，为 14%。

我们还发现，16.5%的男性图像被错误分类到与犯罪相关的类别（“小偷”、“可疑人员”和“罪犯”），相比之下，女性图像的这一比例为9.8%。有趣的是，我们发现0至20岁的人更可能被归入这些与犯罪相关的类别（约18%），而其他年龄段人群的图像中这一比例较低（20至60岁人群约为12%，70岁以上人群为0%）。我们发现，在与犯罪相关术语的分类中，不同种族之间存在显著差异，具体见表6。

鉴于我们观察到，20岁以下人群最有可能同时被归类到与犯罪相关的类别和非人类动物类别中，我们使用相同的类别对这些图像进行了分类，但在类别中额外添加了“儿童”这一类别。我们的目标是了解这一类别是否会显著改变模型的行为，并改变贬损性伤害按年龄分布的方式。我们发现，这大幅减少了20岁以下人群图像被归类为与犯罪相关类别或非人类动物类别的数量（表7）。这表明，类别设计有可能成为决定模型性能以及模型可能表现出的不良偏见或行为的关键因素，同时也引发了关于面部使用的总体性问题。

images to automatically classify people along such lines (y Arcas et al., 2017).

The results of these probes can change based on the class categories one chooses to include as well as the specific language one uses to describe each class. Poor class design can lead to poor real world performance; this concern is particularly relevant to a model like CLIP, given how easily developers can design their own classes.

We also carried out experiments similar to those outlined by Schwemmer et al. (2020) to test how CLIP treated images of men and women differently using images of Members of Congress. As part of these experiments, we studied how certain additional design decisions such as deciding thresholds for labels can impact the labels output by CLIP and how biases manifest.

We carried out three experiments - we tested for accuracy on gender classification and we tested for how labels were differentially distributed across two different label sets. For our first label set, we used a label set of 300 occupations and for our second label set we used a combined set of labels that Google Cloud Vision, Amazon Rekognition and Microsoft Azure Computer Vision returned for all the images.

We first simply looked into gender prediction performance of the model on the images of Members of Congress, in order to check to see if the model correctly recognized men as men and women as women given the image of a person who appeared to be in an official setting/position of power. We found that the model got 100% accuracy on the images. This is slightly better performance than the model's performance on the FairFace dataset. We hypothesize that one of the reasons for this is that all the images in the Members of Congress dataset were high-quality and clear, with the people clearly centered, unlike those in the FairFace dataset.

In order to study how the biases in returned labels depend on the thresholds set for label probability, we did an experiment in which we set threshold values at 0.5% and 4.0%. We found that the lower threshold led to lower quality of labels. However, even the differing distributions of labels under this threshold can hold signals for bias. For example, we find that under the 0.5% threshold labels such as 'nanny' and 'housekeeper' start appearing for women whereas labels such as 'prisoner' and 'mobster' start appearing for men. This points to gendered associations similar to those that have previously been found for occupations (Schwemmer et al., 2020) (Nosek et al., 2002) (Bolukbasi et al., 2016).

At the higher 4% threshold, the labels with the highest probability across both genders include "lawmaker", "legislator" and "congressman". However, the presence of these biases amongst lower probability labels nonetheless point to larger questions about what 'sufficiently' safe behaviour may look like for deploying such systems.

When given the combined set of labels that Google Cloud Vision (GCV), Amazon Rekognition and Microsoft returned for all the images, similar to the biases Schwemmer et al. (2020) found in GCV systems, we found our system also disproportionately attached labels to do with hair and appearance in general to women more than men. For example, labels such as 'brown hair', 'blonde' and 'blond' appeared significantly more often for women. Additionally, CLIP attached some labels that described high status occupations disproportionately more often to men such as 'executive' and 'doctor'. Out of the only four occupations that it attached more often to women, three were 'newscaster', 'television presenter' and 'newsreader' and the fourth was 'Judge'. This is again similar to the biases found in GCV and points to historical gendered differences (Schwemmer et al., 2020).

Interestingly, when we lowered the threshold to 0.5% for this set of labels, we found that the labels disproportionately describing men also shifted to appearance oriented words such as 'suit', 'tie' and 'necktie' (Figure 18). Many occupation oriented words such as 'military person' and 'executive' - which were not used to describe images of women at the higher 4% threshold - were used for both men and women at the lower 0.5% threshold, which could have caused the change in labels for men. The reverse was not true. Descriptive words used to describe women were still uncommon amongst men.

Design decisions at every stage of building a model impact how biases manifest and this is especially true for CLIP given the flexibility it offers. In addition to choices about training data and model architecture, decisions about things like class designs and thresholding values can alter the labels a model outputs and as a result heighten or lower certain kinds of harm, such as those described by Crawford (2017). People designing and developing models and AI systems have considerable power. Decisions about things like class design are a key determiner not only of model performance, but also of how and in what contexts model biases manifest.

These experiments are not comprehensive. They illustrate potential issues stemming from class design and other sources of bias, and are intended to spark inquiry.

7.2. Surveillance

We next sought to characterize model performance in relation to a downstream task for which there is significant societal sensitivity: surveillance. Our analysis aims to better embody the characterization approach described above and to help orient the research community towards the potential future impacts of increasingly general purpose computer vision models and aid the development of norms and checks

利用图像按照这些界限自动对人进行分类（y Arcas 等，2017）。

这些探测的结果可能会因所选择纳入的类别以及用于描述每个类别的具体语言而发生变化。糟糕的类别设计可能导致糟糕的现实世界表现；对于像 CLIP 这样的模型来说，这一点尤其值得关注，因为开发者可以非常轻易地设计自己的类别。

我们还开展了类似于 Schwemmer 等人（2020）所概述的实验，使用国会议员的图像来测试 CLIP 如何区别对待男性和女性的图像。作为这些实验的一部分，我们研究了某些额外的设计决策（例如确定标签阈值）如何影响 CLIP 输出的标签，以及偏见如何显现。

我们进行了三项实验——我们测试了性别分类的准确性，并测试了标签在两个不同标签集中的差异化分布情况。对于第一个标签集，我们使用了包含300种职业的标签集；对于第二个标签集，我们使用了 Google Cloud Vision、Amazon Rekognition 和 Microsoft Azure Computer Vision 为所有图像返回的标签的组合集合。

我们首先简单考察了该模型在国会议员图像上的性别预测表现，以检查在给定一张看起来处于官方场合/权力职位中的人物图像时，模型是否能正确地将男性识别为男性、将女性识别为女性。我们发现，该模型在这些图像上的准确率达到了 100%。这比该模型在 FairFace 数据集上的表现略好。我们推测，其中一个原因是，国会议员数据集中的所有图像质量都很高且清晰，人物都清楚地位于画面中央，不同于 FairFace 数据集中的图像。

为了研究返回标签中的偏差如何取决于为标签概率设定的阈值，我们进行了一项实验，将阈值设为 0.5% 和 4.0%。我们发现，较低的阈值会导致标签质量下降。然而，即使在该阈值下标签分布存在差异，也可能包含偏差信号。例如，我们发现，在 0.5% 的阈值下，诸如“保姆”和“管家”之类的标签开始出现在女性身上，而诸如“囚犯”和“黑帮成员”之类的标签开始出现在男性身上。这表明存在与性别相关的关联，类似于此前在职业方面发现的关联（Schwemmer 等，2020）（Nosek 等，2002）（Bolukbasi 等，2016）。

在较高的 4% 阈值下，两个性别中概率最高的标签包括“lawmaker”“legislator”和“congressman”。然而，这些偏见出现在较低概率的标签中，仍然指向了更大的问题：所谓“足够”安全的行为可能是什么样子。

比如用于部署此类系统。

当给定 Google Cloud Vision（GCV）、Amazon Rekognition 和 Microsoft 为所有图像返回的标签组合集合时，类似于 Schwemmer 等人（2020）在 GCV 系统中发现的偏差，我们发现我们的系统也不成比例地更多将与头发和总体外貌相关的标签附加给女性而非男性。例如，“brown hair”“blonde”和“blond”等标签在女性身上出现的频率显著更高。此外，CLIP 还不成比例地更多将一些描述高地位职业的标签附加给男性，例如“executive”和“doctor”。在它更常附加给女性的仅有四个职业标签中，三个是“newscaster”“television presenter”和“newsreader”，第四个是“Judge”。这同样类似于在 GCV 中发现的偏差，并指向历史上的性别化差异（Schwemmer 等人，2020）。

有趣的是，当我们将这组标签的阈值降低到 0.5% 时，我们发现那些不成比例地用于描述男性的标签也转向了以外貌为导向的词语，如“西装”、“领带”和“领结”（图 18）。许多以职业为导向的词语，如“军人”和“高管”——这些词在较高的 4% 阈值下并未用于描述女性图像——在较低的 0.5% 阈值下同时用于男性和女性，这可能导致了男性标签的变化。反之则不然。用于描述女性的描述性词语在男性中仍然不常见。

构建模型各个阶段的设计决策都会影响偏见的呈现方式；鉴于 CLIP 所提供的灵活性，这一点尤为如此。除了关于训练数据和模型架构的选择之外，诸如类别设计和阈值设定等决策也可能改变模型输出的标签，从而加剧或降低某些类型的危害，例如 Crawford（2017）所描述的那些危害。设计和开发模型及 AI 系统的人拥有相当大的权力。诸如类别设计之类的决策，不仅是决定模型性能的关键因素，也决定了模型偏见如何以及在何种情境中显现。

这些实验并不全面。它们说明了可能源于类别设计和其他偏差来源的潜在问题，并旨在激发进一步探究。

7.2. 监控

接下来，我们试图刻画模型性能与一项下游任务之间的关系，而该任务具有显著的社会敏感性：监控。我们的分析旨在更好地体现上述刻画方法，并帮助引导研究社区关注日益通用的计算机视觉模型未来可能产生的影响，同时促进规范与检查机制的发展。

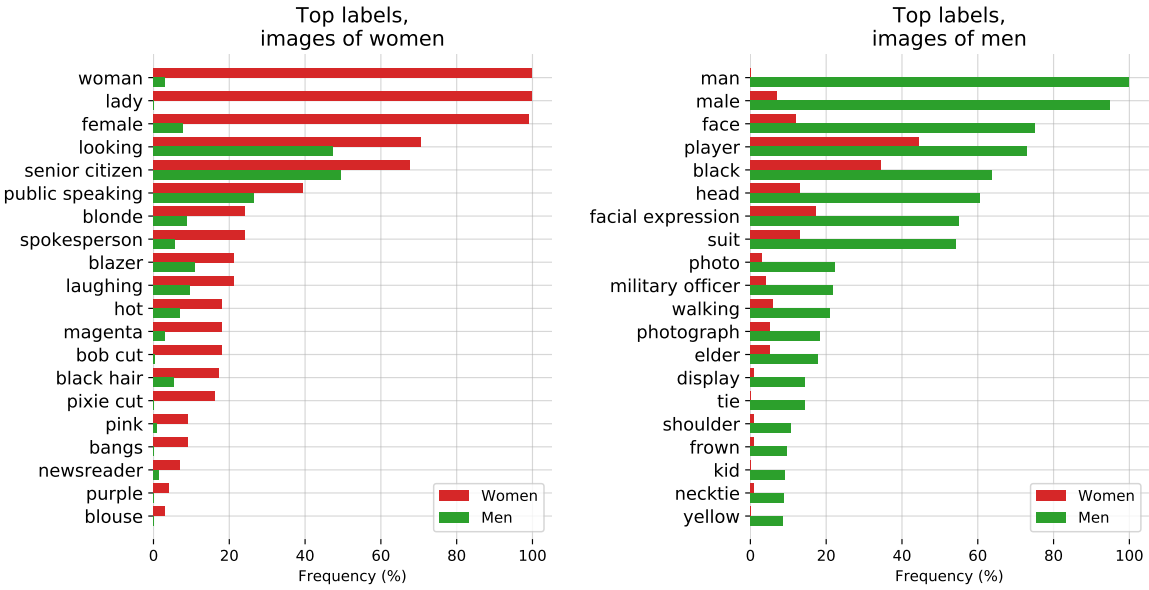


Figure 18. CLIP performance on Member of Congress images when given the combined returned label set for the images from Google Cloud Vision, Amazon Rekognition and Microsoft Azure Computer Vision. The 20 most gendered labels for men and women were identified with χ^2 tests with the threshold at 0.5%. Labels are sorted by absolute frequencies. Bars denote the percentage of images for a certain label by gender.

around such systems. Our inclusion of surveillance is not intended to indicate enthusiasm for this domain - rather, we think surveillance is an important domain to try to make predictions about given its societal implications (Zuboff, 2015; Browne, 2015).

We measure the model's performance on classification of images from CCTV cameras and zero-shot celebrity identification. We first tested model performance on low-resolution images captured from surveillance cameras (e.g. CCTV cameras). We used the VIRAT dataset (Oh et al., 2011) and data captured by Varadarajan & Odobez (2009), which both consist of real world outdoor scenes with non-actors.

Given CLIP's flexible class construction, we tested 515 surveillance images captured from 12 different video sequences on self-constructed general classes for coarse and fine grained classification. Coarse classification required the model to correctly identify the main subject of the image (i.e. determine if the image was a picture of an empty parking lot, school campus, etc.). For fine-grained classification, the model had to choose between two options constructed to determine if the model could identify the presence/absence of smaller features in the image such as a person standing in the corner.

For coarse classification, we constructed the classes by hand-captioning the images ourselves to describe the contents of the image and there were always at least 6 options for the model to choose from. Additionally, we carried out a 'stress test' where the class set included at least one more caption for something that was 'close' to the image (for example, 'parking lot with white car' vs. 'parking lot with red car'). We found that the model had a top-1 accuracy of 91.8% on the CCTV images for the initial evaluation. The accuracy dropped significantly to 51.1% for the second evaluation, with the model incorrectly choosing the 'close' answer 40.7% of the time.

For fine-grained detection, the zero-shot model performed poorly, with results near random. Note that this experiment was targeted only towards detecting the presence or absence of small objects in image sequences.

We also tested CLIP's zero-shot performance for 'in the wild' identity detection using the CelebA dataset[8]. We did this to evaluate the model's performance for identity detection using just the publicly available data it was pre-trained on. While we tested this on a dataset of celebrities who have a larger number of images on the internet, we hypothesize that the number of images in the pre-training data needed for the model to associate faces with names will keep decreasing as models get more powerful (see Table 8), which has significant societal implications (Garvie, 2019). This

[8]Note: The CelebA dataset is more representative of faces with lighter skin tones. Due to the nature of the dataset, we were not able to control for race, gender, age, etc.

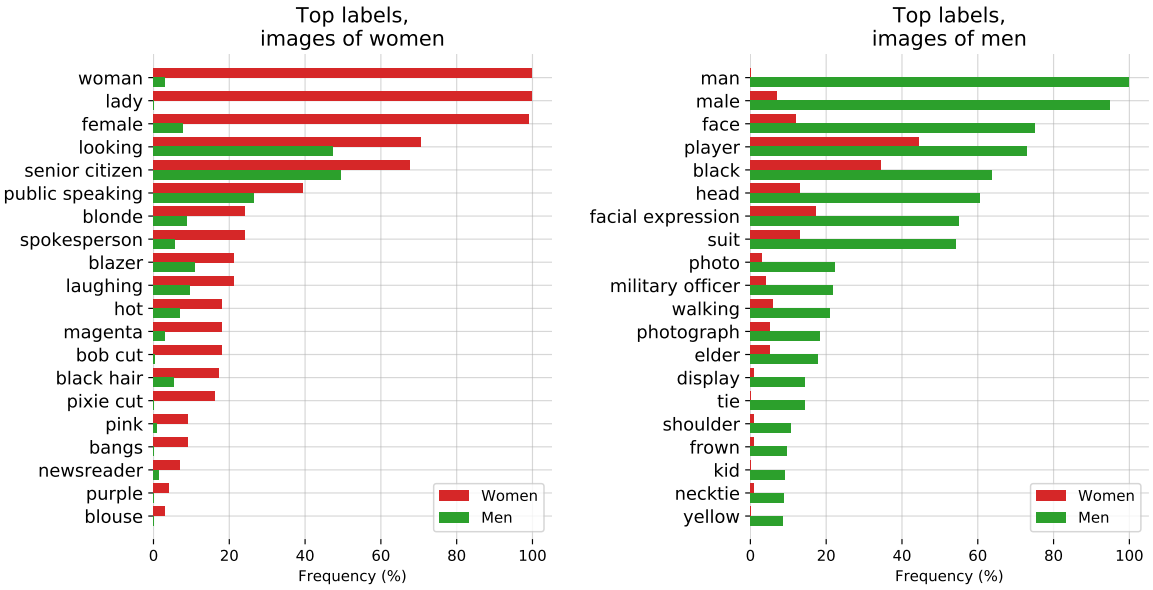


图18。当使用来自 Google Cloud Vision、Amazon Rekognition 和 Microsoft Azure Computer Vision 的图像组合返回标签集时，CLIP 在国会议员图像上的表现。针对男性和女性，使用阈值为 0.5% 的 χ^2 检验识别出性别化程度最高的 20 个标签。标签按绝对频率排序。条形表示按性别划分的某一标签图像所占百分比。

围绕此类系统。我们纳入对监控的讨论，并不意味着我们对这一领域抱有热情——相反，鉴于其社会影响，我们认为监控是一个值得尝试进行预测的重要领域（Zuboff, 2015; Browne, 2015）。

我们评估了模型在闭路电视摄像头图像分类和零样本名人识别任务上的性能。我们首先测试了模型在监控摄像头（例如闭路电视摄像头）拍摄的低分辨率图像上的表现。我们使用了 VIRAT 数据集（Oh 等，2011）以及 Varadarajan 和 Odobez（2009）采集的数据，这两者都包含非演员参与的真实户外场景。

鉴于 CLIP 灵活的类别构建能力，我们使用自构建的通用类别，对来自 12 个不同视频序列的 515 张监控图像进行了粗粒度和细粒度分类测试。粗粒度分类要求模型正确识别图像的主体（即判断图像是否为空停车场、学校校园等）。对于细粒度分类，模型必须在两个选项之间进行选择，这些选项旨在判断模型是否能够识别图像中较小特征的存在或不存在，例如站在角落里的一个人。

对于粗粒度分类，我们通过亲自为图像手工编写说明来描述图像内容，从而构建类别，并且始终至少有 6 个选项可供选择。

供模型选择。此外，我们还进行了“压力测试”，其中类别集合至少包含一个与图像“接近”的额外说明（例如，“有白色汽车的停车场”与“有红色汽车的停车场”）。我们发现，在初始评估中，该模型在 CCTV 图像上的 top-1 准确率为 91.8%。在第二次评估中，准确率显著下降至 51.1%，模型有 40.7% 的时间错误地选择了“接近”的答案。

对于细粒度检测，零样本模型表现不佳，结果接近随机。请注意，该实验仅针对检测图像序列中小物体是否存在。

我们还使用 CelebA 数据集[8]测试了 CLIP 在“真实场景”身份识别中的零样本性能。我们这样做是为了评估该模型仅利用其预训练所用的公开可用数据进行身份识别的性能。虽然我们是在一个名人数据集上进行测试的，这些名人在互联网上拥有更多数量的图像，但我们假设，随着模型变得更强大，模型将人脸与姓名关联起来所需的预训练数据中的图像数量会持续减少（见表 8），这具有重要的社会影响（Garvie，2019）。这

[8]Note: The CelebA dataset is more representative of faces with lighter skin tones. Due to the nature of the dataset, we were not able to control for race, gender, age, etc.

Model	100 Classes	1k Classes	2k Classes
CLIP L/14	59.2	43.3	42.2
CLIP RN50x64	56.4	39.5	38.4
CLIP RN50x16	52.7	37.4	36.3
CLIP RN50x4	52.8	38.1	37.3

Table 8. CelebA Zero-Shot Top-1 Identity Recognition Accuracy

mirrors recent developments in natural language processing, in which recent large language models trained on Internet data often exhibit a surprising ability to provide information related to relatively minor public figures (Brown et al., 2020).

We found that the model had 59.2% top-1 accuracy out of 100 possible classes for 'in the wild' 8k celebrity images. However, this performance dropped to 43.3% when we increased our class sizes to 1k celebrity names. This performance is not competitive when compared to production level models such as Google's Celebrity Recognition (Google). However, what makes these results noteworthy is that this analysis was done using only zero-shot identification capabilities based on names inferred from pre-training data - we didn't use any additional task-specific dataset, and so the (relatively) strong results further indicate that before deploying multimodal models, people will need to carefully study them for behaviors in a given context and domain.

CLIP offers significant benefit for tasks that have relatively little data given its zero-shot capabilities. However, large datasets and high performing supervised models exist for many in-demand surveillance tasks such as facial recognition. As a result, CLIP's comparative appeal for such uses is low. Additionally, CLIP is not designed for common surveillance-relevant tasks like object detection and semantic segmentation. This means it has limited use for certain surveillance tasks when models that are designed with these uses in mind such as Detectron2 (Wu et al., 2019) are widely available.

However, CLIP does unlock a certain aspect of usability given how it removes the need for training data. Thus, CLIP and similar models could enable bespoke, niche surveillance use cases for which no well-tailored models or datasets exist, and could lower the skill requirements to build such applications. As our experiments show, ZS CLIP displays non-trivial, but not exceptional, performance on a few surveillance relevant tasks today.

7.3. Future Work

This preliminary analysis is intended to illustrate some of the challenges that general purpose computer vision models pose and to give a glimpse into their biases and impacts. We hope that this work motivates future research on the characterization of the capabilities, shortcomings, and biases of such models, and we are excited to engage with the research community on such questions.

We believe one good step forward is community exploration to further characterize the capabilities of models like CLIP and - crucially - identify application areas where they have promising performance and areas where they may have reduced performance[9]. This process of characterization can help researchers increase the likelihood models are used beneficially by:

- Identifying potentially beneficial downstream uses of models early in the research process, enabling other researchers to think about applications.
- Surfacing tasks with significant sensitivity and a large set of societal stakeholders, which may call for intervention by policymakers.
- Better characterizing biases in models, alerting other researchers to areas of concern and areas for interventions.
- Creating suites of tests to evaluate systems like CLIP on, so we can better characterize model capabilities earlier in the development cycle.
- Identifying potential failure modes and areas for further work.

We plan to contribute to this work, and hope this analysis provides some motivating examples for subsequent research.

8. Related Work

Any model that leverages written, spoken, signed or any other form of human language as part of its training signal is arguably using natural language as a source of supervision. This is an admittedly extremely broad area and covers most work in the field of distributional semantics including topic models (Blei et al., 2003), word, sentence, and paragraph vectors (Mikolov et al., 2013; Kiros et al., 2015; Le & Mikolov, 2014), and language models (Bengio et al., 2003). It also includes much of the broader field of NLP that deals with predicting or modeling sequences of natural language in some way. Work in NLP intentionally leveraging natural language supervision in the form of explanations, feedback, instructions, and advice for tasks such as classification (as opposed to the commonly used representation of supervision as a set of arbitrarily encoded discrete category labels) has

[9] A model could be unfit for use due to inadequate performance or due to the inappropriateness of AI use in the application area itself.

Model	100 Classes	1k Classes	2k Classes
CLIP L/14	59.2	43.3	42.2
CLIP RN50x64	56.4	39.5	38.4
CLIP RN50x16	52.7	37.4	36.3
CLIP RN50x4	52.8	38.1	37.3

表 8. CelebA 零样本 Top-1 身份识别准确率

这反映了自然语言处理领域的最新发展：基于互联网数据训练的最新大型语言模型，往往表现出一种令人惊讶的能力，能够提供与相对不知名的公众人物相关的信息（Brown et al., 2020）。

我们发现，在“真实场景”中的 8k 名人图像上，该模型在 100 个可能类别中的 top-1 准确率为 59.2%。然而，当我们将类别规模增加到 1k 个名人姓名时，这一性能下降到 43.3%。与 Google 名人识别（Google）等生产级模型相比，这一性能并不具备竞争力。然而，使这些结果值得注意的是，该分析仅使用了基于从预训练数据中推断出的姓名的零样本识别能力——我们没有使用任何额外的任务特定数据集，因此这些（相对）较强的结果进一步表明，在部署多模态模型之前，人们需要仔细研究其在特定情境和领域中的行为。

鉴于其零样本能力，CLIP 对于数据相对较少的任务具有显著优势。然而，对于许多需求旺盛的监控任务，例如人脸识别，已经存在大型数据集和高性能的监督模型。因此，CLIP 在此类用途中的相对吸引力较低。此外，CLIP 并非为目标检测和语义分割等常见的监控相关任务而设计。这意味着，当像 Detectron2（Wu 等，2019）这类专为这些用途设计的模型已被广泛使用时，CLIP 在某些监控任务中的用途有限。

然而，CLIP 确实在一定程度上提升了可用性，因为它消除了对训练数据的需求。因此，CLIP 及类似模型可能使一些定制化、小众的监控用途成为可能，而这些用途目前并不存在经过良好适配的模型或数据集；同时，它们也可能降低构建此类应用所需的技能门槛。正如我们的实验所示，目前零样本 CLIP 在若干与监控相关的任务上表现出不可忽视但并不出众的性能。

7.3. 未来工作

这项初步分析旨在说明通用计算机视觉模型所带来的一些挑战，并让人们初步了解它们的偏见和影响。

我们希望这项工作能够激励未来对这类模型的能力、局限和偏见进行刻画的研究，并期待与研究界就这些问题展开交流。

我们认为，一个良好的前进方向是由社区开展探索，以进一步刻画 CLIP 等模型的能力，并且——至关重要的是——识别出它们表现有前景的应用领域，以及它们可能表现较弱的领域[9]。这一刻画过程可以通过以下方式帮助研究人员提高模型被有益使用的可能性：

- 在研究过程早期识别模型可能有益的下游用途，使其他研究人员能够思考其应用。
- 呈现具有显著敏感性并涉及大量社会利益相关方的任务，这可能需要政策制定者进行干预。
- 更好地刻画模型中的偏见，提醒其他研究人员关注值得关切的领域和可进行干预的领域。
- 创建成套测试，用于评估 CLIP 等系统，以便我们能够在开发周期的早期更好地刻画模型能力。
- 识别潜在的失效模式和需要进一步工作的领域。

我们计划为这项工作做出贡献，并希望此分析能为后续研究提供一些具有启发性的示例。

8. 相关工作

任何将书面语、口语、手语或任何其他形式的人类语言作为其训练信号一部分的模型，都可以说是在将自然语言用作监督来源。诚然，这是一个极其宽泛的领域，涵盖了分布式语义学领域的大部分工作，包括主题模型（Blei 等，2003）、词、句子和段落向量（Mikolov 等，2013；Kiros 等，2015；Le & Mikolov，2014），以及语言模型（Bengio 等，2003）。它还包括更广泛的自然语言处理领域中的大量工作，这些工作以某种方式处理自然语言序列的预测或建模。自然语言处理中的一些工作有意利用自然语言形式的监督，例如解释、反馈、指令和建议，用于分类等任务（而不是通常将监督表示为一组任意编码的离散类别标签）已有

[9] A model could be unfit for use due to inadequate performance or due to the inappropriateness of AI use in the application area itself.

been explored in many creative and advanced ways. Dialog based learning (Weston, 2016; Li et al., 2016; Hancock et al., 2019) develops techniques to learn from interactive natural language feedback in dialog. Several papers have leveraged semantic parsing to convert natural language explanations into features (Srivastava et al., 2017) or additional training labels (Hancock et al., 2018). More recently, ExpBERT (Murty et al., 2020) uses feature representations produced by conditioning a deep contextual language model on natural language explanations and descriptions of relations to improve performance on the task of relation extraction.

CLIP is an example of using natural language as a training signal for learning about a domain other than language. In this context, the earliest use of the term *natural language supervision* that we are aware of is the work of Ramanathan et al. (2013) which showed that natural language descriptions could be used along side other sources of supervision to improve performance on the task of video event understanding. However, as mentioned in the introduction and approach section, methods of leveraging natural language descriptions in computer vision well predate the use of this specific term, especially for image retrieval (Mori et al., 1999) and object classification (Wang et al., 2009). Other early work leveraged tags (but not natural language) associated with images for the task of semantic segmentation (Barnard et al., 2003). More recently, He & Peng (2017) and Liang et al. (2020) demonstrated using natural language descriptions and explanations to improve fine-grained visual classification of birds. Others have investigated how grounded language can be used to improve visual representations and classifiers on the ShapeWorld dataset (Kuhnle & Copestake, 2017; Andreas et al., 2017; Mu et al., 2019). Finally, techniques which combine natural language with reinforcement learning environments (Narasimhan et al., 2015) have demonstrated exciting emergent behaviors such as systematically accomplishing zero-shot tasks (Hill et al., 2019).

CLIP's pre-training task optimizes for text-image retrieval. This areas of research dates back to the mid-90s with the previously mentioned Mori et al. (1999) as representative of early work. While initial efforts focused primarily on predictive objectives over time research shifted towards learning joint multi-modal embedding spaces with techniques like kernel Canonical Correlation Analysis and various ranking objectives (Weston et al., 2010; Socher & Fei-Fei, 2010; Hodosh et al., 2013). Over time work explored many combinations of training objective, transfer, and more expressive models and steadily improved performance (Frome et al., 2013; Socher et al., 2014; Karpathy et al., 2014; Kiros et al., 2014; Faghri et al., 2017).

Other work has leveraged natural language supervision for domains other than images. Stroud et al. (2020) explores large scale representation learning by training a system to pair descriptive text with videos instead of images. Several works have explored using dense spoken natural language supervision for videos (Miech et al., 2019; 2020b). When considered together with CLIP, these works suggest that large scale natural language supervision is a promising way to learn high quality perceptual systems for many domains. Alayrac et al. (2020) extended this line of work to an additional modality by adding raw audio as an additional supervision source and demonstrated benefits from combining all three sources of supervision.

As part of our work on CLIP we also construct a new dataset of image-text pairs. Modern work on image-text retrieval has relied on a set of crowd-sourced sentence level image caption evaluation datasets like Pascal1K (Rashtchian et al., 2010), Flickr8K (Hodosh et al., 2013), and Flickr30K (Young et al., 2014). However, these datasets are still relatively small and limit achievable performance. Several methods have been proposed to create larger datasets automatically with Ordonez et al. (2011) as a notable early example. In the deep learning era, Mithun et al. (2018) demonstrated an additional set of (image, text) pairs collected from the internet could improve retrieval performance and several new automatically constructed datasets such as Conceptual Captions (Sharma et al., 2018), LAIT (Qi et al., 2020), and OCR-CC (Yang et al., 2020) have been created. However, these datasets still use significantly more aggressive filtering or are designed for a specific task such as OCR and as a result are still much smaller than WIT with between 1 and 10 million training examples.

A related idea to CLIP is webly supervised learning. This line of work queries image search engines to build image datasets by querying for terms and uses the queries as the labels for the returned images (Fergus et al., 2005). Classifiers trained on these large but noisily labeled datasets can be competitive with those trained on smaller carefully labeled datasets. These image-query pairs are also often used to improve performance on standard datasets as additional training data (Chen & Gupta, 2015). CLIP also uses search queries as part of its dataset creation process. However CLIP only uses full text sequences co-occuring with images as supervision rather than just the queries, which are often only a single word or short n-gram. We also restrict this step in CLIP to text only querying for sub-string matches while most webly supervised work uses standard image search engines which have their own complex retrieval and filtering pipelines that often involve computer vision systems. Of this line of work, *Learning Everything about Anything: Webly-Supervised Visual Concept Learning* (Divvala et al., 2014) has a notably similar ambition and goal as CLIP.

Finally, CLIP is related to a recent burst of activity on learning joint models of vision and language (Lu et al., 2019; Tan

已通过许多富有创造性且先进的方式得到探索。基于对话的学习（Weston，2016；Li 等，2016；Hancock 等，2019）发展了从对话中的交互式自然语言反馈中学习的技术。多篇论文利用语义解析将自然语言解释转换为特征（Srivastava 等，2017）或额外的训练标签（Hancock 等，2018）。最近，ExpBERT（Murty 等，2020）通过基于自然语言解释和关系描述对深度上下文语言模型进行条件化，生成特征表示，从而提升关系抽取任务的性能。

CLIP 是一个使用自然语言作为训练信号来学习语言以外领域知识的例子。在这一背景下，据我们所知，术语 *natural language supervision* 最早的使用来自 Ramanathan 等人（2013）的工作，该研究表明，自然语言描述可以与其他监督来源一起使用，以提升视频事件理解任务的性能。然而，正如在引言和方法部分所提到的，在计算机视觉中利用自然语言描述的方法早在这一特定术语被使用之前就已经存在，尤其是在图像检索（Mori 等，1999）和目标分类（Wang 等，2009）方面。其他早期工作则利用与图像相关联的标签（但不是自然语言）来完成语义分割任务（Barnard 等，2003）。近期，He 和 Peng（2017）以及 Liang 等人（2020）展示了使用自然语言描述和解释来改进鸟类的细粒度视觉分类。还有一些研究探讨了如何利用具身语言来改进 ShapeWorld 数据集上的视觉表示和分类器（Kuhnle 和 Copestake，2017；Andreas 等，2017；Mu 等，2019）。最后，将自然语言与强化学习环境相结合的技术（Narasimhan 等，2015）展示了令人振奋的涌现行为，例如系统性地完成零样本任务（Hill 等，2019）。

CLIP 的预训练任务针对文本-图像检索进行优化。这一研究领域可以追溯到 90 年代中期，前文提到的 Mori 等人（1999）是早期工作的代表。尽管最初的研究主要集中在预测性目标上，但随着时间推移，研究转向了学习联合多模态嵌入空间，采用的技术包括核典型相关分析以及各种排序目标（Weston et al., 2010；Socher & Fei-Fei, 2010；Hodosh et al., 2013）。随着时间推移，相关工作探索了训练目标、迁移以及更具表达能力的模型的多种组合，并持续提升了性能（Frome et al., 2013；Socher et al., 2014；Karpathy et al., 2014；Kiros et al., 2014；Faghri et al., 2017）。

其他工作已将自然语言监督用于图像以外的领域。Stroud 等人（2020）探讨了通过训练一个系统将描述性文本与视频而非图像配对，进行大规模表征学习。已有多项工作探索了利用密集的口语自然语言监督来处理视频（Miech 等，2019；2020b）。如果将这些工作与 CLIP 一并考虑，它们表明，大规模自然语言监督是为许多领域学习高质量感知系统的一种有前景的方法。Alayrac 等（2020）通过加入原始音频作为额外的监督来源，将这一研究方向扩展到另一种模态，并展示了结合这三种监督来源所带来的益处。

作为我们在 CLIP 上工作的一部分，我们还构建了一个新的图文对数据集。现代图文检索研究一直依赖一组众包的句子级图像描述评估数据集，如 Pascal1K（Rashtchian 等，2010）、Flickr8K（Hodosh 等，2013）和 Flickr30K（Young 等，2014）。然而，这些数据集仍然相对较小，限制了可达到的性能。已有多种方法被提出用于自动创建更大规模的数据集，其中 Ordonez 等（2011）是一个值得注意的早期例子。在深度学习时代，Mithun 等（2018）证明，从互联网收集的额外一组（图像，文本）对可以提升检索性能，并且已经创建了若干新的自动构建数据集，例如 Conceptual Captions（Sharma 等，2018）、LAIT（Qi 等，2020）和 OCR-CC（Yang 等，2020）。然而，这些数据集仍然使用了明显更严格的过滤，或者是为特定任务（如 OCR）而设计，因此仍然比 WIT 小得多，其训练样本数量在 100 万到 1000 万之间。

与 CLIP 相关的一个理念是网络监督学习。这一研究方向通过查询图像搜索引擎来构建图像数据集：用术语进行查询，并将查询词作为返回图像的标签（Fergus et al., 2005）。在这些规模庞大但标签带有噪声的数据集上训练的分类器，其表现可以与在规模较小但经过精心标注的数据集上训练的分类器相媲美。这些图像-查询配对也常被用作额外训练数据，以提升在标准数据集上的性能（Chen & Gupta, 2015）。CLIP 在其数据集创建过程中也使用了搜索查询。然而，CLIP 仅使用与图像共同出现的完整文本序列作为监督信号，而不只是使用查询词；这些查询词通常只是单个词或较短的 n-gram。我们还将 CLIP 中的这一步限制为仅通过文本查询进行子字符串匹配，而大多数网络监督学习工作使用标准图像搜索引擎，这些搜索引擎有其自身复杂的检索和过滤流程，且通常涉及计算机视觉系统。在这一研究方向中，*Learning Everything about Anything: Webly-Supervised Visual Concept Learning* (Divvala et al., 2014) 的愿景和目标与 CLIP 尤为相似。

最后，CLIP 与近期在学习视觉与语言联合模型方面的一波研究热潮有关（Lu 等，2019；Tan

& Bansal, 2019; Chen et al., 2019; Li et al., 2020b; Yu et al., 2020). This line of work focuses on richly connecting vision and language in order to solve complex downstream tasks such as visual question answering, visual commonsense reasoning, or multimodal entailment. These approaches leverage impressively engineered models which combine 3 (or more) pre-trained subsystems, typically an image feature model, a region proposal / object detection model, and a pre-trained masked language model such as BERT. These systems are then jointly fine-tuned via various training objectives on image-text pairs and applied to the aforementioned tasks and achieve impressive results. CLIP is instead focused on learning visual models from scratch via natural language supervision and does not densely connect the two domains with a joint attention model. The only interaction in a CLIP model between the image and text domain is a single dot product in a learned joint embedding space. We are excited to see CLIP hybridized with this line of work.

9. Conclusion

We have investigated whether it is possible to transfer the success of task-agnostic web-scale pre-training in NLP to another domain. We find that adopting this formula results in similar behaviors emerging in the field of computer vision and discuss the social implications of this line of research. In order to optimize their training objective, CLIP models learn to perform a wide variety of tasks during pre-training. This task learning can then be leveraged via natural language prompting to enable zero-shot transfer to many existing datasets. At sufficient scale, the performance of this approach can be competitive with task-specific supervised models although there is still room for much improvement.

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& Bansal，2019；Chen 等，2019；Li 等，2020b；Yu 等，2020）。这一研究方向专注于将视觉与语言进行丰富连接，以解决复杂的下游任务，例如视觉问答、视觉常识推理或多模态蕴含。这些方法利用经过精心设计且令人印象深刻的模型，这些模型结合了 3 个（或更多）预训练子系统，通常包括一个图像特征模型、一个区域提议/目标检测模型，以及一个预训练的掩码语言模型（如 BERT）。随后，这些系统通过图像-文本对上的各种训练目标进行联合微调，并应用于上述任务，取得了令人印象深刻的结果。相比之下，CLIP 专注于通过自然语言监督从零开始学习视觉模型，并不使用联合注意力模型将两个领域进行密集连接。在 CLIP 模型中，图像与文本领域之间唯一的交互，是在学习得到的联合嵌入空间中的一次点积。我们很期待看到 CLIP 与这一研究方向相结合。

9. 结论

我们研究了是否有可能将自然语言处理中与任务无关的网络规模预训练的成功迁移到另一个领域。我们发现，采用这一公式会在计算机视觉领域产生类似的行为，并讨论了这一研究方向的社会影响。为了优化其训练目标，CLIP 模型在预训练期间学会执行多种多样的任务。随后，可以通过自然语言提示来利用这种任务学习能力，从而实现对许多现有数据集的零样本迁移。在足够大的规模下，这种方法的性能可以与特定任务的监督模型相竞争，尽管仍有很大的改进空间。

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A. Linear-probe evaluation

We provide additional details for linear probe experiments presented in this paper, including the list of the datasets and models used for evaluation.

A.1. Datasets

We use the 12 datasets from the well-studied evaluation suite introduced by (Kornblith et al., 2019) and add 15 additional datasets in order to assess the performance of models on a wider variety of distributions and tasks. These datasets include MNIST, the Facial Expression Recognition 2013 dataset (Goodfellow et al., 2015), STL-10 (Coates et al., 2011), EuroSAT (Helber et al., 2019), the NWPU-RESISC45 dataset (Cheng et al., 2017), the German Traffic Sign Recognition Benchmark (GTSRB) dataset (Stallkamp et al., 2011), the KITTI dataset (Geiger et al., 2012), PatchCamelyon (Veeling et al., 2018), the UCF101 action recognition dataset (Soomro et al., 2012), Kinetics 700 (Carreira et al., 2019), 2,500 random samples of the CLEVR dataset (Johnson et al., 2017), the Hateful Memes dataset (Kiela et al., 2020), and the ImageNet-1k dataset (Deng et al., 2012). For the two video datasets (UCF101 and Kinetics700), we use the middle frame of each video clip as the input image. STL-10 and UCF101 have multiple predefined train/validation/test splits, 10 and 3 respectively, and we report the average over all splits. Details on each dataset and the corresponding evaluation metrics are provided in Table 9.

Additionally, we created two datasets that we call Country211 and Rendered SST2. The Country211 dataset is designed to assess the geolocation capability of visual representations. We filtered the YFCC100m dataset (Thomee et al., 2016) to find 211 countries (defined as having an ISO-3166 country code) that have at least 300 photos with GPS coordinates, and we built a balanced dataset with 211 categories, by sampling 200 photos for training and 100 photos for testing, for each country.

The Rendered SST2 dataset is designed to measure the optical character recognition capability of visual representations. To do so, we used the sentences from the Stanford Sentiment Treebank dataset (Socher et al., 2013) and rendered them into images, with black texts on a white background, in a 448×448 resolution. Two example images from this dataset are shown in Figure 19.

A.2. Models

In combination with the datasets listed above, we evaluate the following series of models using linear probes.

LM RN50 This is a multimodal model that uses an autoregressive loss instead of a contrastive loss, while using the ResNet-50 architecture as in the smallest contrastive model. To do so, the output from the CNN is projected into four tokens, which are then fed as a prefix to a language model autoregressively predicting the text tokens. Apart from the training objective, the model was trained on the same dataset for the same number of epochs as other CLIP models.

CLIP-RN Five ResNet-based contrastive CLIP models are included. As discussed in the paper, the first two models follow ResNet-50 and ResNet-101, and we use EfficientNet-style (Tan & Le, 2019) scaling for the next three models which simultaneously scale the model width, the number of layers, and the input resolution to obtain models with roughly 4x, 16x, and 64x computation.

CLIP-ViT We include four CLIP models that use the Vision Transformer (Dosovitskiy et al., 2020) architecture as the image encoder. We include three models trained on 224-by-224 pixel images: ViT-B/32, ViT-B/16, ViT-L/14, and the ViT-L/14 model fine-tuned on 336-by-336 pixel input images.

EfficietNet We use the nine models (B0-B8) from the original EfficientNet paper (Tan & Le, 2019), as well as the noisy-student variants (B0-B7, L2-475, and L2-800) (Tan & Le, 2019). The largest models (L2-475 and L2-800) take the input resolutions of 475x475 and 800x800 pixels, respectively.

Instagram-pretrained ResNeXt We use the four models (32x8d, 32x16d, 32x32d, 32x48d) released by (Mahajan et al., 2018), as well as their two FixRes variants which use higher input resolutions (Touvron et al., 2019).

Big Transfer (BiT) We use BiT-S and BiT-M models (Kolesnikov et al., 2019), trained on the ImageNet-1k and ImageNet-21k datasets. The model weights for BiT-L is not publicly available.

Vision Transformer (ViT) We also include four ViT (Dosovitskiy et al., 2020) checkpoints pretrained on the ImageNet-21k dataset, namely ViT-B/32, ViT-B/16, ViT-L/16, and ViT-H/14. We note that their best-performing models, trained on the JFT-300M dataset, are not available publicly.

SimCLRv2 The SimCLRv2 (Chen et al., 2020c) project released pre-trained and fine-tuned models in various settings. We use the seven pretrain-only checkpoints with selective kernels.

BYOL We use the recently released model weights of BYOL (Grill et al., 2020), specifically their 50x1 and 200x2

A. 线性探针评估

我们提供了本文中线性探针实验的更多细节，包括用于评估的数据集和模型列表。

A.1. 数据集

我们使用由（Kornblith et al., 2019）提出并被广泛研究的评估套件中的 12 个数据集，并额外加入 15 个数据集，以评估模型在更广泛的分布和任务上的性能。这些数据集包括 MNIST、Facial Expression Recognition 2013 数据集（Goodfellow et al., 2015）、STL-10（Coates et al., 2011）、EuroSAT（Helber et al., 2019）、NWPU-RESISC45 数据集（Cheng et al., 2017）、German Traffic Sign Recognition Benchmark（GTSRB）数据集（Stallkamp et al., 2011）、KITTI 数据集（Geiger et al., 2012）、PatchCamelyon（Veeling et al., 2018）、UCF101 动作识别数据集（Soomro et al., 2012）、Kinetics 700（Carreira et al., 2019）、CLEVR 数据集的 2,500 个随机样本（Johnson et al., 2017）、Hateful Memes 数据集（Kiela et al., 2020）以及 ImageNet-1k 数据集（Deng et al., 2012）。对于两个视频数据集（UCF101 和 Kinetics 700），我们使用每个视频片段的中间帧作为输入图像。STL-10 和 UCF101 分别具有 10 个和 3 个预定义的训练/验证/测试划分，我们报告所有划分上的平均结果。每个数据集的详细信息及其对应的评估指标见表 9。

此外，我们创建了两个数据集，称为 Country211 和 Rendered SST2。Country211 数据集旨在评估视觉表示的地理定位能力。我们对 YFCC100m 数据集（Thomee 等，2016）进行了筛选，找到了 211 个国家（定义为具有 ISO-3166 国家代码），这些国家至少有 300 张带 GPS 坐标的照片；并构建了一个包含 211 个类别的均衡数据集，其中每个国家抽取 200 张照片用于训练、100 张照片用于测试。

Rendered SST2 数据集旨在衡量视觉表征的光学字符识别能力。为此，我们使用了 Stanford Sentiment Treebank 数据集（Socher 等，2013）中的句子，并将其渲染为图像，即白色背景上的黑色文本，分辨率为 448×448。该数据集中的两个示例图像如图 19 所示。

A.2. 模型

结合上述数据集，我们使用线性探针对以下一系列模型进行评估。

LM RN50 这是一个多模态模型，它使用自回归损失而不是对比损失，同时使用采用与最小的对比模型相同的 ResNet-50 架构。为此，将 CNN 的输出投影为四个 token，然后将其作为前缀输入到语言模型中，由该模型以自回归方式预测文本 token。除训练目标外，该模型在与其他 CLIP 模型相同的数据集上训练了相同的轮数。

CLIP-RN 包含五个基于 ResNet 的对比式 CLIP 模型。如论文中所述，前两个模型遵循 ResNet-50 和 ResNet-101，接下来的三个模型则采用 EfficientNet 风格（Tan & Le, 2019）的缩放方式，同时缩放模型宽度、层数和输入分辨率，以获得计算量约为 4 倍、16 倍和 64 倍的模型。

CLIP-ViT 我们包含四个使用视觉 Transformer（Dosovitskiy 等，2020）架构作为图像编码器的 CLIP 模型。我们包含三个在 224×224 像素图像上训练的模型：ViT-B/32、ViT-B/16、ViT-L/14，以及在 336×336 像素输入图像上微调的 ViT-L/14 模型。

EfficientNet 我们使用了原始 EfficientNet 论文（Tan & Le，2019）中的九个模型（B0-B8），以及 noisy-student 变体（B0-B7、L2-475 和 L2-800）（Tan & Le，2019）。最大的模型（L2-475 和 L2-800）分别采用 475x475 和 800x800 像素的输入分辨率。

Instagram 预训练的 ResNeXt 我们使用了（Mahajan 等，2018）发布的四个模型（32x8d、32x16d、32x32d、32x48d），以及它们的两个使用更高输入分辨率的 FixRes 变体（Touvron 等，2019）。

大迁移（BiT）我们使用在 ImageNet-1k 和 ImageNet-21k 数据集上训练的 BiT-S 和 BiT-M 模型（Kolesnikov 等，2019）。BiT-L 的模型权重未公开提供。

视觉 Transformer（ViT）我们还包括四个在 ImageNet-21k 数据集上预训练的 ViT（Dosovitskiy 等，2020）检查点，即 ViT-B/32、ViT-B/16、ViT-L/16 和 ViT-H/14。我们注意到，他们在 JFT-300M 数据集上训练的表现最佳的模型并未公开提供。

SimCLRv2 SimCLRv2（Chen 等，2020c）项目发布了在各种设置下预训练和微调的模型。我们使用了七个带有选择性卷积核的仅预训练检查点。

BYOL 我们使用了最近发布的 BYOL（Grill 等，2020）的模型权重，具体来说是其 50x1 和 200x2

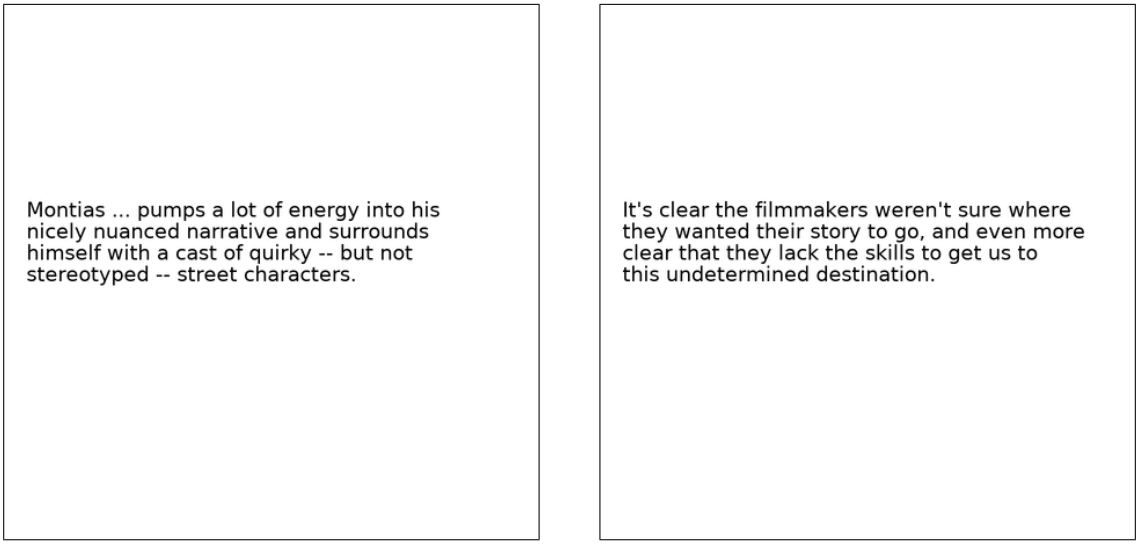


Figure 19. Two example images from the Rendered SST2 dataset

checkpoints.

Momentum Contrast (MoCo) We include the MoCo-v1 (He et al., 2020) and the MoCo-v2 (Chen et al., 2020d) checkpoints.

VirTex We use the pretrained model of VirTex (Desai & Johnson, 2020). We note that VirTex has a similar model design to CLIP-AR but is trained on a 1000x smaller dataset of high-quality captions from MSCOCO.

ResNet We add the original ResNet checkpoints released by (He et al., 2016b), namely ResNet-50, ResNet-101, and ResNet152.

A.3. Evaluation

We use image features taken from the penultimate layer of each model, ignoring any classification layer provided. For CLIP-ViT models, we used the features before the linear projection to the embedding space, which corresponds to `I_f` in Figure 3. We train a logistic regression classifier using scikit-learn's L-BFGS implementation, with maximum 1,000 iterations, and report the corresponding metric for each dataset. We determine the L2 regularization strength λ using a hyperparameter sweep on the validation sets over the range between 10^{-6} and 10^{6}, with 96 logarithmically spaced steps. To save compute required for the sweeps, we perform a parametric binary search that starts with $\lambda = [10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}, 10^{4}, 10^{6}]$ and iteratively halves the interval around the peak until it reaches a resolution of 8 steps per decade. The hyperparameter sweeps are performed on a validation split of each dataset. For the datasets that contain a validation split in addition to a test split, we use the provided validation set to perform the hyperparameter search, and for the datasets that do not provide a validation split or have not published labels for the test data, we split the training dataset to perform the hyperparameter search. For the final result, we combine the validation split back with the training split and report the performance on the unused split.

A.4. Results

The individual linear probe scores are provided in Table 10 and plotted in Figure 20. The best-performing CLIP model, using ViT-L/14 archiecture and 336-by-336 pixel images, achieved the state of the art in 21 of the 27 datasets, i.e. included in the Clopper-Pearson 99.5% confidence interval around each dataset's top score. For many datasets, CLIP performs significantly better than other models, demonstrating the advantage of natural language supervision over traditional pre-training approaches based on image classification. See Section 3.2 for more discussions on the linear probe results.

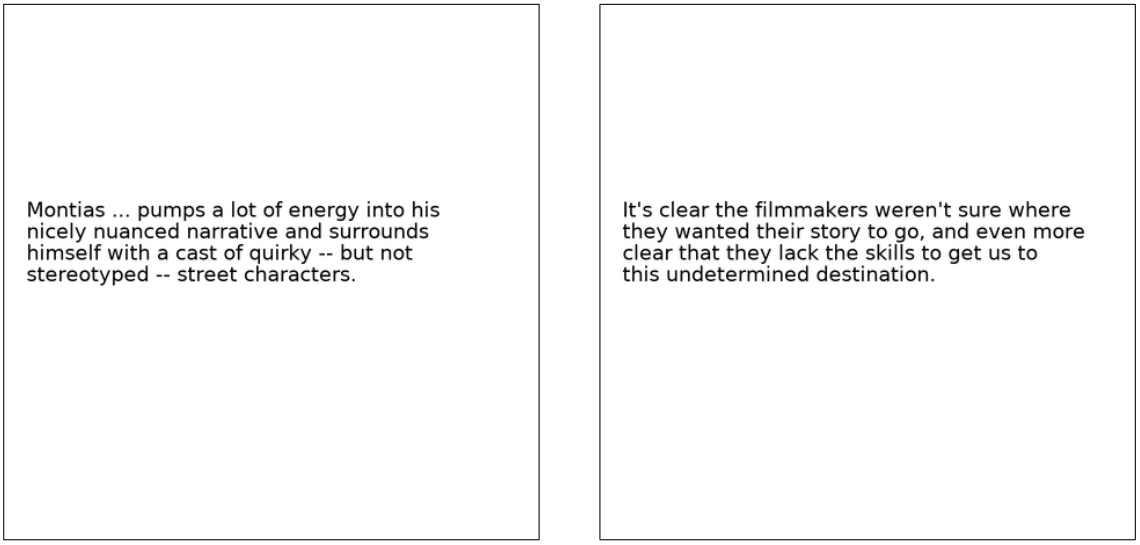


图 19. Rendered SST2 数据集中的两个示例图像

检查点。

动量对比（MoCo）我们包含 MoCo-v1（He 等，2020）和 MoCo-v2（Chen 等，2020d）检查点。

VirTex 我们使用 VirTex 的预训练模型（Desai & Johnson, 2020）。我们注意到，VirTex 的模型设计与 CLIP-AR 相似，但其训练所用的数据集规模小 1000 倍，来自 MSCOCO 的高质量图像描述。

ResNet 我们添加了（He 等人，2016b）发布的原始 ResNet 检查点，即 ResNet-50、ResNet-101 和 ResNet152。

A.3. 评估

我们使用取自每个模型倒数第二层的图像特征，忽略任何提供的分类层。对于 CLIP-ViT 模型，我们使用线性投影到嵌入空间之前的特征，这对应于图 3 中的 I f。我们使用 scikit-learn 的 L-BFGS 实现训练逻辑回归分类器，最大迭代次数为 1,000，并报告每个数据集对应的指标。我们通过在验证集上进行超参数扫描，在 10^{-6} 到 10^{6} 的范围内以 96 个按对数间隔分布的步长，确定 L2 正则化强度 λ。为节省扫描所需的计算量，我们执行参数化二分搜索，从 $\lambda = [10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}, 10^{4}, 10^{6}]$ 开始，并迭代地将峰值周围的区间减半，直到达到每十倍区间 8 个步长的分辨率。超参数扫描在每个数据集的验证划分上进行。对于除以下之外还包含验证划分的数据集

对于测试划分，我们使用提供的验证集进行超参数搜索；对于未提供验证划分或未公布测试数据标签的数据集，我们划分训练数据集来进行超参数搜索。对于最终结果，我们将验证划分重新合并回训练划分，并报告在未使用划分上的性能。

A.4. 结果

各个线性探针得分见表 10，并绘制于图 20 中。表现最佳的 CLIP 模型采用 ViT-L/14 架构和 336×336 像素图像，在 27 个数据集中的 21 个上达到了最先进水平，即落入各数据集最高得分周围的 Clopper-Pearson 99.5 % 置信区间内。对于许多数据集，CLIP 的表现显著优于其他模型，展示了自然语言监督相较于基于图像分类的传统预训练方法的优势。有关线性探针结果的更多讨论，请参见第 3.2 节。

Dataset	Classes	Train size	Test size	Evaluation metric
Food-101	102	75,750	25,250	accuracy
CIFAR-10	10	50,000	10,000	accuracy
CIFAR-100	100	50,000	10,000	accuracy
Birdsnap	500	42,283	2,149	accuracy
SUN397	397	19,850	19,850	accuracy
Stanford Cars	196	8,144	8,041	accuracy
FGVC Aircraft	100	6,667	3,333	mean per class
Pascal VOC 2007 Classification	20	5,011	4,952	11-point mAP
Describable Textures	47	3,760	1,880	accuracy
Oxford-IIIT Pets	37	3,680	3,669	mean per class
Caltech-101	102	3,060	6,085	mean-per-class
Oxford Flowers 102	102	2,040	6,149	mean per class
MNIST	10	60,000	10,000	accuracy
Facial Emotion Recognition 2013	8	32,140	3,574	accuracy
STL-10	10	1000	8000	accuracy
EuroSAT	10	10,000	5,000	accuracy
RESISC45	45	3,150	25,200	accuracy
GTSRB	43	26,640	12,630	accuracy
KITTI	4	6,770	711	accuracy
Country211	211	43,200	21,100	accuracy
PatchCamelyon	2	294,912	32,768	accuracy
UCF101	101	9,537	1,794	accuracy
Kinetics700	700	494,801	31,669	mean(top1, top5)
CLEVR Counts	8	2,000	500	accuracy
Hateful Memes	2	8,500	500	ROC AUC
Rendered SST2	2	7,792	1,821	accuracy
ImageNet	1000	1,281,167	50,000	accuracy

Table 9. Datasets examined for linear probes. We note that, for the Birdsnap and Kinetics700 datasets, we used the resources that are available online at the time of this writing.

Dataset	Classes	Train size	Test size	Evaluation metric
Food-101	102	75,750	25,250	accuracy
CIFAR-10	10	50,000	10,000	accuracy
CIFAR-100	100	50,000	10,000	accuracy
Birdsnap	500	42,283	2,149	accuracy
SUN397	397	19,850	19,850	accuracy
Stanford Cars	196	8,144	8,041	accuracy
FGVC Aircraft	100	6,667	3,333	mean per class
Pascal VOC 2007 Classification	20	5,011	4,952	11-point mAP
Describable Textures	47	3,760	1,880	accuracy
Oxford-IIIT Pets	37	3,680	3,669	mean per class
Caltech-101	102	3,060	6,085	mean-per-class
Oxford Flowers 102	102	2,040	6,149	mean per class
MNIST	10	60,000	10,000	accuracy
Facial Emotion Recognition 2013	8	32,140	3,574	accuracy
STL-10	10	1000	8000	accuracy
EuroSAT	10	10,000	5,000	accuracy
RESISC45	45	3,150	25,200	accuracy
GTSRB	43	26,640	12,630	accuracy
KITTI	4	6,770	711	accuracy
Country211	211	43,200	21,100	accuracy
PatchCamelyon	2	294,912	32,768	accuracy
UCF101	101	9,537	1,794	accuracy
Kinetics700	700	494,801	31,669	mean(top1, top5)
CLEVR Counts	8	2,000	500	accuracy
Hateful Memes	2	8,500	500	ROC AUC
Rendered SST2	2	7,792	1,821	accuracy
ImageNet	1000	1,281,167	50,000	accuracy

表 9. 用于线性探针评估的数据集。我们注意到，对于 Birdsnap 和 Kinetics700 数据集，我们使用了撰写本文时可在线获取的资源。

		Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech101	Flowers	MNIST	FER2013	STL10*	EuroSAT	RESISC45	GTSRB	KITTI	Country211	PCAM	UCF101	Kinetics700	CLEVR	HatefulMemes	SST	ImageNet
LM RN50		81.3	82.8	61.7	44.2	69.6	74.9	44.9	85.5	71.5	82.8	85.5	91.1	96.6	60.1	95.3	93.4	84.0	73.8	70.2	19.0	82.9	76.4	51.9	51.2	65.2	76.8	65.2
CLIP-RN	50	86.4	88.7	70.3	56.4	73.3	78.3	49.1	87.1	76.4	88.2	89.6	96.1	98.3	64.2	96.6	95.2	87.5	82.4	70.2	25.3	82.7	81.6	57.2	53.6	65.7	72.6	73.3
	101	88.9	91.1	73.5	58.6	75.1	84.0	50.7	88.0	76.3	91.0	92.0	96.4	98.4	65.2	97.8	95.9	89.3	82.4	**73.6**	26.6	82.8	84.0	60.3	50.3	68.2	73.3	75.7
	50x4	91.3	90.5	73.0	65.7	77.0	85.9	57.3	88.4	79.5	91.9	92.5	97.8	98.5	68.1	97.8	96.4	89.7	85.5	59.4	30.3	83.0	85.7	62.6	52.5	68.0	76.6	78.2
	50x16	93.3	92.2	74.9	72.8	79.2	88.7	62.7	**89.0**	79.1	93.5	93.7	98.3	**98.9**	68.7	98.6	97.0	91.4	89.0	69.2	34.8	83.5	88.0	66.3	53.8	71.1	**80.0**	81.5
	50x64	94.8	94.1	78.6	77.2	81.1	90.5	67.7	**88.9**	**82.0**	94.5	95.4	98.9	**98.9**	**71.3**	99.1	97.1	92.8	90.2	69.2	40.7	83.7	89.5	69.1	55.0	**75.0**	**81.2**	83.6
CLIP-ViT	B/32	88.8	95.1	80.5	58.5	76.6	81.8	52.0	87.7	76.5	90.0	93.0	96.9	**99.0**	69.2	98.3	97.0	90.5	85.3	66.2	27.8	83.9	85.5	61.7	52.1	66.7	70.8	76.1
	B/16	92.8	96.2	83.1	67.8	78.4	86.7	59.5	**89.2**	79.2	93.1	94.7	98.1	**99.0**	69.5	99.0	97.1	92.7	86.6	67.8	33.3	83.5	88.4	66.1	**57.1**	70.3	75.5	80.2
	L/14	95.2	98.0	87.5	77.0	**81.8**	**90.9**	69.4	**89.6**	**82.1**	**95.1**	**96.5**	99.2	**99.2**	**72.2**	**99.7**	**98.2**	94.1	**92.5**	64.7	42.9	85.8	**91.5**	72.0	**57.8**	**76.2**	**80.8**	83.9
	L/14-336px	**95.9**	97.9	87.4	**79.9**	**82.2**	**91.5**	**71.6**	**89.9**	**83.0**	**95.1**	**96.0**	99.2	**99.2**	**72.9**	**99.7**	**98.1**	**94.9**	**92.4**	69.2	**46.4**	85.6	**92.0**	**73.0**	**60.3**	**77.3**	**80.5**	85.4
EfficientNet	B0	74.3	92.5	76.5	59.7	62.0	62.5	55.7	84.4	71.2	93.0	93.3	91.7	98.2	57.2	97.1	97.3	85.5	80.0	**73.8**	12.4	83.1	74.4	47.6	47.9	55.7	53.4	76.9
	B1	74.2	93.2	77.2	61.3	62.6	62.5	56.1	84.7	74.2	93.4	93.6	92.4	98.3	57.0	97.5	96.8	84.5	75.9	**75.5**	12.5	82.7	74.7	48.5	44.3	54.5	54.4	78.6
	B2	75.8	93.6	77.9	64.4	64.0	63.2	57.0	85.3	73.5	93.9	93.5	92.9	98.5	56.6	97.7	96.9	84.4	76.4	**73.1**	12.6	84.3	75.1	49.4	42.6	55.4	55.2	79.7
	B3	77.4	94.0	78.0	66.5	64.4	66.0	59.3	85.8	73.1	94.1	93.7	93.3	98.5	57.1	98.2	97.3	85.0	75.8	**76.1**	13.4	83.3	78.1	50.9	45.1	53.8	54.8	81.0
	B4	79.7	94.1	78.7	70.1	65.4	66.4	60.4	86.5	73.4	94.7	93.5	93.2	98.8	57.9	98.6	96.8	85.0	78.3	**72.3**	13.9	83.1	79.1	52.5	46.5	54.4	55.4	82.9
	B5	81.5	93.6	77.9	72.4	67.1	72.7	68.9	86.7	73.9	**95.0**	94.7	94.5	98.4	58.5	98.7	96.8	86.0	78.5	69.6	14.9	84.7	80.9	54.5	46.6	53.3	56.3	83.7
	B6	82.4	94.0	78.0	73.5	65.8	71.1	68.2	87.6	73.9	**95.0**	94.1	93.7	98.4	60.2	98.7	96.8	85.4	78.1	**72.7**	15.3	84.2	80.0	54.1	51.1	53.3	57.0	84.0
	B7	84.5	94.9	80.1	74.7	69.0	77.1	**72.3**	87.2	76.8	**95.2**	94.7	95.9	98.6	61.3	99.1	96.3	86.8	80.8	**75.8**	16.4	85.2	81.9	56.8	51.9	54.4	57.8	84.8
	B8	84.5	95.0	80.7	75.2	69.6	76.8	**71.5**	87.4	77.1	**94.9**	95.2	96.3	98.6	61.4	99.2	97.0	87.4	80.4	70.9	17.4	85.2	82.4	57.7	51.4	51.7	55.8	85.3
EfficientNet Noisy Student	B0	78.1	94.0	78.6	63.5	65.5	57.2	53.7	85.6	75.6	93.8	93.1	94.5	98.1	55.6	98.2	97.0	84.3	74.0	71.6	14.0	83.1	76.7	51.7	47.3	55.7	55.0	78.5
	B1	80.4	95.1	80.2	66.6	67.6	59.6	53.7	86.2	77.0	94.6	94.4	95.1	98.0	56.1	98.6	96.9	84.3	73.1	67.1	14.5	83.9	79.9	54.5	46.1	54.3	54.9	81.1
	B2	80.9	95.3	81.3	67.6	67.9	60.9	55.2	86.3	77.7	**95.0**	94.7	94.4	98.0	55.5	98.8	97.3	84.6	71.7	70.0	14.6	82.9	80.1	55.1	46.1	54.1	55.3	82.2
	B3	82.6	95.9	82.1	68.6	68.8	60.6	55.4	86.5	77.2	**95.0**	94.8	95.2	98.1	56.0	99.1	96.5	85.0	70.5	69.5	15.1	83.1	81.8	56.8	45.1	55.7	52.0	83.8
	B4	85.2	95.6	81.0	72.5	69.7	56.1	52.6	87.0	78.7	**94.8**	95.2	95.3	98.2	56.0	99.3	95.3	84.8	61.9	64.8	16.0	82.8	83.4	59.8	43.2	55.3	53.0	85.4
	B5	87.6	96.3	82.4	75.3	71.6	64.7	64.8	87.8	79.6	**95.5**	95.6	96.6	98.8	60.9	99.4	96.1	87.0	68.5	**73.7**	16.4	83.5	86.4	61.6	46.3	53.4	55.8	85.8
	B6	87.3	97.0	83.9	75.8	71.4	67.6	65.6	87.3	78.5	**95.2**	**96.4**	97.2	98.6	61.9	**99.5**	96.6	86.1	70.7	**72.4**	17.6	84.2	85.5	61.0	49.6	54.6	55.7	86.4
	B7	88.4	96.0	82.0	76.9	72.6	72.2	**71.2**	88.1	**80.5**	**95.5**	95.5	96.6	98.5	62.7	99.4	96.2	88.5	73.4	**73.0**	18.5	83.8	86.6	63.2	50.5	57.2	56.7	87.0
	L2-475	91.6	**99.0**	**91.0**	74.8	76.4	75.1	66.8	**89.5**	**81.9**	**95.6**	**96.5**	97.7	**98.9**	67.5	**99.6**	97.0	89.5	73.4	68.9	22.2	86.3	89.4	68.2	**58.3**	58.6	55.2	**88.3**
	L2-800	92.0	**98.7**	89.0	**78.5**	75.7	75.5	68.4	**89.4**	**82.5**	**95.6**	94.7	97.9	98.5	68.4	**99.7**	97.2	89.9	77.7	66.9	23.7	**86.8**	88.9	66.7	**62.7**	58.4	56.9	**88.4**
Instagram	32x8d	84.8	95.9	80.9	63.8	69.0	74.2	56.0	88.0	75.4	**95.4**	93.9	91.7	97.4	60.7	99.1	95.7	82.1	72.3	69.2	16.7	82.3	80.1	56.8	42.2	53.3	55.2	83.3
	32x16d	85.7	96.5	80.9	64.8	70.5	77.5	56.7	87.9	76.2	**95.6**	94.9	92.5	97.4	61.6	99.3	95.5	82.8	73.8	66.1	17.5	83.4	81.1	58.2	41.3	54.2	56.1	84.4
	32x32d	86.7	96.8	82.7	67.1	71.5	77.5	55.4	88.3	78.5	**95.8**	95.3	94.4	97.9	62.4	99.3	95.7	85.4	71.2	66.8	18.0	83.7	82.1	58.8	39.7	55.3	56.7	85.0
	32x48d	86.9	96.8	83.4	65.9	72.2	76.6	53.2	88.0	77.2	**95.5**	95.8	93.6	98.1	63.7	99.4	95.3	85.4	73.0	67.2	18.5	82.7	82.8	59.2	41.3	55.5	56.7	85.2
	FixRes-v1	88.5	95.7	81.1	67.4	72.9	80.5	57.6	88.0	77.9	**95.8**	**96.1**	94.5	97.9	62.2	99.4	96.2	86.6	76.5	64.8	19.3	82.5	83.4	59.8	43.5	56.6	59.0	86.0
	FixRes-v2	88.5	95.7	81.1	67.3	72.9	80.7	57.5	88.0	77.9	**95.0**	**96.0**	94.5	98.0	62.1	99.4	96.5	86.6	76.3	64.8	19.5	82.3	83.5	59.8	44.2	56.6	59.0	86.0
BiT-S	R50x1	72.5	91.7	74.8	57.7	61.1	53.5	52.5	83.7	72.4	92.3	91.2	92.0	98.4	56.1	96.4	97.4	85.0	70.0	66.0	12.5	83.0	72.3	47.5	48.3	54.1	55.3	75.2
	R50x3	75.1	93.7	79.0	61.1	63.7	55.2	54.1	84.8	74.6	92.5	91.6	92.8	98.8	58.7	97.0	**97.8**	86.4	73.1	**73.8**	14.0	84.2	76.4	50.0	49.2	54.7	54.2	77.2
	R101x1	73.5	92.8	77.4	58.4	61.3	54.0	52.4	84.4	73.5	92.5	91.8	90.6	98.3	56.5	96.8	97.3	84.6	69.4	68.9	12.6	82.0	73.5	48.6	45.4	52.6	55.5	76.0
	R101x3	74.7	93.9	79.8	57.8	62.9	54.7	53.3	84.7	75.5	92.3	91.2	92.6	98.8	59.7	97.3	**98.0**	85.5	71.8	60.2	14.1	83.1	75.9	50.4	49.7	54.1	54.6	77.4
	R152x2	74.9	94.3	79.7	58.7	62.7	55.9	53.6	85.3	74.9	93.0	92.0	91.7	98.6	58.3	97.1	**97.8**	86.2	71.8	71.6	13.9	84.1	76.2	49.9	48.2	53.8	55.9	77.1
	R152x4	74.7	94.2	79.2	57.8	62.9	51.2	50.8	85.4	75.4	93.1	91.2	91.4	**98.9**	61.4	97.2	**98.0**	85.5	72.8	67.9	14.9	83.1	76.0	50.3	42.9	53.6	56.0	78.5
BiT-M	R50x1	83.3	94.9	82.2	70.9	69.9	59.0	55.6	86.8	77.3	91.5	93.9	**99.4**	98.0	60.6	98.4	97.5	87.4	68.6	68.2	16.6	82.5	79.4	53.2	49.4	54.5	53.4	76.7
	R50x3	86.9	96.7	86.2	75.7	74.6	60.6	54.2	87.7	78.5	93.2	95.3	**99.4**	98.6	64.6	99.3	**98.0**	88.1	69.9	59.6	19.6	83.4	83.5	57.8	51.3	55.8	55.6	80.7
	R101x1	85.5	95.7	84.4	73.0	72.5	59.8	55.0	87.3	78.1	92.2	95.0	**99.5**	98.1	62.5	99.0	**97.6**	87.8	68.7	67.7	18.0	84.0	82.3	55.9	53.4	54.8	53.1	79.4
	R101x3	87.2	97.4	87.5	72.4	75.0	57.4	47.4	87.5	79.6	93.2	95.4	**99.6**	98.6	64.3	99.4	**98.2**	87.7	68.8	64.1	20.7	80.4	84.0	58.7	52.6	54.9	54.3	81.2
	R152x2	88.0	97.5	87.8	75.8	75.9	61.5	55.3	88.1	79.8	93.6	**95.9**	**99.5**	98.5	64.3	**99.5**	**97.9**	89.0	70.0	70.3	20.7	82.6	85.5	59.6	50.8	54.9	55.1	81.9
	R152x4	87.2	97.6	88.2	72.4	75.0	49.1	43.4	87.1	79.9	92.4	95.4	99.3	98.5	65.7	**99.5**	**97.8**	87.7	68.2	57.1	20.6	80.4	84.6	59.0	49.7	57.2	55.1	81.5
ViT	B/32	81.8	96.7	86.3	65.2	70.7	49.1	42.7	85.3	73.1	90.4	94.5	98.7	97.8	59.0	99.0	96.3	83.0	68.1	65.1	15.7	82.6	79.1	51.7	38.9	57.1	54.6	76.6
	B/16	86.7	96.9	86.4	74.0	74.2	54.7	46.0	86.7	74.3	92.7	94.1	99.2	97.4	61.3	99.5	96.4	84.5	63.1	61.5	17.5	85.4	82.7	56.6	40.0	57.0	56.1	80.9
	L/16	87.4	97.9	89.0	76.5	74.9	62.5	52.2	86.1	75.0	92.9	94.7	99.3	98.0	64.0	**99.6**	96.5	85.7	70.4	58.8	17.7	85.7	84.1	58.0	38.4	58.4	52.8	81.9
	H/14	83.4	95.8	84.5	70.2	69.2	62.3	54.8	84.7	75.4	91.7	93.7	98.9	98.5	62.4	98.4	97.3	87.0	73.9	63.4	15.4	**87.0**	79.4	52.1	41.1	55.9	54.1	75.9
SimCLRv2	R50x1	76.4	93.2	77.9	48.6	64.1	56.3	51.7	84.4	77.0	88.3	91.8	92.9	97.6	59.7	96.7	97.5	85.8	71.1	69.1	15.8	84.8	78.4	51.0	56.2	53.9	53.8	73.8
	R50x3	81.0	95.6	82.4	56.5	67.0	65.6	61.1	85.9	78.8	90.9	94.1	95.4	98.7	62.6	98.2	**97.9**	88.2	78.2	**74.7**	17.6	85.4	82.6	54.6	55.4	54.2	55.2	77.3
	R101x1	77.9	94.8	79.9	51.9	65.2	57.1	52.0	85.4	77.2	90.0	91.6	92.7	97.2	59.4	97.6	96.8	84.6	65.7	70.6	16.1	84.3	78.8	52.4	53.6	55.1	55.7	76.1
	R101x3	82.2	96.4	83.4	57.5	68.2	64.6	60.0	86.2	78.9	91.8	95.0	95.4	98.4	63.0	98.5	**97.9**	88.0	77.5	69.1	18.3	85.5	82.9	55.9	52.2	54.5	56.3	78.8
	R152x1	78.6	95.0	79.9	50.3	65.6	55.6	52.2	85.8	77.3	90.1	92.5	91.8	97.6	59.8	98.1	96.6	84.3	64.8	70.3	16.6	83.9	79.4	53.1	**57.2**	55.8	54.8	76.9
	R152x2	82.3	96.7	83.9	58.1	68.5	64.9	58.7	86.6	79.1	92.2	94.1	96.0	98.2	64.1	98.5	**98.0**	88.1	77.0	69.8	18.4	85.3	82.7	56.2	53.6	56.0	56.5	79.2
	R152x3	83.6	96.8	84.5	60.3	69.1	68.5	63.1	86.7	**80.5**	92.6	94.9	96.3	98.7	65.4	98.8	**98.1**	89.5	78.4	68.5	19.4	85.2	83.5	57.0	54.4	54.6	54.2	80.0
BYOL	50x1	74.0	93.6	79.1	47.6	63.7	61.6	62.3	82.6	77.0	88.3	93.7	94.3	98.7	58.8	96.4	**97.6**	88.2	80.1	71.4	14.1	84.8	77.3	49.3	56.1	53.8	54.4	73.3
	200x2	78.5	96.2	83.3	53.4	68.5	61.7	55.4	86.6	77.4	91.9	95.5	93.9	98.7	62.6	98.6	**97.7**	87.4	77.1	**76.4**	16.4	84.0	82.6	55.1	54.1	52.5	52.4	79.2
MoCo	v1	65.9	85.0	63.1	27.5	52.6	35.9	43.5	75.7	70.0	70.4	78.1	85.4	97.6	54.3	85.6	97.1	82.9	62.6	60.2	12.6	85.7	64.2	40.7	54.7	55.6	53.5	57.2
	v2	72.2	93.4	76.3	39.6	60.2	48.3	51.1	82.6	75.1	84.4	89.9	90.7	98.4	58.3	95.7	97.2	85.4	75.7	**75.4**	13.2	85.6	72.7	47.8	**56.9**	53.9	53.8	69.1
VirTex		57.9	83.9	57.5	17.0	49.8	22.4	34.5	83.8	58.2	53.6	70.6	74.7	98.1	56.5	86.7	94.8	74.1	69.5	71.3	8.7	83.1	61.5	39.9	45.5	53.5	55.8	50.7
ResNet	50	71.3	91.8	74.5	52.7	60.5	49.9	48.5	83.8	72.3	92.4	90.8	90.8	98.3	54.9	96.4	96.7	83.6	70.6	67.1	11.7	82.5	71.2	46.8	43.0	56.5	55.5	74.3
	101	72.7	93.0	77.2	53.7	60.8	50.1	47.0	84.4	71.6	92.3	91.9	90.4	98.5	56.6	97.0	97.1	83.4	72.5	63.6	11.9	83.3	72.7	48.3	43.2	53.0	54.7	75.8
	152	73.7	93.5	78.0	55.1	61.6	52.8	48.4	84.5	71.9	93.0	92.1	89.6	98.2	57.0	97.6	97.0	83.1	70.1	70.2	12.3	82.9	75.3	49.2	42.4	53.2	53.9	77.1

Table 10. Linear probe performance of various pre-trained models over 27 datasets. Scores within the 99.5% Clopper-Pearson confidence interval of each dataset's top score are shown in bold.

*We updated the STL10 scores from the previous version of this paper after fixing a CUDA-related bug.

		Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech101	Flowers	MNIST	FER2013	STL10	CIFAR100	CLEVR	PCAM	CLEVR	Resisc45	KITTI	Country211	Mean
	LM RN50	81.3	82.8	61.7	44.2	69.6	74.9	44.9	85.5	71.5	82.8	85.5	91.1	96.6	60.1	95.3	93.4	84.0	73.8	70.2	19.0	82.9	76.4	51.9	51.2	65.2	76.8	65.2
NPLC	50	86.4	88.7	70.3	56.4	73.3	78.3	49.1	87.1	76.4	88.2	89.6	96.1	98.3	64.2	96.6	95.2	87.5	82.4	70.2	25.3	82.7	81.6	57.2	53.6	65.7	72.6	73.3
	101	88.9	91.1	73.5	58.6	75.1	84.0	50.7	88.0	76.3	91.0	92.0	96.4	98.4	65.2	97.8	95.9	89.3	82.4	**73.6**	26.6	82.8	84.0	60.3	50.3	68.2	73.3	75.7
	50x4	91.3	90.5	73.0	65.7	77.0	85.9	57.3	88.4	79.5	91.9	92.5	97.8	98.5	68.1	97.8	96.4	89.7	85.5	59.4	30.3	83.0	85.7	62.6	52.5	68.0	76.6	78.2
	50x16	93.3	92.2	74.9	72.8	79.2	88.7	62.7	**89.0**	79.1	93.5	93.7	98.3	**98.9**	68.7	98.6	97.0	91.4	89.0	69.2	34.8	83.5	88.0	66.3	53.8	71.1	**80.0**	81.5
	50x64	94.8	94.1	78.6	77.2	81.1	90.5	67.7	**88.9**	**82.0**	94.5	95.4	98.9	**98.9**	**71.3**	99.1	97.1	92.8	90.2	69.2	40.7	83.7	89.5	69.1	55.0	**75.0**	**81.2**	83.6
TVPLC	B/32	88.8	95.1	80.5	58.5	76.6	81.8	52.0	87.7	76.5	90.0	93.0	96.9	**99.0**	69.2	98.3	97.0	90.5	85.3	66.2	27.8	83.9	85.5	61.7	52.1	66.7	70.8	76.1
	B/16	92.8	96.2	83.1	67.8	78.4	86.7	59.5	**89.2**	79.2	93.1	94.7	98.1	**99.0**	69.5	99.0	97.1	92.7	86.6	67.8	33.3	83.5	88.4	66.1	**57.1**	70.3	75.5	80.2
	L/14	95.2	98.0	87.5	77.0	**81.8**	**90.9**	69.4	**89.6**	**82.1**	**95.1**	**96.5**	99.2	**99.2**	**72.2**	**99.7**	**98.2**	94.1	**92.5**	64.7	42.9	85.8	**91.5**	72.0	**57.8**	**76.2**	**80.8**	83.9
	L/14-336px	**95.9**	97.9	87.4	**79.9**	**82.2**	**91.5**	**71.6**	**89.9**	**83.0**	**95.1**	**96.0**	99.2	**99.2**	**72.9**	**99.7**	**98.1**	**94.9**	**92.4**	69.2	**46.4**	85.6	**92.0**	**73.0**	**60.3**	**77.3**	**80.5**	85.4
EfficientNet	B0	74.3	92.5	76.5	59.7	62.0	62.5	55.7	84.4	71.2	93.0	93.3	91.7	98.2	57.2	97.1	97.3	85.5	80.0	**73.8**	12.4	83.1	74.4	47.6	47.9	55.7	53.4	76.9
	B1	74.2	93.2	77.2	61.3	62.6	62.5	56.1	84.7	74.2	93.4	93.6	92.4	98.3	57.0	97.5	96.8	84.5	75.9	**75.5**	12.5	82.7	74.7	48.5	44.3	54.5	54.4	78.6
	B2	75.8	93.6	77.9	64.4	64.0	63.2	57.0	85.3	73.5	93.9	93.5	92.9	98.5	56.6	97.7	96.9	84.4	76.4	**73.1**	12.6	84.3	75.1	49.4	42.6	55.4	55.2	79.7
	B3	77.4	94.0	78.0	66.5	64.4	66.0	59.3	85.8	73.1	94.1	93.7	93.3	98.5	57.1	98.2	97.3	85.0	75.8	**76.1**	13.4	83.3	78.1	50.9	45.1	53.8	54.8	81.0
	B4	79.7	94.1	78.7	70.1	65.4	66.4	60.4	86.5	73.4	94.7	93.5	93.2	98.8	57.9	98.6	96.8	85.0	78.3	**72.3**	13.9	83.1	79.1	52.5	46.5	54.4	55.4	82.9
	B5	81.5	93.6	77.9	72.4	67.1	72.7	68.9	86.7	73.9	**95.0**	94.7	94.5	98.4	58.5	98.7	96.8	86.0	78.5	69.6	14.9	84.7	80.9	54.5	46.6	53.3	56.3	83.7
	B6	82.4	94.0	78.0	73.5	65.8	71.1	68.2	87.6	73.9	**95.0**	94.1	93.7	98.4	60.2	98.7	96.8	85.4	78.1	**72.7**	15.3	84.2	80.0	54.1	51.1	53.3	57.0	84.0
	B7	84.5	94.9	80.1	74.7	69.0	77.1	**72.3**	87.2	76.8	**95.2**	94.7	95.9	98.6	61.3	99.1	96.3	86.8	80.8	**75.8**	16.4	85.2	81.9	56.8	51.9	54.4	57.8	84.8
	B8	84.5	95.0	80.7	75.2	69.6	76.8	**71.5**	87.4	77.1	**94.9**	95.2	96.3	98.6	61.4	99.2	97.0	87.4	80.4	70.9	17.4	85.2	82.4	57.7	51.4	51.7	55.8	85.3
EfficientNet-NoisyStudent	B0	78.1	94.0	78.6	63.5	65.5	57.2	53.7	85.6	75.6	93.8	93.1	94.5	98.1	55.6	98.2	97.0	84.3	74.0	71.6	14.0	83.1	76.7	51.7	47.3	55.7	55.0	78.5
	B1	80.4	95.1	80.2	66.6	67.6	59.6	53.7	86.2	77.0	94.6	94.4	95.1	98.0	56.1	98.6	96.9	84.3	73.1	67.1	14.5	83.9	79.9	54.5	46.1	54.3	54.9	81.1
	B2	80.9	95.3	81.3	67.6	67.9	60.9	55.2	86.3	77.7	**95.0**	94.7	94.4	98.0	55.5	98.8	97.3	84.6	71.7	70.0	14.6	82.9	80.1	55.1	46.1	54.1	55.3	82.2
	B3	82.6	95.9	82.1	68.6	68.8	60.6	55.4	86.5	77.2	**95.0**	94.8	95.2	98.1	56.0	99.1	96.5	85.0	70.5	69.5	15.1	83.1	81.8	56.8	45.1	55.7	52.0	83.8
	B4	85.2	95.6	81.0	72.5	69.7	56.1	52.6	87.0	78.7	**94.8**	95.2	95.3	98.2	56.0	99.3	95.3	84.8	61.9	64.8	16.0	82.8	83.4	59.8	43.2	55.3	53.0	85.4
	B5	87.6	96.3	82.4	75.3	71.6	64.7	64.8	87.8	79.6	**95.5**	95.6	96.6	98.8	60.9	99.4	96.1	87.0	68.5	**73.7**	16.4	83.5	86.4	61.6	46.3	53.4	55.8	85.8
	B6	87.3	97.0	83.9	75.8	71.4	67.6	65.6	87.3	78.5	**95.2**	**96.4**	97.2	98.6	61.9	**99.5**	96.6	86.1	70.7	**72.4**	17.6	84.2	85.5	61.0	49.6	54.6	55.7	86.4
	B7	88.4	96.0	82.0	76.9	72.6	72.2	**71.2**	88.1	**80.5**	**95.5**	95.5	96.6	98.5	62.7	99.4	96.2	88.5	73.4	**73.0**	18.5	83.8	86.6	63.2	50.5	57.2	56.7	87.0
	L2-475	91.6	**99.0**	**91.0**	74.8	76.4	75.1	66.8	**89.5**	**81.9**	**95.6**	**96.5**	97.7	**98.9**	67.5	**99.6**	97.0	89.5	73.4	68.9	22.2	86.3	89.4	68.2	**58.3**	58.6	55.2	**88.3**
	L2-800	92.0	**98.7**	89.0	**78.5**	75.7	75.5	68.4	**89.4**	**82.5**	**95.6**	94.7	97.9	98.5	68.4	**99.7**	97.2	89.9	77.7	66.9	23.7	**86.8**	88.9	66.7	**62.7**	58.4	56.9	**88.4**
MAWS	32x8d	84.8	95.9	80.9	63.8	69.0	74.2	56.0	88.0	75.4	**95.4**	93.9	91.7	97.4	60.7	99.1	95.7	82.1	72.3	69.2	16.7	82.3	80.1	56.8	42.2	53.3	55.2	83.3
	32x16d	85.7	96.5	80.9	64.8	70.5	77.5	56.7	87.9	76.2	**95.6**	94.9	92.5	97.4	61.6	99.3	95.5	82.8	73.8	66.1	17.5	83.4	81.1	58.2	41.3	54.2	56.1	84.4
	32x32d	86.7	96.8	82.7	67.1	71.5	77.5	55.4	88.3	78.5	**95.8**	95.3	94.4	97.9	62.4	99.3	95.7	85.4	71.2	66.8	18.0	83.7	82.1	58.8	39.7	55.3	56.7	85.0
	32x48d	86.9	96.8	83.4	65.9	72.2	76.6	53.2	88.0	77.2	**95.5**	95.8	93.6	98.1	63.7	99.4	95.3	85.4	73.0	67.2	18.5	82.7	82.8	59.2	41.3	55.5	56.7	85.2
	FixRes-v1	88.5	95.7	81.1	67.4	72.9	80.5	57.6	88.0	77.9	**95.8**	**96.1**	94.5	97.9	62.2	99.4	96.2	86.6	76.5	64.8	19.3	82.5	83.4	59.8	43.5	56.6	59.0	86.0
	FixRes-v2	88.5	95.7	81.1	67.3	72.9	80.7	57.5	88.0	77.9	**95.0**	**96.0**	94.5	98.0	62.1	99.4	96.5	86.6	76.3	64.8	19.5	82.3	83.5	59.8	44.2	56.6	59.0	86.0
STB	R50x1	72.5	91.7	74.8	57.7	61.1	53.5	52.5	83.7	72.4	92.3	91.2	92.0	98.4	56.1	96.4	97.4	85.0	70.0	66.0	12.5	83.0	72.3	47.5	48.3	54.1	55.3	75.2
	R50x3	75.1	93.7	79.0	61.1	63.7	55.2	54.1	84.8	74.6	92.5	91.6	92.8	98.8	58.7	97.0	**97.8**	86.4	73.1	**73.8**	14.0	84.2	76.4	50.0	49.2	54.7	54.2	77.2
	R101x1	73.5	92.8	77.4	58.4	61.3	54.0	52.4	84.4	73.5	92.5	91.8	90.6	98.3	56.5	96.8	97.3	84.6	69.4	68.9	12.6	82.0	73.5	48.6	45.4	52.6	55.5	76.0
	R101x3	74.7	93.9	79.8	57.8	62.9	54.7	53.3	84.7	75.5	92.3	91.2	92.6	98.8	59.7	97.3	**98.0**	85.5	71.8	60.2	14.1	83.1	75.9	50.4	49.7	54.1	54.6	77.4
	R152x2	74.9	94.3	79.7	58.7	62.7	55.9	53.6	85.3	74.9	93.0	92.0	91.7	98.6	58.3	97.1	**97.8**	86.2	71.8	71.6	13.9	84.1	76.2	49.9	48.2	53.8	55.9	77.1
	R152x4	74.7	94.2	79.2	57.8	62.9	51.2	50.8	85.4	75.4	93.1	91.2	91.4	**98.9**	61.4	97.2	**98.0**	85.5	72.8	67.9	14.9	83.1	76.0	50.3	42.9	53.6	56.0	78.5
MTB	R50x1	83.3	94.9	82.2	70.9	69.9	59.0	55.6	86.8	77.3	91.5	93.9	**99.4**	98.0	60.6	98.4	97.5	87.4	68.6	68.2	16.6	82.5	79.4	53.2	49.4	54.5	53.4	76.7
	R50x3	86.9	96.7	86.2	75.7	74.6	60.6	54.2	87.7	78.5	93.2	95.3	**99.4**	98.6	64.6	99.3	**98.0**	88.1	69.9	59.6	19.6	83.4	83.5	57.8	51.3	55.8	55.6	80.7
	R101x1	85.5	95.7	84.4	73.0	72.5	59.8	55.0	87.3	78.1	92.2	95.0	**99.5**	98.1	62.5	99.0	**97.6**	87.8	68.7	67.7	18.0	84.0	82.3	55.9	53.4	54.8	53.1	79.4
	R101x3	87.2	97.4	87.5	72.4	75.0	57.4	47.4	87.5	79.6	93.2	95.4	**99.6**	98.6	64.3	99.4	**98.2**	87.7	68.8	64.1	20.7	80.4	84.0	58.7	52.6	54.9	54.3	81.2
	R152x2	88.0	97.5	87.8	75.8	75.9	61.5	55.3	88.1	79.8	93.6	**95.9**	**99.5**	98.5	64.3	**99.5**	**97.9**	89.0	70.0	70.3	20.7	82.6	85.5	59.6	50.8	54.9	55.1	81.9
	R152x4	87.2	97.6	88.2	72.4	75.0	49.1	43.4	87.1	79.9	92.4	95.4	99.3	98.5	65.7	**99.5**	**97.8**	87.7	68.2	57.1	20.6	80.4	84.6	59.0	49.7	57.2	55.1	81.5
TV	B/32	81.8	96.7	86.3	65.2	70.7	49.1	42.7	85.3	73.1	90.4	94.5	98.7	97.8	59.0	99.0	96.3	83.0	68.1	65.1	15.7	82.6	79.1	51.7	38.9	57.1	54.6	76.6
	B/16	86.7	96.9	86.4	74.0	74.2	54.7	46.0	86.7	74.3	92.7	94.1	99.2	97.4	61.3	99.5	96.4	84.5	63.1	61.5	17.5	85.4	82.7	56.6	40.0	57.0	56.1	80.9
	L/16	87.4	97.9	89.0	76.5	74.9	62.5	52.2	86.1	75.0	92.9	94.7	99.3	98.0	64.0	**99.6**	96.5	85.7	70.4	58.8	17.7	85.7	84.1	58.0	38.4	58.4	52.8	81.9
	H/14	83.4	95.8	84.5	70.2	69.2	62.3	54.8	84.7	75.4	91.7	93.7	98.9	98.5	62.4	98.4	97.3	87.0	73.9	63.4	15.4	**87.0**	79.4	52.1	41.1	55.9	54.1	75.9
2vRLCms	R50x1	76.4	93.2	77.9	48.6	64.1	56.3	51.7	84.4	77.0	88.3	91.8	92.9	97.6	59.7	96.7	97.5	85.8	71.1	69.1	15.8	84.8	78.4	51.0	56.2	53.9	53.8	73.8
	R50x3	81.0	95.6	82.4	56.5	67.0	65.6	61.1	85.9	78.8	90.9	94.1	95.4	98.7	62.6	98.2	**97.9**	88.2	78.2	**74.7**	17.6	85.4	82.6	54.6	55.4	54.2	55.2	77.3
	R101x1	77.9	94.8	79.9	51.9	65.2	57.1	52.0	85.4	77.2	90.0	91.6	92.7	97.2	59.4	97.6	96.8	84.6	65.7	70.6	16.1	84.3	78.8	52.4	53.6	55.1	55.7	76.1
	R101x3	82.2	96.4	83.4	57.5	68.2	64.6	60.0	86.2	78.9	91.8	95.0	95.4	98.4	63.0	98.5	**97.9**	88.0	77.5	69.1	18.3	85.5	82.9	55.9	52.2	54.5	56.3	78.8
	R152x1	78.6	95.0	79.9	50.3	65.6	55.6	52.2	85.8	77.3	90.1	92.5	91.8	97.6	59.8	98.1	96.6	84.3	64.8	70.3	16.6	83.9	79.4	53.1	**57.2**	55.8	54.8	76.9
	R152x2	82.3	96.7	83.9	58.1	68.5	64.9	58.7	86.6	79.1	92.2	94.1	96.0	98.2	64.1	98.5	**98.0**	88.1	77.0	69.8	18.4	85.3	82.7	56.2	53.6	56.0	56.5	79.2
	R152x3	83.6	96.8	84.5	60.3	69.1	68.5	63.1	86.7	**80.5**	92.6	94.9	96.3	98.7	65.4	98.8	**98.1**	89.5	78.4	68.5	19.4	85.2	83.5	57.0	54.4	54.6	54.2	80.0
LOYO	50x1	74.0	93.6	79.1	47.6	63.7	61.6	62.3	82.6	77.0	88.3	93.7	94.3	98.7	58.8	96.4	**97.6**	88.2	80.1	71.4	14.1	84.8	77.3	49.3	56.1	53.8	54.4	73.3
	200x2	78.5	96.2	83.3	53.4	68.5	61.7	55.4	86.6	77.4	91.9	95.5	93.9	98.7	62.6	98.6	**97.7**	87.4	77.1	**76.4**	16.4	84.0	82.6	55.1	54.1	52.5	52.4	79.2
CoM	v1	65.9	85.0	63.1	27.5	52.6	35.9	43.5	75.7	70.0	70.4	78.1	85.4	97.6	54.3	85.6	97.1	82.9	62.6	60.2	12.6	85.7	64.2	40.7	54.7	55.6	53.5	57.2
	v2	72.2	93.4	76.3	39.6	60.2	48.3	51.1	82.6	75.1	84.4	89.9	90.7	98.4	58.3	95.7	97.2	85.4	75.7	**75.4**	13.2	85.6	72.7	47.8	**56.9**	53.9	53.8	69.1
	VirTex	57.9	83.9	57.5	17.0	49.8	22.4	34.5	83.8	58.2	53.6	70.6	74.7	98.1	56.5	86.7	94.8	74.1	69.5	71.3	8.7	83.1	61.5	39.9	45.5	53.5	55.8	50.7
ResNet	50	71.3	91.8	74.5	52.7	60.5	49.9	48.5	83.8	72.3	92.4	90.8	90.8	98.3	54.9	96.4	96.7	83.6	70.6	67.1	11.7	82.5	71.2	46.8	43.0	56.5	55.5	74.3
	101	72.7	93.0	77.2	53.7	60.8	50.1	47.0	84.4	71.6	92.3	91.9	90.4	98.5	56.6	97.0	97.1	83.4	72.5	63.6	11.9	83.3	72.7	48.3	43.2	53.0	54.7	75.8
	152	73.7	93.5	78.0	55.1	61.6	52.8	48.4	84.5	71.9	93.0	92.1	89.6	98.2	57.0	97.6	97.0	83.1	70.1	70.2	12.3	82.9	75.3	49.2	42.4	53.2	53.9	77.1

表 10. 各种预训练模型在 27 个数据集上的线性探针性能。每个数据集最高分的 99.5% Clopper-Pearson 置信区间内的分数以粗体显示。

*We updated the STL10 scores from the previous version of this paper after fixing a CUDA-related bug.

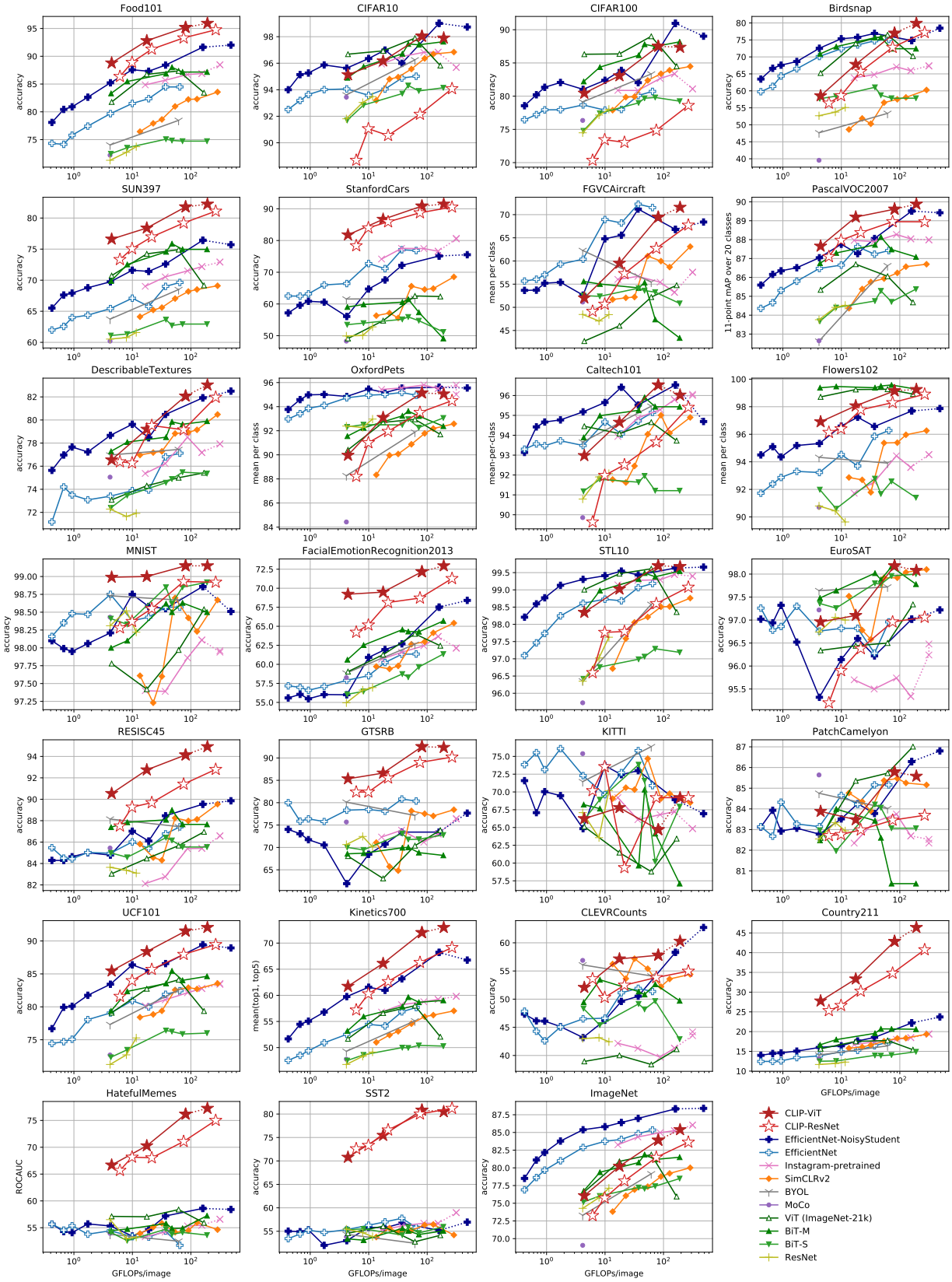


Figure 20. Linear probe performance plotted for each of the 27 datasets, using the data from Table 10.

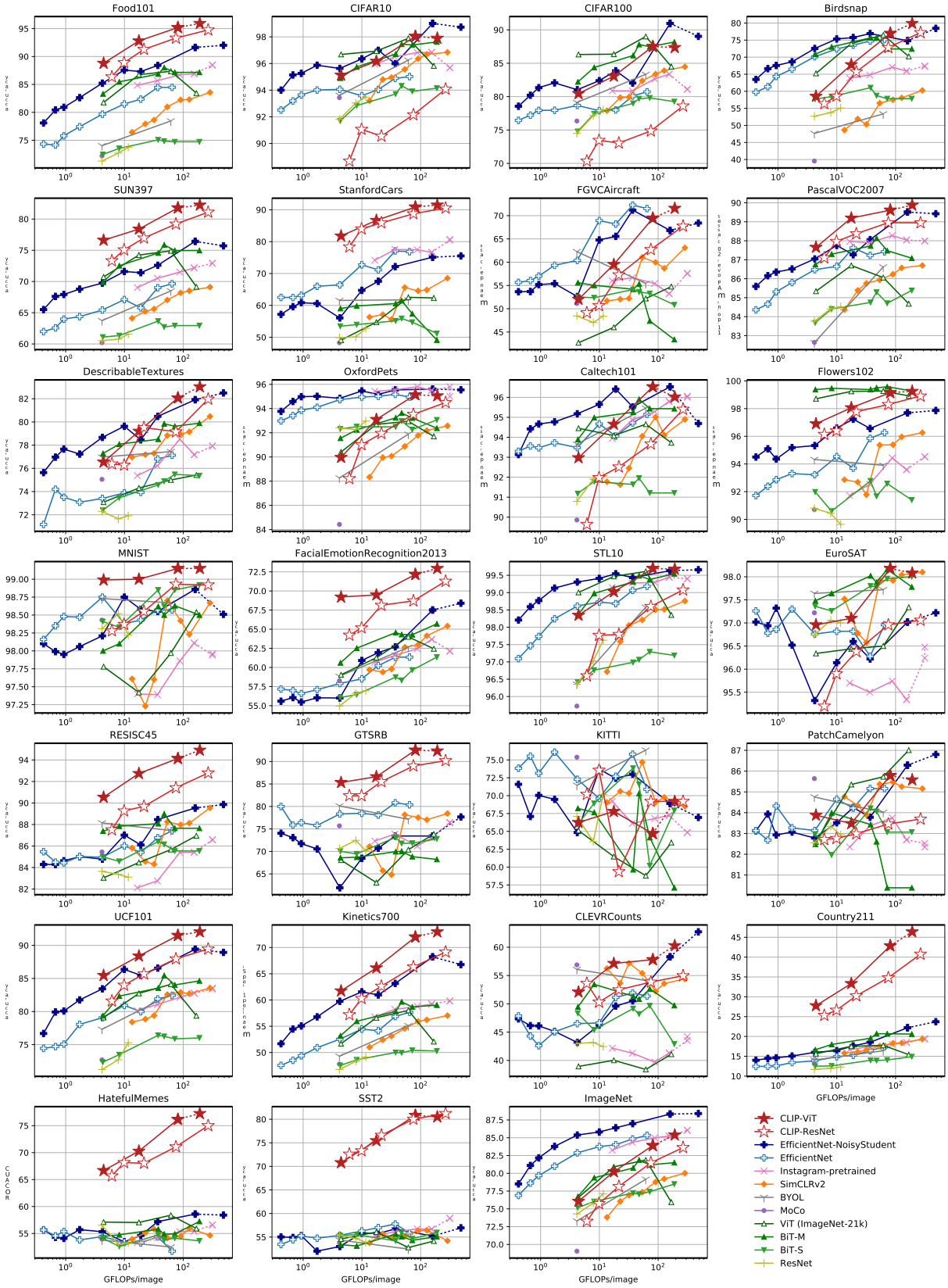


图20. 使用表10中的数据，绘制了27个数据集中每个数据集的线性探针性能。



Figure 21. Visualization of predictions from 36 CLIP zero-shot classifiers. All examples are random with the exception of reselecting Hateful Memes to avoid offensive content. The predicted probability of the top 5 classes is shown along with the text used to represent the class. When more than one template is used, the first template is shown. The ground truth label is colored green while an incorrect prediction is colored orange.



图 21. 36 个 CLIP 零样本分类器预测结果的可视化。除重新选择 Hateful Memes 以避免冒犯性内容外，所有示例均为随机选取。图中显示了前 5 个类别的预测概率，以及用于表示该类别的文本。当使用多个模板时，显示第一个模板。真实标签以绿色标出，而错误预测以橙色标出。

		Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Stanford Cars	FGVC Aircraft	VOC2007	DTD	Oxford Pets	Caltech101	Flowers102	MNIST	FER2013	STL10	EuroSAT	RESISC45	GTSRB	KITTI	Country211	PCam	UCF101	Kinetics700	CLEVR	HatefulMemes	Rendered SST2	ImageNet
CLIP-ResNet	RN50	81.1	75.6	41.6	32.6	59.6	55.8	19.3	82.1	41.7	85.4	82.1	65.9	66.6	42.2	94.3	41.1	54.2	35.2	42.2	16.1	57.6	63.6	43.5	20.3	59.7	56.9	59.6
	RN101	83.9	81.0	49.0	37.2	59.9	62.3	19.5	82.4	43.9	86.2	85.1	65.7	59.3	45.6	96.7	33.1	58.5	38.3	33.3	16.9	55.2	62.2	46.7	28.1	61.1	64.2	62.2
	RN50x4	86.8	79.2	48.9	41.6	62.7	67.9	24.6	83.0	49.3	88.1	86.0	68.0	75.2	51.1	96.4	35.0	59.2	35.7	26.0	20.2	57.5	65.5	49.0	17.0	58.3	66.6	65.8
	RN50x16	90.5	82.2	54.2	45.9	65.0	72.3	30.3	82.9	52.8	89.7	87.6	71.9	80.0	56.0	97.8	40.3	64.4	39.6	33.9	24.0	62.5	68.7	53.4	17.6	58.9	67.6	70.5
	RN50x64	91.8	86.8	61.3	48.9	66.9	76.0	35.6	83.8	53.4	93.4	90.6	77.3	90.8	61.0	98.3	59.4	69.7	47.9	33.2	29.6	65.0	74.1	56.8	27.5	62.1	70.7	73.6
CLIP-ViT	B/32	84.4	91.3	65.1	37.8	63.2	59.4	21.2	83.1	44.5	87.0	87.9	66.7	51.9	47.3	97.2	49.4	60.3	32.2	39.4	17.8	58.4	64.5	47.8	24.8	57.6	59.6	63.2
	B/16	89.2	91.6	68.7	39.1	65.2	65.6	27.1	83.9	46.0	88.9	89.3	70.4	56.0	52.7	98.2	54.1	65.5	43.3	44.0	23.3	48.1	69.8	52.4	23.4	61.7	59.8	68.6
	L/14	92.9	96.2	77.9	48.3	67.7	77.3	36.1	84.1	55.3	93.5	92.6	78.7	87.2	57.5	99.3	59.9	71.6	50.3	23.1	32.7	58.8	76.2	60.3	24.3	63.3	64.0	75.3
	L/14-336px	93.8	95.7	77.5	49.5	68.4	78.8	37.2	84.3	55.7	93.5	92.8	78.3	88.3	57.7	99.4	59.6	71.7	52.3	21.9	34.9	63.0	76.9	61.3	24.8	63.3	67.9	76.2

Table 11. Zero-shot performance of CLIP models over 27 datasets.

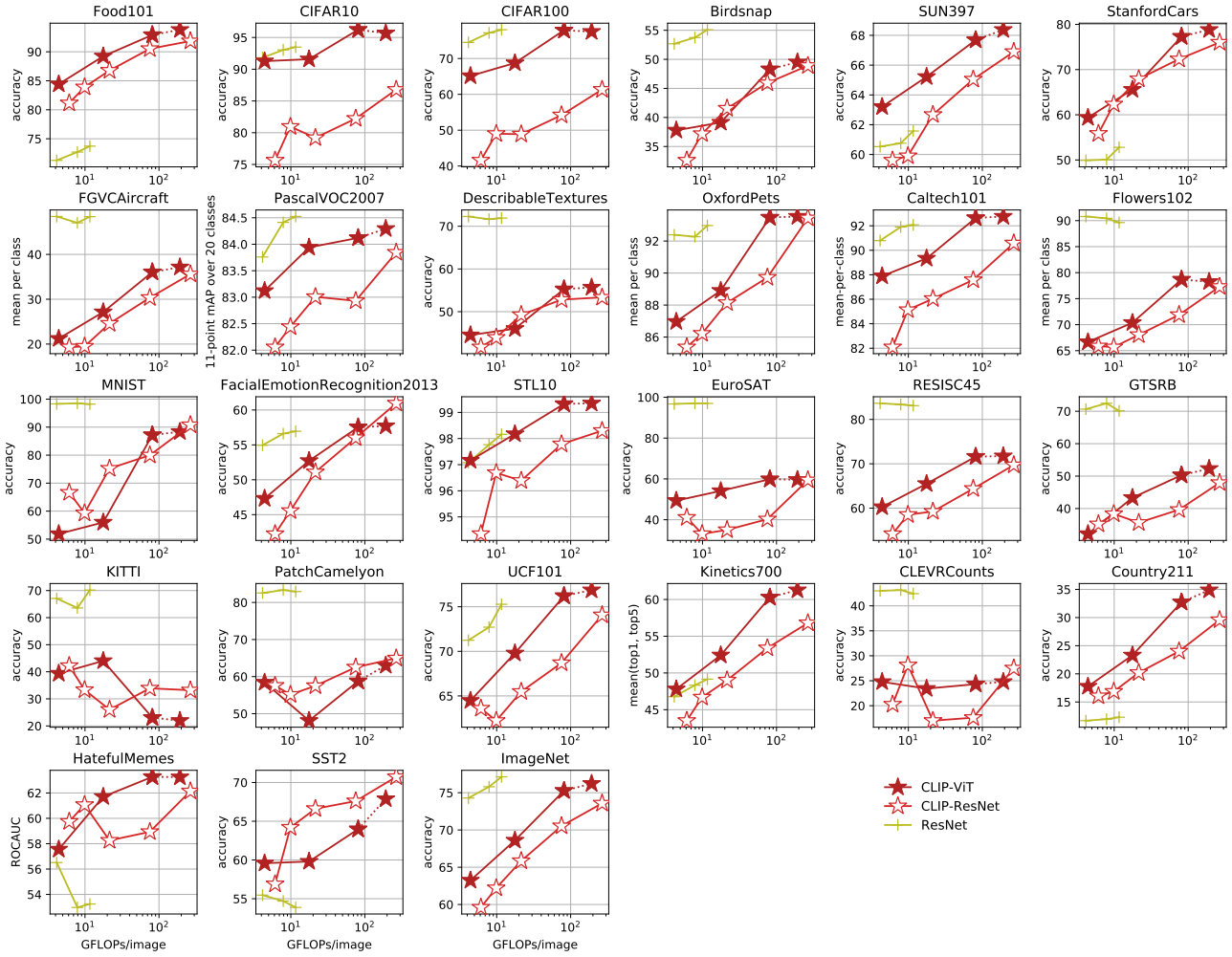


Figure 22. CLIP's zero-shot performance compared to linear-probe ResNet performance

		Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Stanford Cars	FGVC Aircraft	VOC2007	DTD	Oxford Pets	Caltech101	Flowers102	MNIST	FER2013	STL10	EuroSAT	RESISC45	GTSRB	KITTI	Pcam	UCF101	Kinetics700	CLEVR Counts	Country211	HatefulMemes	SST2	ImageNet	mean
RN	RN50	81.1	75.6	41.6	32.6	59.6	55.8	19.3	82.1	41.7	85.4	82.1	65.9	66.6	42.2	94.3	41.1	54.2	35.2	42.2	16.1	57.6	63.6	43.5	20.3	59.7	56.9	59.6
	RN101	83.9	81.0	49.0	37.2	59.9	62.3	19.5	82.4	43.9	86.2	85.1	65.7	59.3	45.6	96.7	33.1	58.5	38.3	33.3	16.9	55.2	62.2	46.7	28.1	61.1	64.2	62.2
	RN50x4	86.8	79.2	48.9	41.6	62.7	67.9	24.6	83.0	49.3	88.1	86.0	68.0	75.2	51.1	96.4	35.0	59.2	35.7	26.0	20.2	57.5	65.5	49.0	17.0	58.3	66.6	65.8
	RN50x16	90.5	82.2	54.2	45.9	65.0	72.3	30.3	82.9	52.8	89.7	87.6	71.9	80.0	56.0	97.8	40.3	64.4	39.6	33.9	24.0	62.5	68.7	53.4	17.6	58.9	67.6	70.5
	RN50x64	91.8	86.8	61.3	48.9	66.9	76.0	35.6	83.8	53.4	93.4	90.6	77.3	90.8	61.0	98.3	59.4	69.7	47.9	33.2	29.6	65.0	74.1	56.8	27.5	62.1	70.7	73.6
ViT	B/32	84.4	91.3	65.1	37.8	63.2	59.4	21.2	83.1	44.5	87.0	87.9	66.7	51.9	47.3	97.2	49.4	60.3	32.2	39.4	17.8	58.4	64.5	47.8	24.8	57.6	59.6	63.2
	B/16	89.2	91.6	68.7	39.1	65.2	65.6	27.1	83.9	46.0	88.9	89.3	70.4	56.0	52.7	98.2	54.1	65.5	43.3	44.0	23.3	48.1	69.8	52.4	23.4	61.7	59.8	68.6
	L/14	92.9	96.2	77.9	48.3	67.7	77.3	36.1	84.1	55.3	93.5	92.6	78.7	87.2	57.5	99.3	59.9	71.6	50.3	23.1	32.7	58.8	76.2	60.3	24.3	63.3	64.0	75.3
	L/14-336px	93.8	95.7	77.5	49.5	68.4	78.8	37.2	84.3	55.7	93.5	92.8	78.3	88.3	57.7	99.4	59.6	71.7	52.3	21.9	34.9	63.0	76.9	61.3	24.8	63.3	67.9	76.2

表11. CLIP 模型在 27 个数据集上的零样本性能。

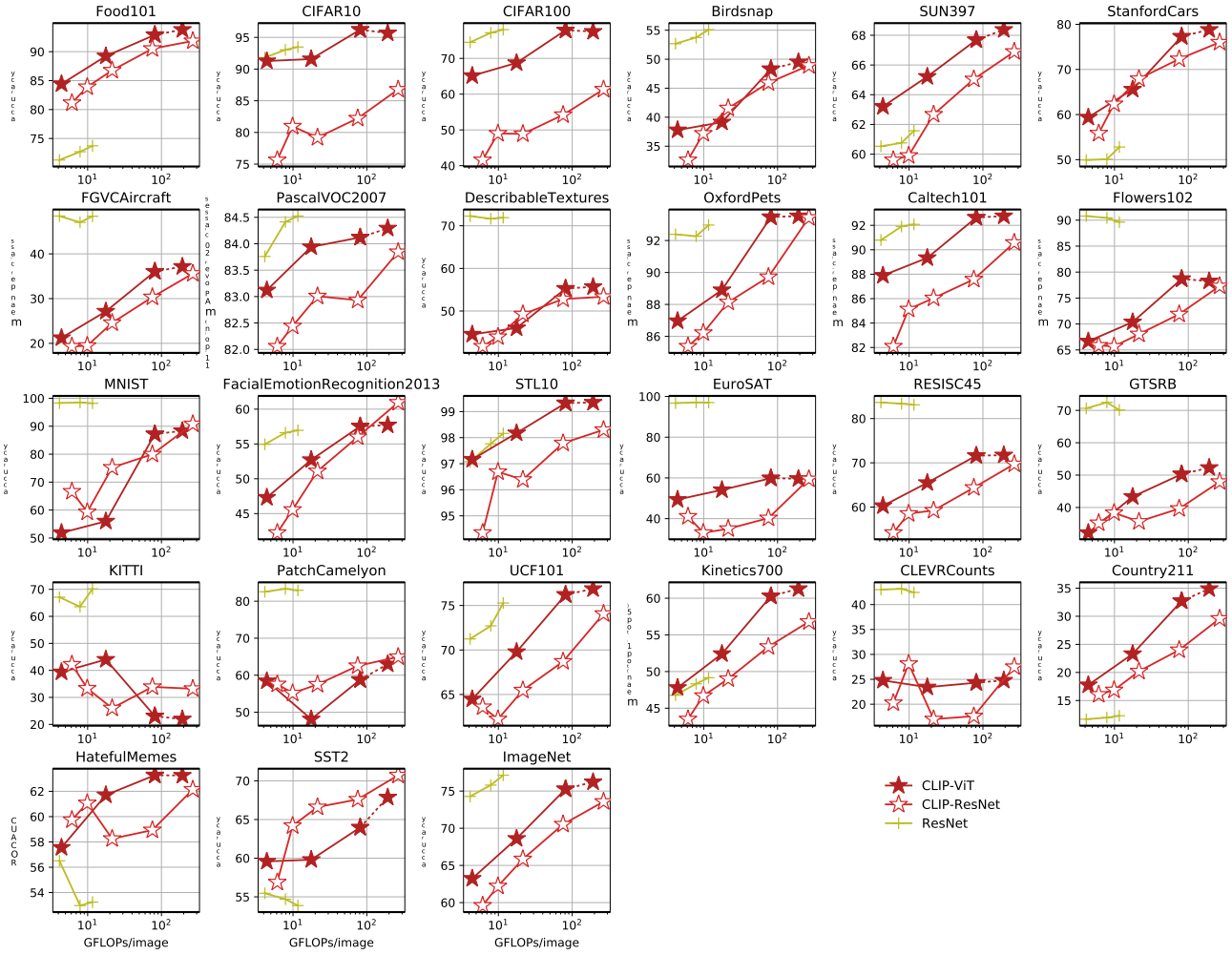


图 22. CLIP 的零样本性能与线性探测 ResNet 性能的比较

B. Zero-Shot Prediction

To provide a qualitative summary / overview of CLIP's zero-shot performance we visualize a randomly selected prediction for 36 different zero-shot CLIP classifiers in Figure 21. In addition, Table 11 and Figure 22 show the individual zero-shot performance scores for each dataset.

C. Duplicate Detector

Our early attempts at duplicate detection and analysis used nearest neighbors in the model's learned embedding space. While it is intuitive to use a model's own notion of similarity, we encountered issues. We found the model's feature space is weighted very heavily towards semantic similarity. Many false positives occurred due to distinct objects that would be described similarly (soccer balls, flowers of the same species, etc...) having almost perfect similarity. We also observed the model was quite poor at assigning certain kinds of near-duplicates high similarity scores. We noticed repeatedly that images with high-frequency textures (such as fur or stripe patterns) pre-processed by different resizing algorithms (nearest neighbor vs bi-linear) could have surprisingly low similarity. This resulted in many false negatives.

We built our own near-duplicate detector to fix this issue. We created a synthetic data augmentation pipeline that combined a variety of common image manipulations. The augmentation pipeline combines random cropping and zooming, aspect ratio distortion, downsizing and upscaling to different resolutions, minor rotations, jpeg compression, and HSV color jitter. The pipeline also randomly selects from different interpolation algorithms for all relevant steps. We then trained a model to maximize the similarity of an image and its transformed variant while minimizing similarity to all other images in a training batch. We used the same n-pair / InfoNCE loss as CLIP but with a fixed temperature of 0.07.

We selected a ResNet-50 as the model architecture. We modified the base ResNet-50 with the anti-alias improvements from (Zhang, 2019) and used weight norm (Salimans & Kingma, 2016) instead of batch norm (Ioffe & Szegedy, 2015) to avoid leaking information about duplicates via batch statistics - a problem previously noted in (Henaff, 2020). We also found the GELU activation function (Hendrycks & Gimpel, 2016) to perform better for this task. We trained the model with a total batch size of 1,712 for approximately 30 million images sampled from our pre-training dataset. At the end of training it achieves nearly 100% accuracy on its proxy training task.

	Linear Classifier			*Zero Shot*		
Dataset	YFCC	WIT	Δ	YFCC	WIT	Δ
Birdsnap	47.4	35.3	+12.1	19.9	4.5	+15.4
Country211	23.1	17.3	+5.8	5.2	5.3	+0.1
Flowers102	94.4	89.8	+4.6	48.6	21.7	+26.9
GTSRB	66.8	72.5	−5.7	6.9	7.0	−0.1
UCF101	69.2	74.9	−5.7	22.9	32.0	−9.1
Stanford Cars	31.4	50.3	−18.9	3.8	10.9	−7.1
ImageNet	**62.0**	60.8	+1.2	**31.3**	27.6	+3.7
Dataset Average	65.5	**66.6**	−1.1	29.6	**30.0**	−0.4
Dataset "Wins"	10	**15**	−5	**19**	18	+1

Table 12. **CLIP performs similarly when trained on only YFCC100M.** Comparing a ResNet-50 trained on only YFCC100M with a same sized subset of WIT shows similar average performance and number of wins on zero shot and linear classifier evals. However, large differences in dataset specific performance occur. We include performance on the 3 datasets where YFCC does best and worst compared to WIT according to a linear probe in order to highlight this as well as aggregate performance across all linear and zero-shot evals and the canonical ImageNet dataset.

D. Dataset Ablation on YFCC100M

To study whether our custom dataset is critical to the performance of CLIP, we trained a model on a filtered subset of the YFCC100M dataset (details described in Section 2.2) and compared its performance to the same model trained on an equally sized subset of WIT. We train each model for 32 epochs at which point transfer performance begins to plateau due to overfitting. Results are shown in Table 12. Across our whole eval suite, YFCC and WIT perform similarly on average for both zero-shot and linear probe settings. However, performance on specific fine-grained classification datasets can vary widely - sometimes by over 10%. Our speculation is that these differences in performance reflect the relative density of relevant data in each pre-training dataset. For instance, pre-training on YFCC100M, which might contain many photos of birds and flowers (common subjects for photographers), results in better performance on Birdsnap and Flowers102, while pre-training on WIT results in better car and pet classifiers (which appear common in our dataset).

Overall, these results are encouraging as they suggest our approach can use any reasonably filtered collection of paired (text, image) data. This mirrors recent work which reported positive results using the same contrastive pre-training objective on the relatively different domain of medical imaging (Zhang et al., 2020). It also is similar to the findings of noisy student self-training which reported only slight improvements when using their JFT300M dataset over YFCC100M (Xie et al., 2020). We suspect the major advantage of our dataset over the already existing YFCC100M is its much larger size.

B. 零样本预测

为了对 CLIP 的零样本性能进行定性总结/概览，我们在图 21 中可视化了 36 个不同零样本 CLIP 分类器的随机选取预测结果。此外，表 11 和图 22 展示了每个数据集的单独零样本性能得分。

C. 重复检测器

我们早期的重复检测与分析尝试使用了模型学习到的嵌入空间中的最近邻。虽然使用模型自身对相似性的理解很直观，但我们遇到了一些问题。我们发现，模型的特征空间非常偏向语义相似性。许多误报是由于一些不同的物体会被以相似的方式描述（如足球、同一物种的花等），从而具有几乎完美的相似度。我们还观察到，模型在为某些类型的近重复样本分配高相似度分数方面表现相当差。我们反复注意到，具有高频纹理的图像（如毛发或条纹图案）如果经过不同缩放算法（最近邻与双线性）预处理，可能会得到出人意料地低的相似度。这导致了许多漏报。

我们构建了自己的近重复检测器来解决这个问题。我们创建了一条合成数据增强流水线，结合了多种常见的图像处理操作。该增强流水线结合了随机裁剪和缩放、宽高比扭曲、降采样和上采样到不同分辨率、轻微旋转、JPEG 压缩以及 HSV 颜色抖动。该流水线还会在所有相关步骤中随机选择不同的插值算法。随后，我们训练了一个模型，使图像及其变换版本之间的相似度最大化，同时使其与训练批次中所有其他图像的相似度最小化。我们使用了与 CLIP 相同的 n-pair / InfoNCE 损失，但温度固定为 0.07。

我们选择 ResNet-50 作为模型架构。我们采用（Zhang, 2019）中的抗混叠改进对基础 ResNet-50 进行了修改，并使用权重归一化（Salimans & Kingma, 2016）代替批归一化（Ioffe & Szegedy, 2015），以避免通过批统计信息泄露有关重复样本的信息——这一问题此前已在（Henaff, 2020）中指出。我们还发现，GELU 激活函数（Hendrycks & Gimpel, 2016）在该任务上表现更好。我们使用总批量大小为 1,712，对从预训练数据集中采样的约 3000 万张图像训练了该模型。在训练结束时，它在其代理训练任务上达到了近 100% 的准确率。

	Linear Classifier			*Zero Shot*		
Dataset	YFCC	WIT	Δ	YFCC	WIT	Δ
Birdsnap	47.4	35.3	+12.1	19.9	4.5	+15.4
Country211	23.1	17.3	+5.8	5.2	5.3	+0.1
Flowers102	94.4	89.8	+4.6	48.6	21.7	+26.9
GTSRB	66.8	72.5	−5.7	6.9	7.0	−0.1
UCF101	69.2	74.9	−5.7	22.9	32.0	−9.1
Stanford Cars	31.4	50.3	−18.9	3.8	10.9	−7.1
ImageNet	**62.0**	60.8	+1.2	**31.3**	27.6	+3.7
Dataset Average	65.5	**66.6**	−1.1	29.6	**30.0**	−0.4
Dataset “Wins”	10	**15**	−5	**19**	18	+1

表 12. 仅在 YFCC100M 上训练时，CLIP 的表现相近。将仅在 YFCC100M 上训练的 ResNet-50 与 WIT 中相同规模的子集进行比较，显示其在零样本和线性分类器评估中的平均性能和获胜次数相近。然而，在特定数据集上的性能存在较大差异。我们列出了根据线性探针评估，YFCC 相较于 WIT 表现最好和最差的 3 个数据集上的性能，以突出这一点，同时也列出了所有线性和零样本评估的总体性能以及标准 ImageNet 数据集上的性能。

D. YFCC100M 上的数据集消融

为了研究我们的自定义数据集是否对 CLIP 的性能至关重要，我们在 YFCC100M 数据集的一个过滤子集上训练了一个模型（细节见第 2.2 节），并将其性能与在同等规模的 WIT 子集上训练的相同模型进行了比较。我们将每个模型训练 32 个 epoch，此时迁移性能由于过拟合而开始趋于平稳。结果如表 12 所示。在我们的整个评估套件中，YFCC 和 WIT 在零样本和线性探测设置下的平均表现相近。然而，在特定的细粒度分类数据集上的性能可能差异很大——有时超过 10%。我们的推测是，这些性能差异反映了每个预训练数据集中相关数据的相对密度。例如，在 YFCC100M 上进行预训练可能包含许多鸟类和花卉的照片（摄影师常拍的主题），因此在 Birdsnap 和 Flowers102 上表现更好；而在 WIT 上进行预训练则能得到更好的汽车和宠物分类器（这些类别似乎在我们的数据集中很常见）。

总体而言，这些结果令人鼓舞，因为它们表明我们的方法可以使用任何经过合理过滤的成对（文本、图像）数据集合。这与近期的一项工作相呼应，该工作在医学影像这一相对不同的领域中使用相同的对比预训练目标并报告了积极结果（Zhang et al., 2020）。这也类似于噪声学生自训练的发现，该方法报告称，与使用 YFCC100M 相比，使用其 JFT300M 数据集仅带来了轻微改进（Xie et al., 2020）。我们认为，我们的数据集相较于已有的 YFCC100M 的主要优势在于其规模大得多。

Finally, we caution that WIT includes this filtered subset of YFCC100M. This could result in our ablation underestimating the size of performance differences between YFCC100M and the rest of WIT. We do not think this is likely as YFCC100M is only 3.7% of the overall WIT data blend and it did not noticeably change the performance of models when it was added to the existing data blend during the creation of WIT.

E. Selected Task and Dataset Results

Due to the large variety of datasets and experiments considered in this work, the main body focuses on summarizing and analyzing overall results. In the following subsections we report details of performance for specific groups of tasks, datasets, and evaluation settings.

E.1. Image and Text Retrieval

CLIP pre-trains for the task of image-text retrieval on our noisy web-scale dataset. Although the focus of this paper is on representation learning and task learning for the purpose of transfer to a wide variety of downstream datasets, validating that CLIP is able to achieve high transfer performance transfer on exactly what it is pre-trained for is an important sanity check / proof of concept. In Table 13 we check the zero-shot transfer performance of CLIP for both text and image retrieval on the Flickr30k and MSCOCO datsets. Zero-shot CLIP matches or outperforms all prior zero-shot results on these two datasets. Zero-shot CLIP is also competitive with the current overall SOTA for the task of text retrieval on Flickr30k. On image retrieval, CLIP's performance relative to the overall state of the art is noticeably lower. However, zero-shot CLIP is still competitive with a fine-tuned Unicoder-VL. On the larger MS-COCO dataset fine-tuning improves performance significantly and zero-shot CLIP is not competitive with the most recent work. For both these datasets we prepend the prompt "`a photo of`" to the description of each image which we found boosts CLIP's zero-shot R@1 performance between 1 and 2 points.

E.2. Optical Character Recognition

Although visualizations have shown that ImageNet models contain features that respond to the presence of text in an image (Zeiler & Fergus, 2014), these representations are not sufficiently fine-grained to use for the task of optical character recognition (OCR). To compensate, models are augmented with the outputs of custom OCR engines and features to boost performance on tasks where this capability is required (Singh et al., 2019; Yang et al., 2020). Early during the development of CLIP, we noticed that CLIP began to learn primitive OCR capabilities which appeared to steadily improve over the course of the project. To evaluate this qualitatively noticed behavior, we measured performance on 5 datasets requiring the direct and indirect use of OCR. Three of these datasets MNIST (LeCun), SVHN (Netzer et al., 2011), and IIIT5K (Mishra et al., 2012) directly check the ability of a model to perform low-level character and word recognition, while Hateful Memes (Kiela et al., 2020) and SST-2 (Socher et al., 2013) check the ability of a model to use OCR to perform a semantic task. Results are reported in Table 14.

CLIP's performance is still highly variable and appears to be sensitive to some combination of the domain (rendered or natural images) and the type of text to be recognized (numbers or words). CLIP's OCR performance is strongest Hateful Memes and SST-2 - datasets where the text is digitally rendered and consists mostly of words. On IIIT5K, which is natural images of individually cropped words, zero-shot CLIP performs a bit more respectively and its performance is similar to Jaderberg et al. (2014) early work combining deep learning and structured prediction to perform open-vocabulary OCR. However, performance is noticeably lower on two datasets involving recognition of hand written and street view numbers. CLIP's 51% accuracy on full number SVHN is well below any published results. Inspection suggests CLIP struggles with repeated characters as well as the low resolution and blurry images of SVHN. CLIP's zero-shot MNIST performance is also poor and is outperformed by supervised logistic regression on raw pixels, one of the simplest possible machine learning baselines.

SST-2 is a sentence level NLP dataset which we render into images. We include SST-2 in order to check whether CLIP is able to convert low level OCR capability into a higher level representation. Fitting a linear classifier on CLIP's representation of rendered sentences achives 80.5% accuracy. This is on par with the 80% accuracy of a continuous bag of words baseline using GloVe word vectors pre-trained on 840 billion tokens (Pennington et al., 2014). While this is a simple NLP baseline by today's standard, and well below the 97.5% of the current SOTA, it is encouraging to see that CLIP is able to turn an image of rendered text into a non-trivial sentence level representation. Fully supervised CLIP is also surprisingly strong on Hateful Meme detection, where CLIP is only 0.7 points behind the current single model SOTA and several points above the best baseline from the original paper. Similar to SST-2, these other results on Hateful Memes use the ground truth text which CLIP does not have access to. Finally, we note that zero-shot CLIP outperforms the best results using fully supervised linear probes across all other 56 models included in our evaluation suite. This suggests CLIP's OCR capability is at least somewhat unique compared to existing work on self-supervised and supervised representation learning.

最后，我们提醒注意，WIT 包含了 YFCC100M 的这一过滤子集。这可能导致我们的消融实验低估了 YFCC100M 与 WIT 其余部分之间的性能差异大小。我们认为这种情况不太可能，因为 YFCC100M 仅占 WIT 整体数据混合的 3.7%，并且在创建 WIT 期间将其加入现有数据混合时，并未明显改变模型的性能。

E. 选定任务和数据集结果

由于本文考虑的数据集和实验种类繁多，正文主要侧重于总结和分析总体结果。在以下小节中，我们报告特定任务组、数据集和评估设置的性能细节。

E.1. 图像和文本检索

CLIP 在我们的噪声网络规模数据集上针对图像-文本检索任务进行预训练。尽管本文的重点是表示学习和任务学习，目的是迁移到各种下游数据集，但验证 CLIP 能否在其预训练任务本身上实现较高的迁移性能，是一项重要的合理性检查/概念验证。在表 13 中，我们检查了 CLIP 在 Flickr30k 和 MSCOCO 数据集上用于文本检索和图像检索的零样本迁移性能。零样本 CLIP 在这两个数据集上达到或超过了此前所有零样本结果。零样本 CLIP 在 Flickr30k 的文本检索任务上也能与当前总体最先进水平相竞争。在图像检索方面，CLIP 相对于总体最先进水平的性能明显较低。不过，零样本 CLIP 仍可与经过微调的 Unicoder-VL 相竞争。在更大的 MS-COCO 数据集上，微调能显著提升性能，而零样本 CLIP 无法与最新工作相竞争。对于这两个数据集，我们都会在每张图像的描述前添加提示语“a photo of ”，我们发现这能将 CLIP 的零样本 R@1 性能提升 1 到 2 个百分点。

E.2. 光学字符识别

尽管可视化结果显示，ImageNet 模型包含会对图像中文本的存在作出响应的特征（Zeiler & Fergus，2014），但这些表示还不够细粒度，无法用于光学字符识别（OCR）任务。为了弥补这一点，模型会加入定制 OCR 引擎的输出和相关特征，以提升在需要这种能力的任务上的性能（Singh 等，2019；Yang 等，2020）。在 CLIP 开发的早期，我们注意到 CLIP 开始学习初级的 OCR 能力，而且这种能力似乎在整个项目过程中稳步提升。为了评估这种定性观察到的行为，我们测量了性能在 5 个需要直接和间接使用 OCR 的数据集上。其中三个数据集 MNIST（LeCun）、SVHN（Netzer 等，2011）和 IIIT5K（Mishra 等，2012）直接检验模型执行低级字符和单词识别的能力，而 Hateful Memes（Kiela 等，2020）和 SST-2（Socher 等，2013）则检验模型使用 OCR 执行语义任务的能力。结果报告见表 14。

CLIP 的性能仍然高度不稳定，并且似乎对领域（渲染图像或自然图像）与待识别文本类型（数字或单词）的某种组合较为敏感。CLIP 的 OCR 性能在 Hateful Memes 和 SST-2 上最强——这些数据集中的文本是数字渲染的，并且主要由单词组成。在 IIIT5K 上，即由单独裁剪出的单词自然图像组成的数据集，零样本 CLIP 的表现略好一些，其性能类似于 Jaderberg 等人（2014）将深度学习与结构化预测相结合以执行开放词汇 OCR 的早期工作。然而，在涉及手写数字和街景数字识别的两个数据集上，其性能明显较低。CLIP 在完整数字 SVHN 上 51% 的准确率远低于任何已发表的结果。检查表明，CLIP 难以处理重复字符，以及 SVHN 中低分辨率且模糊的图像。CLIP 的零样本 MNIST 性能也很差，并且被在原始像素上训练的监督式逻辑回归超越，后者是最简单的机器学习基线之一。

SST-2 是一个句子级别的 NLP 数据集，我们将其渲染成图像。我们纳入 SST-2，是为了检查 CLIP 是否能够将低层次的 OCR 能力转化为更高层次的表示。在 CLIP 对渲染句子的表示上拟合一个线性分类器，可达到 80.5% 的准确率。这与使用在 8400 亿个词元上预训练的 GloVe 词向量的连续词袋基线 80% 的准确率相当（Pennington 等，2014）。尽管按照今天的标准来看，这是一个简单的 NLP 基线，并且远低于当前 SOTA 的 97.5%，但看到 CLIP 能够将渲染文本的图像转化为非平凡的句子级表示，仍然令人鼓舞。全监督的 CLIP 在仇恨 Meme 检测上也出人意料地强，其中 CLIP 仅比当前单模型 SOTA 低 0.7 个百分点，并且比原论文中的最佳基线高出几个百分点。与 SST-2 类似，这些在 Hateful Memes 上的其他结果使用了真实文本，而 CLIP 无法访问这些文本。最后，我们注意到，在我们的评估套件所包含的所有其他 56 个模型中，零样本 CLIP 的表现均优于使用全监督线性探针得到的最佳结果。这表明，与现有的自监督和监督表示学习工作相比，CLIP 的 OCR 能力至少在某种程度上是独特的。

		Text Retrieval						Image Retrieval					
		Flickr30k			MSCOCO			Flickr30k			MSCOCO		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Finetune	Unicoder-VL[a]	86.2	96.3	99.0	62.3	87.1	92.8	71.5	90.9	94.9	46.7	76.0	85.3
	Uniter[b]	87.3	98.0	99.2	65.7	88.6	93.8	75.6	94.1	**96.8**	52.9	79.9	88.0
	VILLA[c]	87.9	97.5	98.8	-	-	-	76.3	**94.2**	**96.8**	-	-	-
	Oscar[d]	-	-	-	**73.5**	**92.2**	**96.0**	-	-	-	**57.5**	**82.8**	**89.8**
	ERNIE-ViL[e]	**88.7**	98.0	99.2	-	-	-	**76.7**	93.6	96.4	-	-	-
Zero-Shot	Visual N-Grams[f]	15.4	35.7	45.1	8.7	23.1	33.3	8.8	21.2	29.9	5.0	14.5	21.9
	ImageBERT[g]	-	-	-	44.0	71.2	80.4	-	-	-	32.3	59.0	70.2
	Unicoder-VL[a]	64.3	86.8	92.3	-	-	-	48.4	76.0	85.2	-	-	-
	Uniter[b]	83.6	95.7	97.7	-	-	-	68.7	89.2	93.9	-	-	-
	CLIP	88.0	**98.7**	**99.4**	58.4	81.5	88.1	68.7	90.6	95.2	37.8	62.4	72.2

Table 13. **CLIP improves zero-shot retrieval and is competitive with the best fine-tuned result on Flickr30k text retrieval.** Bold indicates best overall performance while an underline indicates best in category performance (zero-shot or fine-tuned). For all other models, best results from the paper are reported regardless of model size / variant. MSCOCO performance is reported on the 5k test set. [a](Li et al., 2020a) [b](Chen et al., 2019) [c](Gan et al., 2020) [d](Li et al., 2020b) [e](Yu et al., 2020) [f](Li et al., 2017) [g](Qi et al., 2020)

		MNIST	SVHN	IIIT5K 1k	Hateful Memes	SST-2
Finetune	SOTA	**99.8**[a]	**96.4**[b]	**98.9**[c]	**78.0**[d]	**97.5**[e]
	JOINT[f]	-	-	89.6	-	-
	CBoW[g]	-	-	-	-	80.0
Linear	Raw Pixels	92.5	-	-	-	-
	ES Best	98.9[h]	-	-	58.6[h]	59.0[i]
	CLIP	99.2	-	-	77.3	80.5
ZS	CLIP	88.4	51.0	90.0	63.3	67.9

Table 14. OCR performance on 5 datasets. All metrics are accuracy on the test set except for Hateful Memes which reports ROC AUC on the dev set. Single model SOTA reported to best of knowledge. *ES Best* reports the best performance across the 56 non-CLIP models in our evaluation suite. [a](Assiri, 2020) [b](Jaderberg et al., 2015) [c](Wang et al., 2020) [d](Lippe et al., 2020) [f](Jaderberg et al., 2014) [g](Wang et al., 2018) [h](Xie et al., 2020) [i](Mahajan et al., 2018)

		UCF101	K700	RareAct	
		Top-1	AVG	mWAP	mWSAP
Finetune	R(2+1)D-BERT[a]	**98.7**	-	-	-
	NS ENet-L2[b]	-	**84.8**	-	-
	HT100M S3D[d]	91.3	-	-	-
	Baseline I3D[e]	-	70.2	-	-
Linear	MMV FAC[f]	91.8	-	-	-
	NS ENet-L2[c]	89.4[c]	68.2[c]	-	-
	CLIP	92.0	73.0	-	-
ZS	HT100M S3D[d]	-	-	30.5	34.8
	CLIP	80.3	69.6	**40.7**	**44.8**

Table 15. Action recognition performance on 3 video datasets. Single model SOTA reported to best of knowledge. Note that *linear CLIP* and *linear NS ENet-L2* are trained and evaluated on a single frame subsampled version of each dataset and not directly comparable to prior work. On Kinetics-700, we report the ActivityNet competition metric which is the average of top-1 and top-5 performance. [a](Kalfaoglu et al., 2020) [b](Lu et al., 2020) [c](Xie et al., 2020) [d](Miech et al., 2020b) [e](Carreira et al., 2019) [f](Alayrac et al., 2020)

E.3. Action Recognition in Videos

For the purpose of learning, a potentially important aspect of natural language is its ability to express, and therefore supervise, an extremely wide set of concepts. A CLIP model, since it is trained to pair semi-arbitrary text with images, is likely to receive supervision for a wide range of visual concepts involving both common and proper nouns, verbs, and adjectives. ImageNet-1K, by contrast, only labels common nouns. Does the lack of broader supervision in ImageNet result in weaker transfer of ImageNet models to tasks involving the recognition of visual concepts that are not nouns?

To investigate this, we measure and compare the performance of CLIP and ImageNet models on several video action classification datasets which measure the ability of a model to recognize verbs. In Table 15 we report results on UCF-101 (Soomro et al., 2012) and Kinetics-700 (Carreira et al., 2019), two common datasets for the task. Unfortunately, our CPU based linear classifier takes a prohibitively long time to evaluate on a video dataset due to the very large number of training frames. To deal with this, we aggressively sub-sample each video to only a single center frame, effectively turning it into an image classification dataset. As a result, our reported performance in a linear evaluation setting likely under estimates performance by a moderate amount.

		Text Retrieval						Image Retrieval					
		Flickr30k			MSCOCO			Flickr30k			MSCOCO		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Fine-tune	Unicoder-VL[a]	86.2	96.3	99.0	62.3	87.1	92.8	71.5	90.9	94.9	46.7	76.0	85.3
	Uniter[b]	87.3	98.0	99.2	65.7	88.6	93.8	75.6	94.1	**96.8**	52.9	79.9	88.0
	VILLA[c]	87.9	97.5	98.8	-	-	-	76.3	**94.2**	**96.8**	-	-	-
	Oscar[d]	-	-	-	**73.5**	**92.2**	**96.0**	-	-	-	**57.5**	**82.8**	**89.8**
	ERNIE-ViL[e]	**88.7**	98.0	99.2	-	-	-	**76.7**	93.6	96.4	-	-	-
Zero-Shot	Visual N-Grams[f]	15.4	35.7	45.1	8.7	23.1	33.3	8.8	21.2	29.9	5.0	14.5	21.9
	ImageBERT[g]	-	-	-	44.0	71.2	80.4	-	-	-	32.3	59.0	70.2
	Unicoder-VL[a]	64.3	86.8	92.3	-	-	-	48.4	76.0	85.2	-	-	-
	Uniter[b]	83.6	95.7	97.7	-	-	-	68.7	89.2	93.9	-	-	-
	CLIP	88.0	**98.7**	**99.4**	58.4	81.5	88.1	68.7	90.6	95.2	37.8	62.4	72.2

表 13. CLIP 提升了零样本检索性能，并且在 Flickr30k 文本检索上可与最佳微调结果相媲美。粗体表示总体最佳性能，下划线表示类别内最佳性能（零样本或微调）。对于所有其他模型，均报告论文中的最佳结果，而不考虑模型大小/变体。MSCOCO 性能在 5k 测试集上报告。[a](Li 等，2020a) [b](Chen 等，2019) [c](Gan 等，2020) [d](Li 等，2020b) [e](Yu 等，2020) [f](Li 等，2017) [g](Qi 等，2020)

		MNIST	SVHN	IIIT5K 1k	Hateful Memes	SST-2
Fine-tune	SOTA	**99.8**[a]	**96.4**[b]	**98.9**[c]	**78.0**[d]	**97.5**[e]
	JOINT[f]	-	-	89.6	-	-
	CBoW[g]	-	-	-	-	80.0
Transfer	Raw Pixels	92.5	-	-	-	-
	ES Best	98.9[h]	-	-	58.6[h]	59.0[i]
	CLIP	99.2	-	-	77.3	80.5
Zero-Shot	CLIP	88.4	51.0	90.0	63.3	67.9

表 14. 5 个数据集上的 OCR 性能。除 Hateful Memes 报告开发集上的 ROC AUC 外，所有指标均为测试集上的准确率。单模型 SOTA 为据我们所知的最佳结果。*ES Best* 报告了我们评估套件中 56 个非 CLIP 模型的最佳性能。[a](Assiri，2020) [b](Jaderberg 等，2015) [c](Wang 等，2020) [d](Lippe 等，2020) [f](Jaderberg 等，2014) [g](Wang 等，2018) [h](Xie 等，2020) [i](Mahajan 等，2018)

		UCF101 Top-1	K700 AVG	RareAct mWAP	RareAct mWSAP
Fine-tune	R(2+1)D-BERT[a]	**98.7**	-	-	-
	NS ENet-L2[b]	-	**84.8**	-	-
	HT100M S3D[d]	91.3	-	-	-
	Baseline I3D[e]	-	70.2	-	-
Transfer	MMV FAC[f]	91.8	-	-	-
	NS ENet-L2[c]	89.4[c]	68.2[c]	-	-
	CLIP	92.0	73.0	-	-
Zero-Shot	HT100M S3D[d]	-	-	30.5	34.8
	CLIP	80.3	69.6	**40.7**	**44.8**

表15. 3个视频数据集上的动作识别性能。据我们所知，报告的是单模型SOTA。请注意，*linear CLIP* 和 *linear NS ENet-L2* 是在每个数据集的单帧子采样版本上训练和评估的，因此不能与先前工作直接比较。在 Kinetics-700 上，我们报告 ActivityNet 竞赛指标，即 top-1 和 top-5 性能的平均值。[a](Kalfaoglu 等，2020) [b](Lu 等，2020) [c](Xie 等，2020) [d](Miech 等，2020b) [e](Carreira 等，2019) [f](Alayrac 等，2020)

E.3. 视频中的动作识别

从学习的角度来看，自然语言一个潜在的重要方面在于它能够表达并因此监督极其广泛的概念。由于 CLIP 模型被训练来将半任意文本与图像配对，它很可能会获得涉及普通名词和专有名词、动词以及形容词的广泛视觉概念的监督。相比之下，ImageNet-1K 只标注普通名词。那么，ImageNet 中缺乏更广泛的监督，是否会导致 ImageNet 模型在迁移到涉及识别非名词视觉概念的任务时表现较弱?

为了研究这一点，我们测量并比较了 CLIP 和 ImageNet 模型在多个视频上的性能

动作分类数据集用于衡量模型识别动词的能力。在表 15 中，我们报告了在 UCF-101（Soomro 等，2012）和 Kinetics-700（Carreira 等，2019）上的结果，这两个是该任务常用的数据集。不幸的是，由于训练帧数量非常庞大，我们基于 CPU 的线性分类器在视频数据集上进行评估需要耗费过长时间。为了解决这一问题，我们对每个视频进行了激进的子采样，仅保留单个中心帧，从而有效地将其转化为图像分类数据集。因此，我们在线性评估设置下报告的性能很可能在一定程度上低估了实际性能。

	IN Top-1	IN-V2 Top-1	IN-A Top-1	IN-R Top-1	ObjectNet Top-1	IN-Sketch Top-1	IN-Vid PM0	IN-Vid PM10	YTBB PM0	YTBB PM10
NS EfficientNet-L2[a]	**88.3**	**80.2**	**84.9**	74.7	68.5	47.6	88.0	82.1	67.7	63.5
FixResNeXt101-32x48d V2[b]	86.4	78.0	68.4	80.0	57.8	59.1	85.8	72.2	68.9	57.7
Linear Probe CLIP	85.4	75.9	75.3	84.2	66.2	57.4	89.1	77.2	68.7	63.1
Zero-Shot CLIP	76.2	70.1	77.2	**88.9**	**72.3**	**60.2**	**95.3**	**89.2**	**95.2**	**88.5**

Table 16. Detailed ImageNet robustness performance. IN is used to abbreviate for ImageNet. [a](Xie et al., 2020) [b](Touvron et al., 2019)

Despite this handicap, CLIP features transfer surprisingly well to this task. CLIP matches the best prior result on UCF-101 in a linear probe evaluation setting and also outperforms all other models in our evaluation suite. On Kinetics-700, CLIP also outperforms the fine-tuned I3D baseline from the original paper. Since it does not require a training stage, we report CLIP's zero-shot performance when averaging predictions across all frames. CLIP also performs well in this setting and on Kinetics-700 its performance is within 1% of the fully supervised I3D baseline which is trained on 545000 labeled videos. Encouraged by these results, we also measure CLIP's performance on the recently introduced RareAct dataset (Miech et al., 2020a) which was designed to measure zero-shot recognition of unusual actions like "hammering a phone" and "drilling an egg". CLIP improves over the prior state of the art, a S3D model trained on automatically extracted captions from 100 million instructional videos, by 10 points.

While CLIP has encouragingly strong performance on the task of action recognition, we note that there are many differences between the models being compared beyond just their form of supervision such as model architecture, training data distribution, dataset size, and compute used. Further work is needed to more precisely determine what specific design decisions contribute to achieving high performance on this task.

	1km	25km	200km	750km	2500km
ISNs[a]	**16.9**	**43.0**	**51.9**	**66.7**	**80.2**
CPlaNet[b]	16.5	37.1	46.4	62.0	78.5
CLIP	13.9	32.9	43.0	62.0	79.3
Deep-Ret+[c]	14.4	33.3	47.7	61.6	73.4
PlaNet[d]	8.4	24.5	37.6	53.6	71.3

Table 17. Geolocalization performance on the IM2GPS test set. Metric is percent of images localized within a given radius. Models are ordered by average performance. [a](Muller-Budack et al., 2018) [b](Hongsuck Seo et al., 2018) [c](Vo et al., 2017) [c](Weyand et al., 2016)

E.4. Geolocalization

Another behavior we noticed during the development of CLIP was its ability to recognize many places and locations. To quantify this we created the Country211 dataset as described in Appendix A and report results on it throughout the paper. However it is a new benchmark so to compare with prior work on geolocalization we also report results on the IM2GPS test set from Hays & Efros (2008) in Table 17. Since IM2GPS is a regression benchmark, we guess the GPS coordinates of the nearest image in a set of reference images using CLIP's embedding space. This is not a zero-shot result since it uses nearest-neighbor regression. Despite querying only 1 million images, which is much less than prior work, CLIP performs similarly to several task specific models. It is not, however, competitive with the current state of the art.

E.5. Robustness to Distribution Shift

Section 3.3 provides a high level summary and analysis of ImageNet-related robustness results. We briefly provide some additional numerical details in this appendix. Performance results per dataset are provided in Table 16 and compared with the current state of the art results reported in Taori et al. (2020)'s evaluation suite. Zero-shot CLIP improves the state of the art on 5 of the 7 datasets, ImageNet-R, ObjectNet, ImageNet-Sketch, ImageNet-Vid, and Youtube-BB. CLIP's improvements are largest on ImageNet-Vid and Youtube-BB due to its flexible zero-shot capability and on ImageNet-R, which likely reflects CLIP's pre-training distribution including significant amounts of creative content. A similar behavior has been documented for the Instagram pre-trained ResNeXt models as discussed in Taori et al. (2020).

	IN Top-1	IN-V2 Top-1	IN-A Top-1	IN-R Top-1	ObjectNet Top-1	IN-Sketch Top-1	IN-Vid PM0	IN-Vid PM10	YTBB PM0	YTBB PM10
NS EfficientNet-L2[a]	**88.3**	**80.2**	**84.9**	74.7	68.5	47.6	88.0	82.1	67.7	63.5
FixResNeXt101-32x48d V2[b]	86.4	78.0	68.4	80.0	57.8	59.1	85.8	72.2	68.9	57.7
Linear Probe CLIP	85.4	75.9	75.3	84.2	66.2	57.4	89.1	77.2	68.7	63.1
Zero-Shot CLIP	76.2	70.1	77.2	**88.9**	**72.3**	**60.2**	**95.3**	**89.2**	**95.2**	**88.5**

表 16. 详细的 ImageNet 鲁棒性性能。IN 用作 ImageNet 的缩写。[a](Xie 等，2020) [b](Touvron 等，2019)

尽管存在这一缺陷，CLIP 特征迁移到该任务上的效果却出人意料地好。在使用线性探针的评估设置下，CLIP 在 UCF-101 上达到了此前最佳结果，并且在我们的评估套件中也优于所有其他模型。在 Kinetics-700 上，CLIP 也超过了原论文中经过微调的 I3D 基线。由于它不需要训练阶段，我们报告了 CLIP 在对所有帧的预测取平均时的零样本性能。CLIP 在这一设置下同样表现良好，并且在 Kinetics-700 上，其性能与在 545000 个带标签视频上训练的全监督 I3D 基线相差不到 1%。受这些结果鼓舞，我们还测量了 CLIP 在最近提出的 Rare Act 数据集（Miech 等，2020a）上的性能，该数据集旨在衡量对“锤击手机”和“钻鸡蛋”等不寻常动作的零样本识别能力。与此前最先进的方法相比，CLIP 提高了 10 个百分点；该方法是一个 S3D 模型，使用从 1 亿个教学视频中自动提取的字幕进行训练。

尽管 CLIP 在动作识别任务上表现出了令人鼓舞的强劲性能，但我们注意到，被比较的模型之间除了监督形式之外，还存在许多差异，例如模型架构、训练数据分布、数据集规模以及所使用的计算资源。仍需进一步研究，以更精确地确定哪些具体设计决策有助于在该任务上实现高性能。

	1km	25km	200km	750km	2500km
ISNs[a]	**16.9**	**43.0**	**51.9**	**66.7**	**80.2**
CPlaNet[b]	16.5	37.1	46.4	62.0	78.5
CLIP	13.9	32.9	43.0	62.0	79.3
Deep-Ret+[c]	14.4	33.3	47.7	61.6	73.4
PlaNet[d]	8.4	24.5	37.6	53.6	71.3

表17. IM2GPS 测试集上的地理定位性能。指标为定位在给定半径内的图像百分比。模型按平均性能排序。[a](Muller-Budack 等，2018) [b](Hongsuck Seo 等，2018) [c](Vo 等，2017) [c](Weyand 等，2016)

E.4. 地理定位

在 CLIP 的开发过程中，我们注意到的另一种行为是它能够识别许多地点和位置。为了量化这一点，我们创建了 Country211 数据集（如附录 A 所述），并在整篇论文中报告了其上的结果。不过，由于这是一个新的基准，为了与以往的地理定位工作进行比较，我们还在表 17 中报告了 Hays & Efros（2008）的 IM2GPS 测试集上的结果。由于 IM2GPS 是一个回归基准，我们使用 CLIP 的嵌入空间，在一组参考图像中猜测最近图像的 GPS 坐标。这并不是零样本结果，因为它使用了最近邻回归。尽管只查询了 100 万张图像，远少于以往工作，CLIP 的表现仍与若干任务特定模型相近。然而，它并不能与当前最先进的水平相竞争。

E.5. 对分布偏移的鲁棒性

第 3.3 节对与 ImageNet 相关的鲁棒性结果进行了高层次总结和分析。我们在本附录中简要提供一些额外的数值细节。表 16 给出了各数据集的性能结果，并与 Taori 等人（2020）的评估套件中报告的当前最先进结果进行了比较。零样本 CLIP 在 7 个数据集中的 5 个上提升了当前最先进水平，包括 ImageNet-R、ObjectNet、ImageNet-Sketch、ImageNet-Vid 和 Youtube-BB。CLIP 在 ImageNet-Vid 和 Youtube-BB 上的提升最大，这得益于其灵活的零样本能力；在 ImageNet-R 上的提升也很显著，这可能反映了 CLIP 的预训练分布包含大量创意内容。Taori 等人（2020）讨论的基于 Instagram 预训练的 ResNeXt 模型也记录了类似行为。

F. Model Hyperparameters

Hyperparameter	Value
Batch size	32768
Vocabulary size	49408
Training epochs	32
Maximum temperature	100.0
Weight decay	0.2
Warm-up iterations	2000
Adam β_1	0.9
Adam β_2	0.999 (ResNet), 0.98 (ViT)
Adam ϵ	10^{-8} (ResNet), 10^{-6} (ViT)

Table 18. Common CLIP hyperparameters

Model	Learning rate	Embedding dimension	Input resolution	ResNet blocks	ResNet width	Text Transformer layers	Text Transformer width	Text Transformer heads
RN50	5×10^{-4}	1024	224	(3, 4, 6, 3)	2048	12	512	8
RN101	5×10^{-4}	512	224	(3, 4, 23, 3)	2048	12	512	8
RN50x4	5×10^{-4}	640	288	(4, 6, 10, 6)	2560	12	640	10
RN50x16	4×10^{-4}	768	384	(6, 8, 18, 8)	3072	12	768	12
RN50x64	3.6×10^{-4}	1024	448	(3, 15, 36, 10)	4096	12	1024	16

Table 19. CLIP-ResNet hyperparameters

Model	Learning rate	Embedding dimension	Input resolution	Vision Transformer layers	Vision Transformer width	Vision Transformer heads	Text Transformer layers	Text Transformer width	Text Transformer heads
ViT-B/32	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-B/16	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-L/14	4×10^{-4}	768	224	24	1024	16	12	768	12
ViT-L/14-336px	2×10^{-5}	768	336	24	1024	16	12	768	12

Table 20. CLIP-ViT hyperparameters

F. 模型超参数

Hyperparameter	Value
Batch size	32768
Vocabulary size	49408
Training epochs	32
Maximum temperature	100.0
Weight decay	0.2
Warm-up iterations	2000
Adam β_1	0.9
Adam β_2	0.999 (ResNet), 0.98 (ViT)
Adam ϵ	10^{-8} (ResNet), 10^{-6} (ViT)

表 18. 常见的 CLIP 超参数

Model	Learning rate	Embedding dimension	Input resolution	ResNet blocks	ResNet width	Text Transformer layers	Text Transformer width	Text Transformer heads
RN50	5×10^{-4}	1024	224	(3, 4, 6, 3)	2048	12	512	8
RN101	5×10^{-4}	512	224	(3, 4, 23, 3)	2048	12	512	8
RN50x4	5×10^{-4}	640	288	(4, 6, 10, 6)	2560	12	640	10
RN50x16	4×10^{-4}	768	384	(6, 8, 18, 8)	3072	12	768	12
RN50x64	3.6×10^{-4}	1024	448	(3, 15, 36, 10)	4096	12	1024	16

表 19. CLIP-ResNet 超参数

Model	Learning rate	Embedding dimension	Input resolution	Vision Transformer layers	Vision Transformer width	Vision Transformer heads	Text Transformer layers	Text Transformer width	Text Transformer heads
ViT-B/32	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-B/16	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-L/14	4×10^{-4}	768	224	24	1024	16	12	768	12
ViT-L/14-336px	2×10^{-5}	768	336	24	1024	16	12	768	12

表 20. CLIP-ViT 超参数